北京外国语大学

硕士学位论文

中文题目 中国学习者英语语调习得中外语口音的实验研究

外文题目 An Experimental Study of Foreign Accents in English Intonation Acquisition by Chinese Speakers

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研究方向 语音学

专　　业 外国语言学及应用语言学

系　　别 中国外语与教育研究中心

2025 年 5 月 22 日

An Experimental Study on Foreign Accents in English Intonation Acquisition by Chinese Speakers

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A thesis submitted in conformity with the requirements

For the degree of Master of Arts

In Linguistics and Applied Linguistics

Supervised by Su Jiajia

National Research Centre for Foreign Language Education

Beijing Foreign Studies University

22nd May 2025

摘要

大多数中国学生即使长期学习英语，仍有外语口音。语调偏差是很重要的一个方面，但研究却相对较少。英语是语调语言，而汉语是声调语言。“自主音系-节律”理论（Pierrehumbert 1980）认为，英语语调由音高重音，短语重音和边界调决定。但汉语作为声调语言，其语调受到字调的制约。这些跨语言差异造成了习得困难。

本文在二语语调习得理论（Mennen 2015）框架下，对汉语和英语的语调差异进行了详细对比，并依据对比结果设计了语调产出实验，重点考察总体语调、焦点实现和边界调三个方面。有 34 名中国学生和 5 名英语母语者参与了实验。实验包含两个部分：首先，基于中国学生的朗读语料，由英语母语者进行外语口音评分，将其划分为近母语学习者、高级学习者和中级学习者三组。其次，通过调控句子类型、焦点位置、句末词类型和句子长度这 4 个变量，设计出 36 个句子进行录音。使用语音软件 Praat 和 Prosodypro 脚本文件对英语母语者和英语学习者的基频曲线进行分析，并对音高重音和边界调进行 ToBI 标注，数据处理时长总计超过 1000 小时。研究发现如下：

第一，在总体语调方面，学习者的语调表现与其外语口音显著相关。外语口音越重，学习者则会表现出更多的音高变化，具体表现为基频变化量更大，以及基频曲线的波峰和波谷的数量更多，这与汉语的声调特性密切相关。但值得注意的是，句子的音高范围在各组之间未表现出显著差异。

第二，在焦点实现方面，近母语学习者与母语者在焦点音高重音的实现上更为相似，而中级学习者在多个变量上与高级学习更为接近。随着外语口音的减轻，焦点的音调一致性逐渐增强。句子类型和焦点位置是影响焦点实现的主要因素，句子的音节数对焦点实现的影响较小，而句末词类型则无显著影响。

第三，在边界调方面，各组表现均较为出色，近母语学习者和母语者略有差距，而高级学习者和中级学习者则相对较差。边界调的实现仅和句子类型有关，未发现其它句子变量对边界调的显著影响。

关键词：英语语调；汉语语调；二语语调习得理论；外语口音；产出实验

Abstract

Despite many years of learning English, most learners still leave trace of Foreign Accent, with intonation deviations being a significant contributing factor. English, as an intonation language, differs typologically from Mandarin, a tone language. According to the Autosegmental-Metrical theory, English intonation is determined by pitch accents, phrase accents, and boundary tones, while Mandarin's intonation is constrained by lexical tones. These fundamental differences in intonation between Mandarin and English may lead to negative transfer effects for Mandarin learners of English.

This study provides a detailed comparison of the intonational differences between Mandarin and English through the framework of L2 Intonation Learning theory (LILt). Based on the comparison, an experimental framework was developed, focusing on three aspects: overall sentence intonation, focus realization, and boundary tones. The study involved 34 Chinese learners of English, categorized into near-native, advanced, and intermediate groups, with 5 native English speakers as a control group. The materials consisted of 36 sentences by four variables: Sentence Type, Focus Position, Final Word Type, and Sentence Length (Syllables). Using Praat and ProsodyPro software, the fundamental frequency (F0) curves of native English speakers and learners were analyzed, and the ToBI system was employed to annotate pitch accents and boundary tones. Data analysis took over 1000 hours of work. Negative transfer was used to explain the intonational differences between native English speakers and learners, providing theoretical insights and practical implications for second language intonation teaching and learning.

The results indicated that in terms of overall sentence intonation, learners' intonation was significantly influenced by their accent proficiency. Learners with lower proficiency showed greater pitch variations, reflected in larger pitch change amounts and more F0 peaks and valleys, a pattern closely linked to Mandarin’s tonal system. However, no significant differences were found in pitch range across the groups.

In terms of focus realization, near-native learners were more similar to native speakers, while intermediate learners resembled advanced learners. As proficiency decreased, tonal consistency in focus realization diminished. Sentence Type and Focus Position were the primary factors influencing focus realization, while Syllables had minimal impact, and Final word type showed no significant effect.

In terms of boundary tone, all groups performed well, with near-native learners slightly outperforming native speakers, while advanced and intermediate learners performed relatively worse. Sentence Type was the only variable with a clear effect on boundary tone.

Key terms: English intonation, Mandarin intonation, LILt, Foreign Accent, Production Experiment

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Chapter 1 Introduction

This study investigates how differences in language typology—specifically the contrast between Mandarin, a tone language, and English, an intonation language—affect intonation patterns in both native English speakers and Mandarin learners of English. By comparing variables such as Sentence Type, Focus Position, Sentence Length and Final Word Type, the research aims to uncover the impact of these variables on native English intonation and L2 intonation. As a brief introduction, this chapter will outline research background, research aim and significance, and the organization of the thesis.

1.1 Research Background

Most language learners retain some degree of Foreign Accent (FA) when speaking a second language, with their pronunciation differing from that of native speakers. The accented English, common among non-native speakers, primarily results from the influence of the first language (L1) on the acquisition of the L2 sound system. As a consequence, Foreign Accents persist and are difficult to fully eliminate (Major 2001; Derwing and Munro 2009). In previous research, Foreign Accent can result from several factors, including speaker background factors and phonetic factors, consisting of both segmental and suprasegmental ones. Research has long focused on speaker-related factors such as the age at the beginning of learning (e.g., Piske and MacKay 1999), as well as the amount of L1 and L2 use (e.g. Flege et al. 1997), etc. Segmental factors, such as vowel quality, have also been extensively studied (e.g., Munro et al. 1999). However, Suprasegmental factors, which play a crucial role in the perception of FA, received much less attention. Indeed, Gut (2009) noted that only a very small fraction of second language acquisition (SLA) studies addressed L2 intonation, despite the persistence of non-target-like prosody in advanced learners (e.g., Grosser 1993; Rasier 2003). But it is worth noting that in production, deviations in intonation contours have been found to contribute to the perception of Foreign Accent (FA) (e.g., Jilka 2000; Boula de Mareüil, Marotta and Adda-Decker 2004; Trofimovich and Baker 2006; Ulbrich and Mennen 2016), but the relationship of segments and prosody in the contribution of FA is controversial.

Prosody, suprasegmentals, and intonation are often used interchangeably, but in this study, the first two are treated as equivalent broader concepts with intonation more being a component. Prosody is an umbrella term used for anything relevant for speech and rhetoric, including features like pitch, duration, loudness, and speech rate (Ladd 2008). Suprasegmentals are the features of speech that extend beyond individual segments (consonants and vowels) (Roach 2009). Intonation, as defined in this study, specifically refers to pitch variations across speech. It can be likened to the "melody" of a sentence, with pitch changes shaping the tone and meaning of spoken language. Pitch is the perceived frequency of sound, while F0 (fundamental frequency) is the acoustic measure, both representing the frequency of vocal fold vibration. They are often used interchangeably, as will be the case in this study. Intonation, as the pitch parts of prosody, will be the focus of this study.

The native language, as a key influence on L2 intonation, deserves more attention in understanding its role. Mandarin Chinese is a tone language, whereas English is an intonation language, and there are typological differences between the tone and intonation language in prosody (Yip 2002; Hyman 2006). In Mandarin Chinese, there are four tones (yinping (level), yangping (rise), shangsheng (fall rise), and qusheng (fall)), each of which is used to differentiate meaning. Every syllable requires a specific pitch movement to convey its tone, and as a result, the intonation of a sentence is largely determined by the tonal contours of its individual syllables. In contrast, intonation languages do not assign tone specifications at the word or syllable level, which means the tone of a word or syllable in English does not change the meaning of the word. Instead, they use phrasal tones to assign melodies to words and sentences (Yip 2002; Gussenhoven 2004; Ladd 2008; Féry 2017). The typological differences between Mandarin and English can present challenges for Mandarin learners of

English. As a result, Chinese English speakers may tend to carry over the intonation qualities of Mandarin Chinese to English intonation learning.

Of the few studies on L2 intonation of Chinese English speakers, recent research has largely focused on the impact of differences between L1 and L2 intonation on L2 intonation learning (e.g., McGory 1997; Ding et al. 2016; Liu and Chen 2016; Yuan et al. 2018; Liu et al. 2021). However, there are some common limitations in the current body of research.

First, there has been insufficient in-depth comparison of the intonational differences between English and Mandarin, making it hard to pinpoint the specific reasons for the intonational differences between L2 English and native English. Moreover, the lack of in-depth comparison between intonations affects the target selection of measurements since there are too many measurements related to pitch. Furthermore, a lack of targeted measurements in return weakens the explanatory power of the transfer effects, as it becomes difficult to clearly link them to the L1-L2 differences.

Second, most studies focus on intermediate or advanced learners, or even group all learners together (e.g., Yuan et al. 2018), with little attention given to how near-natives (GroupNN) differ from native speakers in terms of intonation. The importance of GroupNN (near-native speakers) in this study lies in the subtle yet significant differences they exhibit compared to native speakers. Although their Foreign Accent (FA) scores are not on par with native speakers, the differences are often too small to pinpoint precisely. This makes it challenging to identify where the differences lie, which is why experimental investigation is necessary to understand these distinctions more clearly.

Third, there is still limited application of theoretical frameworks for L2 intonation, with Mennen's L2 Intonation Learning theory (LILt) (Mennen 2015) being among the few to address this in depth. The complexity of prosody poses a particular challenge for SLA theories because of the difficulty in isolating the categorical or phonological sources of influences from gradient or phonetic sources (Graham and Post 2018). Current L2 speech models– such as Flege's Speech Learning Model (Flege 1995), Best's Perceptual Assimilation Model (Best 1995; Best et al. 2001; Best and Tyler 2007) and Eckman's Markedness Differential Hypothesis (1996) only attempt to account for segmental errors and the ease or difficulty of L2 speech learning. The inadequacy of current L2 speech models for prosody and the complexity of prosody created a gap in SLA until Mennen (2015) proposed the L2 Intonation Learning theory (LILt), which works on the premise that cross-language differences in intonation can occur along four intonation dimensions and predictions can then be made on the difficulty of L2 intonation learning. There are four dimensions: (i) the systemic dimension, which refers to the inventory and distribution of structural phonological elements; (ii) the realizational dimension, which refers to the way the systemic elements are phonetically implemented; (iii) the semantic dimension, which refers to how systemic elements are used to signal intonation function; and (iv) the frequency dimension, which refers to the frequency of use of the structural elements. Some studies, such as Liu and Chen (2016) and Liu et al. (2021), have already applied Mennen's (2015) framework. However, further research is needed to explore this area in greater depth.

In light of this LILt, this study investigates intonational differences from systemic and realizational dimensions (the more fundamental two) between Mandarin and English. Predictions on L2 English intonation by Chinese English learners will be made based on the investigation and will be testified through experiments.

1.2 Research Aim and Significance

This study aims to examine the intonational differences between Mandarin and English through the lens of LILt. By exploring variables such as Sentence Type, Focus Position, Sentence Length (Syllables), and Final Word Type, the research seeks to discern intonational variations between native English speakers and Chinese English learners through measurements. The study also includes learners at three levels of

Foreign Accent proficiency, including GroupNN (near-natives), an area underexplored in previous research. This work contributes to a deeper understanding of L2 intonation acquisition and offers insights to improve language learning and teaching practices, particularly for learners at varying proficiency levels.

1.3 Organization of the Thesis

This paper compares the intonation patterns of Mandarin and English based on the L2 Intonation Learning theory (LILt), and then subsequently examines how variables such as Sentence Type, Focus Position, Sentence length, and Final word type affect the intonation of both native English speakers and Mandarin learners of English. The study is structured into six chapters.

In **Chapter 1**, we introduce the research background, research aim and significance, and organization of the thesis, laying the foundation for the investigation.

Chapter 2 presents the literature review. This chapter first provides a detailed introduction to LILt and explains how we used it as a basis for comparing Mandarin and English intonation, followed by a brief introduction of the AM model and ToBI system, since they are foundations for English intonation. We then reviewed previous studies and made comparisons between English and Mandarin intonation. Our framework was developed based on the comparisons.

Chapter 3 covers the research design, including the research questions, participants, materials, and procedures. This chapter also provides an in-depth explanation of the data analysis process, detailing the methods employed to process and analyze the collected data.

Chapter 4 is the most significant part of the paper, where we present the results comprehensively according to our framework. In this chapter, we illustrate the findings related to intonation patterns in both native and L2 English speakers, as well as the impact of the investigated variables on these patterns.

Chapter 5 builds on the results and revisits the three main research questions. We discuss how intonational differences between Mandarin and English bring effects to the intonation of Mandarin learners of English, which shows similarities and differences with that of native English. This chapter integrates the findings and draws connections between the results and our framework set out earlier in the paper.

Chapter 6 concludes the paper by summarizing the main research findings. It highlights the key insights drawn from the study. Moreover, we will give our implications for intonation learning, teaching and also LILt itself. Finally, this chapter discusses its limitations and provides suggestions for areas where further exploration could offer valuable contributions.

Chapter 2 Literature Review

In the Literature Review section, we do not follow a traditional routine to write. This section is more of a theoretical framework section for the reason that our research framework is concise though intricate, based on theories and our comparison results of Mandarin and English intonation. It takes a whole section to present our framework at the end. Previous studies on L2 intonation and Mandarin and English intonation will be merged in our comparison in section 2.3.

To introduce, we will first introduce the L2 Intonation Learning theory (LILt), followed by a brief introduction of the AM model and ToBI system, which are foundations for analyzing English intonation. Finally, a detailed comparison of Mandarin and English intonation based on LILt will be given by reviewing previous studies. This comparison will inform our framework for measurements.

2.1 The L2 Intonation Learning Theory

2.1.1 Overview of LILt

As previously mentioned, most learners, even after many years of exposure to the L2, still retain traces of a Foreign Accent (FA) (e.g., Munro 1995; Munro and Derwing 1995; Magen 1998; Jilka 2000; Mennen 2004). Intonation being a part of FA trace will be briefly testified in our pre-experiment. The acquisition of intonation in a second language (L2) poses unique challenges for learners, especially compared to segmental features like consonants and vowels. While much research has focused on segmental aspects of L2 speech, prosodic features such as intonation have received much less attention. Existing models of L2 speech learning, such as Flege's Speech Learning Model (Flege 1995), Best's Perceptual Assimilation Model (Best 1995; Best, McRoberts and Goodell 2001; Best and Tyler 2007), and Eckman's Markedness Differential Hypothesis (1996), primarily address segmental acquisition.

The L2 Intonation Learning Theory (LILt) was introduced to address this gap and aims to explain the difficulties L2 learners face in the production of L2 intonation

(Mennen 2015). LILt proposes that cross-language differences in intonation can occur along four dimensions: **systemic, realizational, semantic, and frequency**. Although these dimensions may overlap, they provide a framework for predicting L2 intonation learning based on the differences in intonation systems between languages.

LILt builds on segmental acquisition models such as SLM and PAM-L2. They propose that the perception of L2 segments depends on their similarity to L1 categories. When L2 segments closely match L1 categories, they are perceptually assimilated (PAM-L2) or equivalence-classified (SLM), leading to production deviations. However, LILt goes further by considering both intonational and phonological differences between languages. As Mennen (2015) notes, "It (LILt) should therefore be treated as an evolving or "working" model, which is subject to change as more data are published." Our goal is to contribute additional data that may help refine and further develop the theory.

2.1.2 Framework in LILt

The L2 Intonation Learning Theory (LILt) is built around four key dimensions that can characterize similarities and differences between L1 and L2 intonation. (modified from Ladd 1996). The L2 Intonation Learning Theory (LILt) distinguishes between **phonological** and **phonetic** aspects of intonation as the Autosegmental-Metrical (AM) approach (Pierrehumbert 1980; Pierrehumbert and Beckman 1988), which is significant in English intonation and will be introduced later.

Systemic Dimension: This dimension refers to the inventory and distribution of **phonological** elements in a language, such as pitch accents and boundary tones. This dimension focuses on the structure of intonation in a language, including the types of intonational elements that are available.

Realizational Dimension: The realizational dimension focuses on how these phonological elements are **phonetically** realized in speech. It includes aspects such as pitch alignment (timing of pitch movements relative to segments), **s**caling (the relative height of pitch), and shape (e.g., steep vs. shallow pitch contours). It is about how the systemic elements are realized acoustically in actual speech.

Semantic Dimension: This dimension involves the function and meaning conveyed by the intonational elements. For example, languages may differ in how they mark focus or interrogativity.

Frequency Dimension: The frequency dimension looks at how often certain intonational patterns are used in speech. This includes understanding how certain pitch patterns are more common in specific languages or contexts.

In all, LILt facilitates the comparison of L1 and L2 intonation patterns by analyzing the systemic, realizational, semantic, and frequency dimensions. As Mennen (2015) suggests, these dimensions overlap; for instance, a deviation in the realizational dimension may lead to a semantic or functional deviation. However, the first two dimensions are more fundamental, representing the phonological elements and their phonetic realization. In our framework, we will focus on the first two dimensions—the **systemic dimension** and **realizational dimension**—to streamline the analysis, while still representing the full model. Although all four dimensions are included in the experiment, these two serve as the primary representatives for measuring intonation.

2.2 Autosegmental-Metrical (AM) Model and Tones and Break Indices (ToBI) System

Before delving into the comparison of Mandarin and English intonation, we would like to briefly introduce Autosegmental-Metrical (AM) model (Pierrehumbert 1980), which lays foundation for analyzing English intonation using autosegmental phonology, and Tones and Break Indices (ToBI) system, developed by Pierrehumbert (1980) and further refined by Beckman and Pierrehumbert (1986), providing a set of conventions for annotating prosodic features, focusing on pitch accents, boundary tones, and prosodic phrasing. Contours and explanations of the set of pitch events will be shown in section 2.2.2—Tones and Break Indices system.

2.2.1 Autosegmental-Metrical Model

To address the question "how is English intonation represented phonologically", Pierrehumbert (1980) first established a set of pitch events in English, categorizing them into three types: pitch accents, phrase accents, and boundary tones (see Table 2.1).

Table 2.1 Tonal Categories in Pierrehumbert's (1980) AM Model

Category	Types
Pitch Accents	H*, L*, H*+L- , H-+L*, L*+H- , L-+H*, H*+H-
Phrase Accents	H-, L-
Boundary Tones	H%, L%
Combinations of Phrase Accents and Boundary Tones	L-L%, H-H%, L-H%, H-L%

Instead of treating intonation as a continuous F0 contour, the AM model decomposes it into three main components: pitch accents, phrase accents, and boundary tones as above, which together form a complete intonational contour. The sequencing of these tones follows a finite-state grammar, ensuring a structured and predictable representation of intonation (see Figure 2.1).

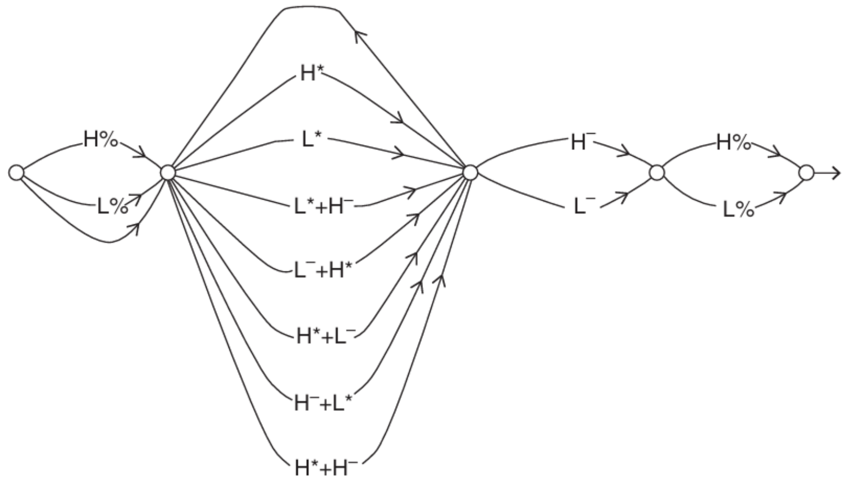


Figure 2.1 Finite State Grammar of Tone Sequences (From Pierrehumbert 1980:29)

A well-formed intonational phrase (IP) in English is composed of one or more pitch accents and it ends with an obligatory phrasal tone (or phrase accent) and an obligatory boundary tone.

In terms of pitch accents, they are used to mark prominent syllables within an IP. Pitch accents align with the primary stressed syllable of a word but do not necessarily appear on every stressed syllable. There are seven types of pitch accents: H*, L*, H*+L- , H-+L*, L*+H- , L-+H* and H*+H-, including two monotonal and five bitonal pitch accent. Starred tone is associated with a metrically strong syllable. For bitonal tones, they always consist of a starred tone followed by another non-starred tone, which can either precede the starred tone (a leading tone) or follow it (a trailing tone).

In terms of phrase accents, they are boundaries of intermediate phrases. An IP consists of one or more intermediate phrases. Every intermediate phrase must contain one pitch accent and one phrase accent. There are two types of phrase accents, including H- and L-, occurring at the edge of an intermediate phrase.

In terms of boundary tones, they signal the end of an intonational phrase, marking intonational phrase boundaries and contributing to the overall sentence meaning (e.g., distinguishing statements from questions). Boundary tones are combined with phrase accents at the end of an IP, consisting of four types: L-L%, H-H%, L-H%, H-L%.

An exemplar below shows how pitch accents, phrase accents and boundary tones are distributed in an IP "There are many intermediate levels. " (see Figure 2.2)

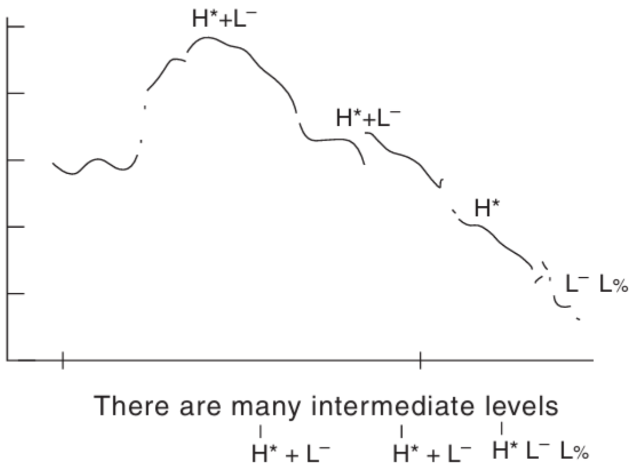


Figure 2.2 "There are MANY INTERMEDIATE LEVELS. " (From Pierrehumbert 1980:329)

2.2.2 Tones and Break Indices System

ToBI (for Tones and Break Indices) is a system (Silverman et al. 1992) for transcribing the intonation patterns and other aspects of the prosody of English utterances (Beckman and Elam 1997). The ToBI system is essentially an application of the Autosegmental-Metrical (AM) theory for intonational annotation. It uses the theoretical concepts of AM theory (such as pitch accents and boundary tones) to provide a practical and standardized method for intonation analysis.

There are four tiers in ToBI, namely, tone tier, break index tier, orthographic tier and miscellaneous tier. The tone tier consists of labels for distinctive pitch events such as pitch accents, phrase accents and boundary tones, which are similar to those in AM model but with some modifications. The break tier marks the prosodic grouping of words in an utterance by labeling the strength of their connection to the following word, ranging from 0 (strongest connection) to 4 (most disjoint). The orthographic tier is a transcription tier and the miscellaneous tier is a "comment" tier (Beckman and Elam

1997). These two tiers are not essential for prosodic analysis and will be skipped. Although the break tier plays an important role in the segmentation of intermediate phrases, we will not discuss it in detail here, as the results presented in this paper do not include break tier annotations. Figure 2.3 is an exemplar of ToBI annotation in our work with an orthographic tier, a phone tier (made by MFA, introduced later) and a tone tier.

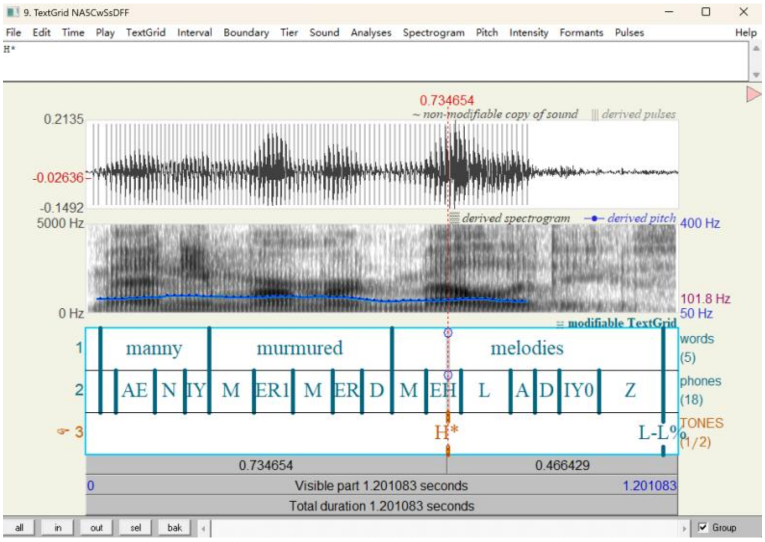


Figure 2.3 An Exemplar of ToBI Annotation

The categories of pitch accents, phrase accents, and boundary tones in ToBI are shown in Table 2.2. Due to space limitations, we will only cover the most significant pitch accents, including H*, L*, L+H*, and L*+H. All boundary tones, however, will be discussed in full.

Table 2.2 Pitch Events in ToBI

Pitch Accents	**Phrase Accents and Boundary Tones**
H* (!H*) ; L*; L+H* (L+!H*); L*+H (L*+!H); H+!H*	L-; H- (!H-) L-L%; H-H%; L-H%; H-L%

Table 2.3 shows four types of pitch accents with their corresponding pitch contours in exemplars, adapted from the figures on the University of Calgary Linguistics website (University of Calgary Linguistics, n.d.).

Table 2.3 Four Types of Pitch Accents in ToBI

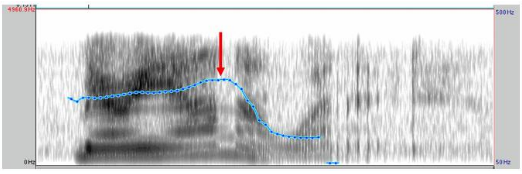
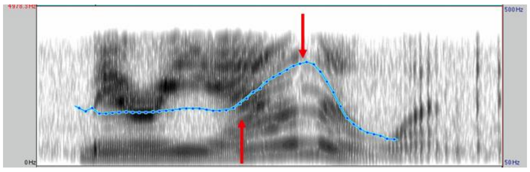
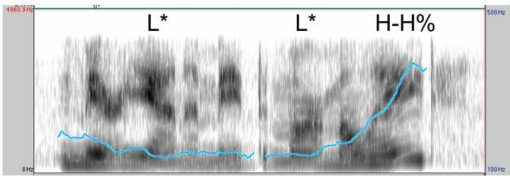
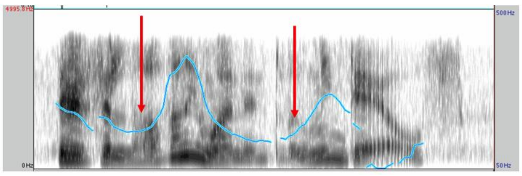
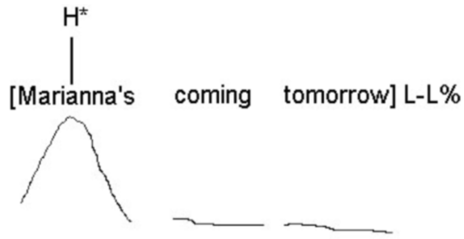
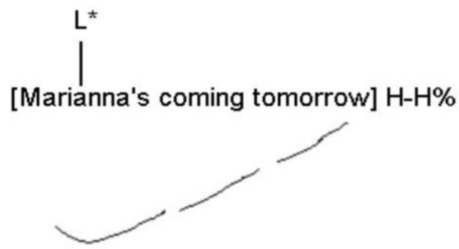
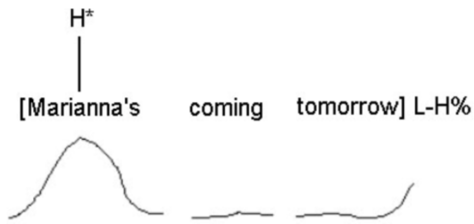
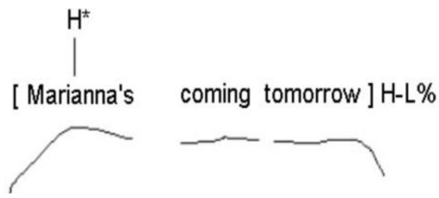
Pitch Accents	Explanations	Exemplars
H*	A pitch peak on, or near, the accented word's primary stressed vowel.	
L+H*	High peak on stressed syllable, preceded by a sharp rise in pitch	
L*	A pitch trough, on, or near, the accented word's primary stressed vowel	
L*+H	Low pitch target on stressed syllable, followed by a sharp rise in pitch.	

Table 2.4 shows the four boundary tones along with their respective pitch contours in exemplars. The figures are adapted from Macquarie University (n.d.).

Table 2.4 Boundary Tones in ToBI

Boundary tones	Explanations	Exemplars
L-L%	This boundary tone typically marks "neutral" statements, with the pitch ending at a low value at the prosodic boundary.	
H-H%	Often found in "yes-no" questions, this tone ends with a high pitch at the boundary.	
L-H%	Known as a continuation-rise, the pitch starts low and rises at the prosodic boundary. The rise in L-H% is not as sharp as that in H-H%.	
H-L%	Common in list recitals, this tone conveys a somewhat indifferent tone, with the pitch starting high and gently falling at the boundary.	

In all, this section introduces the Autosegmental-Metrical (AM) model and the Tones and Break Indices (ToBI) system, both crucial for analyzing intonation. The AM model, proposed by Pierrehumbert (1980), divides intonation into three components: pitch accents, phrase accents, and boundary tones, offering a theoretical framework for English intonation. The ToBI system, an application of the AM model, provides a standardized method for annotating intonation patterns, focusing on pitch accents, phrase accents, and boundary tones, with a detailed four-tier structure. The ToBI system also incorporates a break index tier to mark prosodic groupings of words. However, in our Results, we will not show break index tier and specifically focus on tone tier.

2.3 Comparison of Mandarin and English Intonation

In light of LILt, in this section, we will make an in-depth comparison of Mandarin and English intonation by reviewing previous studies. Our own framework for measuring native English and L2 intonation will be given at the end. As mentioned earlier, cross-language similarity/dissimilarity in the **systemic** or phonological dimension concern typological similarities or differences in the inventory of structural phonological elements, including **pitch accents** and **boundary tone.** Therefore, the order of comparison in this study is as follows: first, the overall sentence intonation in Mandarin and English, then the focus realization (pitch accents) in Mandarin and English, and finally, boundary tones in Mandarin and English. We begin by comparing the **systemic** aspects of these three elements before examining their **realizational** counterparts.

2.3.1 Overall Sentence Intonation in Mandarin and English

2.3.1.1 Systemic Dimension

Mandarin and English belong to different language typology, as it is widely accepted that Mandarin Chinese is a typical tone language, whereas English is a typical non-tone language (Yip 2002; Hyman 2006). Tones are defined in this study as an abstract phonological (discrete) category, which has a concrete F0 component: it can be a level tone, H or L for instance, or a contour tone, rising (LH) or falling (HL), or even a more complex contour. Note here tones are not defined the same as Ladefoged and Johnson wrote (2011:255), "pitch variations that affect the meaning of a word are called tones. " Instead, in this study, both Mandarin and English have their tones, but functions are different. In Mandarin Chinese, there are four tones and neutral tone (toneless). The tones occur on all syllables in a sentence and Mandarin uses pitch to signal lexical contrast (i.e., lexical tones; Duanmu 2007), that is to say, one syllable pronounced in different tones will usually mean different things. For example, "shou" in Tone3 can mean "hand" but "shou" in Tone4 usually means "thin". On the other hand, as a none-tone language, no invariant tonal contour in English is assigned at the syllables. The tones in English defining the sentence melody have their own grammar, as introduced before in AM model (Pierrehumbert 1980), and they associate at different points in the sentence, an important anchor being the lexical stresses (Féry 2017).

In Mandarin Chinese, pitch contour at the utterance level is an interaction of syllable tones and the sentence intonation. The relationship of syllable tones and sentence intonation has its delicate metaphor "small ripples on large waves" described by Chao (1968). By contrast, in English, a sentence may have one or more intermediate phrases, with which stressed syllables usually have a pitch change; but there is at least one stressed syllable that stands out because it carries major pitch change. This syllable carries the pitch accent and the last accented word in any prosodic phrase is nuclear accented (Newman 1946; Bolinger 1958). If every syllable has a lexical tone of its own, it is unrealistic to expect that it will have additional tones at the sentence level (i.e., Mandarin). In the opposite, if a language has no tonal specification at all at the level of the word, sentence intonation could be richer (i.e., English) (Féry 2017). For Chinese English learners, the richer and "uncertain" sentence intonation of English may cause great difficulty in language learning. Thus, it is necessary to investigate the realizational similarities and differences between Mandarin and English overall sentence intonation, and deviations in L2 English found by previous research.

2.3.1.2 Realizational Dimension

The tonal characteristics of Mandarin and none tonal ones of English bring some effects on realizational or acoustic data such as number of F0 peaks and valleys, pitch change amount and pitch range on the utterance level. A body of literature directly compared Mandarin and English intonation. For example, Chen (1972) asked a small amount of English and Mandarin speakers to read words and sentences, in order to compare the mean, standard deviation, and range of F0. The Mandarin speakers, especially the women, have wider F0 ranges and larger standard deviations; the Mandarin women's means are lower, while the men's are the same as the English speakers. In contrast, Eady (1982) compared several measures of F0 from passages read by male Taiwan Mandarin and English speakers. He claimed that F0 patterns of Chinese have higher mean and a greater amount of dynamic movements than those of English, but the standard deviation (taken as the measure of F0 range) is the same in the two languages. Xue et al. (2002) found the younger bilinguals have lower minimum F0 and larger F0 range in their Mandarin valleys than English. Interestingly, Keating and Kuo (2012) found Mandarin has a wider pitch range only in single word utterances but not in prose passage or story voices. And they suggest that the pitch range difference between Mandarin Chinese and English is not due to some generic property of tone languages, but specifically because of Mandarin's high falling tone (Tone 4). Yuan and Liberman (2014) compared broadcast news in Mandarin Chinese and English through copra. The results show that Chinese has wider pitch range and more F0 fluctuations than English, probably due to the effect of lexical tones (see Table 2.5).

Table 2.5 Comparison of Mandarin and English Intonation in Realizational Dimension

Study	Comparison	Findings
Chen (1972)	English and Mandarin	Mandarin speakers, especially women, had wider F0 ranges and larger standard deviations compared to English speakers. Mandarin women's means were lower, while men's were similar to English.

Table 2.5 (Continued)

Study	Comparison	Findings
Eady (1982)	Taiwan Mandarin and English	F0 patterns of Chinese have higher mean and greater dynamic movements than English, but the standard deviation (measure of F0 range) was similar in both languages.
Xue et al. (2002)	Mandarin and English	Younger bilinguals exhibited lower minimum F0 and larger F0 range in Mandarin valleys compared to English.
Han et al. (2011)	Tone languages and non-tonal languages	Tonal languages exhibit more pitch direction changes and larger pitch intervals than non-tonal languages, suggesting a connection between tonal characteristics in music and speech.
Keating and Kuo (2012)	Mandarin and English	Mandarin had a wider pitch range only in single word utterances, not in prose passages or story voices. The pitch range difference between Mandarin and English was attributed specifically to Mandarin's high falling tone (Tone 4).
Yuan and Liberman (2014)	Mandarin and English	Mandarin Chinese showed wider pitch range and more F0 fluctuations than English, possibly due to lexical tones.

Additionally, previous studies also made contribution to direct comparison of L2 English and native English. Ding et al. (2016) made comparison of the F0 patterns in English, Mandarin Chinese and L2 English produced by Chinese speakers and found both pitch range on the phoneme level and pitch change amount on the utterance level are greater in L2 English speech by Chinese subjects than the English speech by native speakers. As for pitch range on the sentence level, no significant difference was found between English and L2 English, except for female speakers. Moreover, Yuan et al. (2018) found that compared to L1, L2 English has even narrower pitch range and slower rate of pitch change through a large-scale study.

Taken above literature, most scholars agree that Mandarin has more peaks and valleys and more F0 movements than English. They may be attributed to the nature of tone and non-tonal language. In order to more accurately capture the potential differences between L2 English and native English, that is to say, L2 English may tend to give more tones on syllables, we plan to measure the pitch change amount on the syllable level and the number of F0 peaks and valleys on the utterance level. The choice of these two variables is deliberate. If only one variable is selected for comparison, a potential paradox arises. For instance, if the number of F0 peaks and valleys is chosen as the sole measurement, a higher count in L2 English may not necessarily indicate larger F0 movements. This could be due to L2 English exhibiting small pitch fluctuations that result in more numerous, yet less prominent, peaks and valleys (see line 1 in Figure 2.4), which does not reflect significant F0 variations. Similarly, if pitch change amount on the syllable level is selected as the sole measurement, a higher value for L2 English (normalized by syllable, given our research design that treats syllables as a variable) does not conclusively demonstrate negative transfer from Mandarin. It might simply reflect substantial F0 movement on individual words, rather than a broader tonal influence (see line 2 in Figure 2.4). Therefore, to establish that L2 English is influenced by Mandarin, it is necessary for both the number of peaks and valleys and the pitch change amount in L2 English to exceed those in native English (see line 3 in Figure 2.4).

Figure 2.4 Assumed F0 Curves Under Different Conditions

However, previous studies were mixed about F0 range with respect to Mandarin, English and L2 English. We agree with the argument made by Keating and Kuo (2012:1059), who suggest that "whether the two languages will appear to have similar or different F0 profiles very much depends on the speech corpus or task, as well as on the acoustic measures." Research needs to delve deeper, especially in the context of L2 English. Previous studies showing differences between languages may have been influenced by factors such as Tone 4 in Mandarin, which could affect the results (Keating and Kuo 2012). Additionally, when Ding, Jokisch, and Hoffman (2012) examined L2 German production by Chinese speakers, they found no significant differences in pitch range at the sentence level among groups, similar to their study on L2 English by Chinese speakers (Ding et al. 2016). Based on these findings, we initially hypothesize that there is no significant difference in pitch range on the utterance level between Mandarin and English, and thus assume no significant difference in pitch range between L2 English and native English. However, this assumption will be further examined once the specific data is available.

In this section, we analyzed the systemic and realizational similarities and differences between Mandarin and English overall sentence intonation. Based on our comparison, we have selected targeted measurements and provided preliminary predictions (see Table 2.6)

Table 2.6 Measurements and Predictions in Overall Sentence Intonation

Systemic dimension	Realizational Dimension		
Tone/non-tonal language	Pitch Change Amount on the Syllable Level	Number of F0 Peaks and Valleys on the Utterance Level	Pitch Range on the Utterance Level
	L2 English > Native English	L2 English > Native English	L2 English ≈ Native English

2.3.2 Focus Realization in Mandarin and English

Comparison of Mandarin and English focus realization is divided into systemic dimension and realizational dimension. Due to space limit, we will only measure systemic dimension in our experiment. ToBI will be used for systemic annotations. Nevertheless, in the Results section, we will include ToBI annotations and F0 curves as a compensation for the lack of experimental data on the realizational dimension. Previous research on L2 focus realization will also be merged into the two dimensions. We will include different Sentence Type (Declarative and Yes-no Questions), Focus Position (Broad Focus and Narrow Focus), and Final Word Type (Content Word and Function Word) in this section.

2.3.2.1 Systemic Dimension

From a phonological perspective, focus in English is mainly achieved by giving prominence to lexically stressed syllables, with pitch accents marking the emphasized items (Gussenhoven 1984). In English, we have a set of pitch accents such as H*, L*, L+H*, L*+H and H+!H* to mark focus. For example, H* can be used to express new information. However, as a typical tone language, Mandarin assigns a lexical tone to each syllable. The existence of pitch accents in Mandarin is debated, and we do not intend to delve into this issue here. However, if we follow Lin (2004), who argues that

Mandarin does have pitch accents, the primary distinction between Mandarin and English lies in the absence of a corresponding phonological element like L* in Mandarin. Wang (2002) found that the increase in the top line is the primary acoustic manifestation of pitch change in Chinese stressed syllables, while the change in the bottom line is more limited, suggesting a tendency to use the H tone on stressed syllables in Mandarin. In contrast, L* H-H% pitch patterns are typical in English yes-no questions.

Of the few research on pitch accent in Mandarin and L2 English, it is noteworthy that L*+H pitch accent, which is often used by native speakers in yes-no questions, is missing in Mandarin speakers (Ji 2009). What is more, in L2 English, it was indeed widely found that LH*L pitch pattern (e.g., !L+H*+L, which means a preceding L boundary tone and then a H*L bitonal pitch accent, with H tone occurring on a stronger position) is overgeneralized by Chinese English learners (McGory 1997; Shack 2000). Learners repeatedly use the LH*L pattern (see Figure 2.5) in all sentence types reveals that their insufficient proficiency in pitch accent in English and it may be due to the negative transfer from Mandarin.

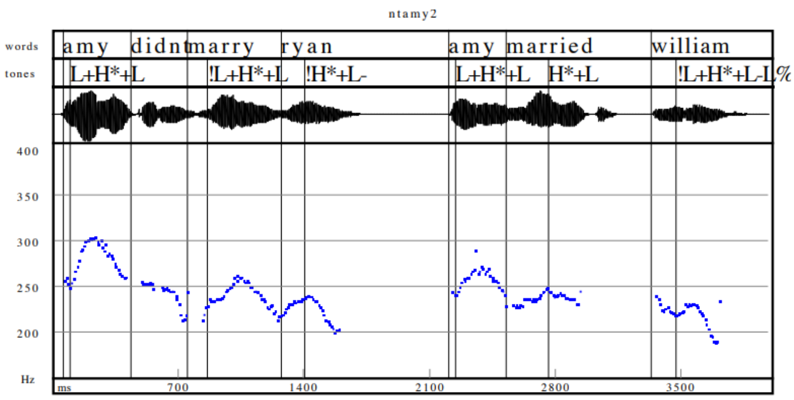


Figure 2.5 Longer Sentence Demonstrating LH*L Pattern in Chinese English Speaker (Shack 2000)

Therefore, we predict that in L2 English, less L*-related pitch patterns will be used in yes-no questions, showing deviations with native English.

2.3.2.2 Realizational Dimension

In the realizational dimension, we will first examine how narrow focus has been specifically realized. Additionally, broad focus will be discussed in conjunction with the Final Word Type, exploring the similarities and differences between Mandarin and English.

Xu (2005) concluded that a narrow focus is realized by expanding the pitch range of the on-focus stressed syllables, suppressing the pitch range of post-focus syllables, and leaving the pitch range of pre-focus syllables largely intact. This has been largely demonstrated in previous research on Mandarin and English (e.g., Liu and Xu 2007; Liu 2009). Moreover, in L2 English, it was found that learners realize focus in the similar way with native speakers do (e.g., Hu et al. 2012). However, in the study by Hu et al. (2012), they found that Beijing L2 English speakers exhibit less F0 variation on the focused word. They explained this from an articulatory strength perspective: articulatory strength determines how quickly a local pitch target is reached. Strong articulatory strength leads to a faster approach to the target, whereas weak strength results in a slower approach (Xu 2005). In tone languages like Mandarin, metrical structure is based on alternating articulatory strengths, rather than pitch targets (Kochanski and Shih 2003). In contrast, English intonation primarily relies on alternations between high and nonhigh pitch targets, with articulatory strength playing a secondary role (Liberman and Prince 1977). Simply put, while English uses pitch contrasts to emphasize items, Mandarin relies on varying articulatory strength. This results in less pitch contrast in L2 English compared to native English. Similarly, Liu et al. (2021) found that L2 speakers exhibit lower peaks and slopes in focus realization compared to native English speakers. By contrast, Liu and Chen (2016) found no significant difference between L2 English and native English in the pitch range of pitch-accented syllables. Therefore, even though our study does not extract realizational data such as peak range or F0 slope, we attempt to directly compare the F0 curves and ToBI results between groups. Predictions on this topic will not be made for varying conditions and results in previous studies.

In terms of broad focus sentences (also serve as control sentences in this study), the placement of nuclear stress is a crucial issue. The Nuclear Stress Rule (NSR) was first proposed by Chomsky and Halle (1968), stating that the main stress is assigned to the rightmost primary stressed syllabled in a phrasal component. Cinque (1993), whose depth stress principle is influential in Chinese syntax, revised the Nuclear Stress Rule (NSR) by proposing in the absence of discourse-related factors, the main stress of a sentence always falls on the most deeply embedded constituent within the branching structure of the syntactic tree. The constituent that carries this main stress is thus identified as the broad focus of the sentence. All major works seem to assume that exceptions to the NSR exist, driven by certain lexical or interpretive properties. Below is an example given by Selkrik (1984):

```
                                    x
                  x                 x
      x           x     x           x
The mayor of Chicago won their support.
                        x
                  x     x
      x           x     x
The mayor of Chicago won it.
```

In Mandarin, the position of broad focus is more complex. Duanmu and Li (2018) found that even in neutral contexts, sentence stress does not uniformly fall at the sentence-final position. Various constituents (subject, adverbial, verb, object) can all be chosen for stress by the speakers. Other studies, such as Wang et al. (2006), also failed to find a fixed focus stress position. This suggests that there may be no clear principle or strategy in Chinese to place the broad focus at the end of the sentence (Duanmu and Li 2018).

In English, exceptions to the NSR are relatively fixed, such as given items, anaphors, indefinites, and functional words. However, in Chinese, the Focus Position may not be as grammaticalized. Therefore, we differentiate between content words and function words using the variable Final Word Type to examine the L2 English performance. Similarly, we do not make any predictions on this topic.

In this section, we analyzed the systemic and realizational similarities and differences between Mandarin and English focus realization. We will mainly pay attention to systemic dimension in our measurements, while in realizational dimension, we do not make a specific prediction and data are supplemented from F0 curves and ToBI annotations (see Table 2.7).

Table 2.7 Measurements and Predictions in Focus Realization

Systemic dimension	Realizational Dimension	
Pitch Patterns (ToBI)	*Peak range and F0 slope of focus*	*Place of broad focus*
L2 English: less L*-related in yes-no questions	?	

2.3.3 Boundary Tones in Mandarin and English

In this section, we focus on the systemic dimension of boundary tones and briefly discuss their realization in L2 English. Given the challenges of accurately measuring boundary tones, the comparison in the realizational dimension will not be included.

2.3.3.1 Systemic Dimension

In English, boundary tones are best captured by the Autosegmental-Metrical (AM) model (Pierrehumbert 1980) The boundary tone distinction between English statement and question sentence (i.e., L-L% in statements and H-H% in yes-no questions) only targets at the right edge (i.e., the end) of an intonational phrase (Ladd 1996). Specifically, Liu and Xu (2006) found that F0 curves in statements and questions start to diverge from the stressed syllable of the first content word, whether or not it is focused. In Mandarin, there are still disputations over whether the boundary tone can be described as the element of intonation. Most scholars hold a negative view (e.g., Chih and Greg 2002; Ho 1976; Shen 1985; Shen X.N. 1990, Shi 1980) while others (e.g., Rumjancev 1972; Hu 1987; Lin 2004) support sentence-final syllables' decisive roles in contrast of Mandarin statements and questions (for an overview see Jiang and Chen 2011) as in English. Specifically, Ho (1997) found a much higher pitch is used throughout a question than in a statement in Mandarin Chinese. Similarly, Yuan (2006) found that the F0 contour of questions is globally higher than that of statements (Phrase Curve Mechanism). He also discovered that sentence-final tones in questions are more intense and longer compared to statements (Strength Mechanism). Furthermore, he observed that the final falling tone flattens, while the rising tone steepens, making question intonation more perceptible when the sentence ends with a falling tone (Tone 4) and harder to identify with a rising tone (Tone 2). Yuan's third finding highlights a key difference between Mandarin, a tonal language, and English. In Mandarin, a question can end with a falling tone (Tone 4), with the tone contour maintained but the starting or ending point of the tone is raised (Lin 2004). This tonal characteristic may cause difficulties for Mandarin speakers when learning English boundary tones. For example, Mandarin learners of English might be influenced by the Tone 4 fall at the end of a yes-no question, leading to a boundary tone like L-H%, which lacks the sharp rise of H-H%.

As for studies on L2 boundary tones, Wang and Meng (2009) showed rather detailed results. They found that, except for the short imperative sentences, L2 learners showed significant differences from native speakers in reading the final boundary tones of the other six sentence patterns, while learners had a relatively good grasp of the sentence-initial boundary tones, and there was no significant difference between high and low groups. When reading long and complex sentences, it is difficult to handle the fluctuation of intonation well. For example, 80% of native speakers use H*L% with a gentle decreasing intonation, while the majority of learners (78.9%) use H*LL% with a larger decreasing intonation when dealing with the final boundary intonation of a statement. When reading yes/no questions, native speakers mainly use L*HH% type with double rising F0(50%), while learners mainly use H*H % type with high-rising F0 trend (39.5%). Moreover, Tang and Zhu (2022) found that in question-final positions, L2 learners exhibit native-like intonational patterns when pitch accent falls on the final syllable, but their F0 boundary contours tend to be shallower than those of native speakers. However, when pitch accent occurs earlier in the utterance, learners adopt more exaggerated pitch movement.

In this section, we analyzed similarities and differences in Mandarin and English boundary tones in terms of systemic dimension. A prediction can be made: L2 English may have less H-H% than native English in yes-no questions (see Table 2.8)

Table 2.8 Measurements and Predictions in Boundary Tones

Systemic dimension
Pitch Patterns (ToBI)
L2 English: less H-H% in yes-no questions

In all, we compared the intonation of Mandarin and English in terms of overall sentence intonation, focus realization, and boundary tones. We provided target measurements and predictions in both the systemic and realizational dimensions, considering variables such as Sentence Type, Focus Position, and Final Word Type. Additionally, we aim to explore the influence of Sentence Length (Syllables) and learner proficiency levels (Groups), such as near-natives with little or mild Foreign Accent (FA) versus native speakers, which has not been addressed in previous studies. The inclusion of varying Syllables helps to better understand how Group differences manifest, as learners may perform similarly on short sentences. Thus, our research design incorporates five variables: Sentence Type, Focus Position, Final Word Type, Syllables, and Groups. A summary of our measurements is provided in Table 2.9. Note that while we broadly use L2 English for predictions, the detailed results will focus on the differences between Groups. In order to provide a better and deeper perspective for explaining L2 intonation learning by Chinese English learners, there is thus a need to conduct an experimental study to investigate what is specifically different between L2 English and native English and the role the intonational differences play.

Table 2.9 Measurements and Preliminary Predictions

Measurements				
	Systemic Dimension	Realizational Dimension		
Overall Sentence Intonation	Tone/Non-tonal language	Pitch Change Amount on the Syllable level(st)	Pitch Range(st)	Number of F0 Peaks and Valleys on the Utterance Level
		L2 English > Native English	L2 English > Native English	L2 English ≈ Native English
Focus Realization	Pitch Pattern	/	/	
	L2 English: less L*-related in yes-no questions			
Boundary Tone	Pitch Pattern	/	/	
	L2 English: less H-H% in yes-no questions			

Chapter 3 Research Design

Chapter 3 provides a detailed description of the design of the phonetic experiment. It is divided into five sections: research questions, materials, participants, procedures, and data analysis.

3.1 Research Questions

Following the L2 Intonation Learning theory (LILt) (Mennen 2015), we made comparison between Mandarin and English intonation (see Chapter 2 for detailed comparison results). Based on the results, our experiment was conducted in the framework below (see Table 3.1):

Table 3.1 Framework for Measurements

Measurements				
	Systemic Dimension	Realizational Dimension		
Overall Sentence Intonation	Tone/None-tone language	Pitch Change Amount on the Syllable Level(st)	Pitch Range(st)	Number of F0 Peaks and Valleys on the Utterance Level
Focus Realization	Pitch Pattern	/	/	
Boundary Tone	Pitch Pattern	/	/	

In this study, we aim to investigate whether and how the following variables—Sentence Type, Focus Position, Sentence Length, and Final Word Type—affect the measurements. Additionally, we seek to explore whether inter-group differences exist, particularly between native Mandarin (a tone language) speakers and native English (an intonation language) speakers. In our comparison, we have showed that these two languages exhibit fundamental differences in intonation patterns, particularly in relation to variables such as Sentence Type, Focus Position, and Final Word Type. Given these differences, we aim to determine if learners show negative transfer from their native language (Mandarin) when acquiring English intonation patterns. Furthermore, we will examine how learners' accent levels influence these patterns. The following Research Questions are made based on the literature and comparison results.

RQ1: What variables influence overall sentence intonation? Do Groups affect overall sentence intonation, and if so, how?

a. What variables influence pitch change amount at the syllable level? Do Groups affect pitch change amount at the syllable level, and if so, how?

b. What variables influence pitch range at the utterance level? Do Groups affect pitch range, and if so, how?

c. What variables influence the number of F0 peaks and valleys at the utterance level? Do Groups affect the number of F0 peaks and valleys at the utterance level, and if so, how?

RQ2: What variables influence focus realization? Do Groups affect focus realization, and if so, how?

RQ3: What variables influence boundary tone? Do Groups affect boundary tone, and if so, how?

3.2 Participants and Classification Standard

3.2.1 Participants

5 native English speakers, form GroupNA (3 male and 2 female), aged between 19 and 29 years (mean age = 23). Two participants are from Canada, and three are from the United States, all being native speakers of North American English. During the data collection period, all participants were living in Beijing and were either students or researchers.

We initially recruited 10 native English speakers, with the recruitment criteria specifying a North American accent. However, three participants were excluded just after the recording because their accents did not align with typical North American varieties. In the process of handling Step 1 of the data analysis, two additional native speakers were excluded due to issues with creaky voice and excessive breathy voice, which caused significant difficulties in pitch extraction. As a result, we retained data from 5 native speakers, which form the final sample for GroupNA. Although recruiting native speakers with a North American accent in Beijing posed some challenges, we ensured the data from GroupNA met high standards by implementing rigorous analysis procedures. We also made efforts to maintain a balanced gender ratio within the group, despite the smaller sample size compared to other Groups.

A total of 37 learners were initially recruited for the study, with 34 participants ultimately retained and categorized into different groups after a pre-experiment. These 34 participants, primarily composed of students from key universities in Beijing, with an additional five participants recruited from technology companies, were categorized into three learner Groups based on their accent levels (see Section 4.1 for detailed results). The participants, aged between 18 and 32 years (mean age = 23), had an average of 15 years of English learning experience, and 85% of them had experience learning a second foreign language. The three Groups are: Near-natives (GroupNN) consisting of 11 participants (9 female), Advanced learners (GroupAD) with 12 participants (8 female), and Intermediate learners (GroupIN) with 11 participants (1 male and 10 female). The categorization of learners was done in pre-experiment through certain classification standard in section 3.2.1.

Therefore, our participants consist of four groups: GroupNA, GroupNN, GroupAD, and GroupIN (see Table 3.2).

Table 3.2 Groups and Speaker Codes

Groups	Speaker Code
GroupNA	NA1-NA5
GroupNN	NN1-NN11
GroupAD	AD1-AD12
GroupIN	IN1-IN11

3.2.2 Classification Standard in Pre-experiment

The classification of learners was carried out through a pre-experiment. Learners were required to read the Rainbow Passage (see Appendix 1) either before or after the recording in main experiment. Classification was determined by three native speakers who evaluated the learners' recordings. The scoring consisted of two aspects: Foreign Accent (FA) score and Intonation Score. Only FA scores were used for classification.

For the FA score, the experimenter explained the concept of Foreign Accent to the raters. The FA score reflects how native-like the learner's speech sounds: the more dissimilar a learner's speech is to that of a native speaker, the lower their FA score. The FA score was rated on a scale from 0 to 10, with increments of 0.5, where 0 represents no exposure to English, and 10 represents a native speaker. Given that all participants were Mandarin learners of English with more than 10 years of language learning experience, scores below 4 were not recommended, as these learners would have been excluded from the study if rated too low. It is important to note that all near-native speakers were identified by native raters, indicating that even those with accents very close to native speech still had discernible differences.

The second part of the evaluation focused solely on intonation. Raters were instructed to assess learners based only on their intonation, which the experimenter defined as the melodic component of prosody. Prosody, along with consonants and vowels, forms the sound system of English. While the FA score evaluated the overall sound system, the Intonation Score was specifically for intonation, rated on the same 0-10 scale with 0.5-point increments, the same as FA score. Below is a summary of judgement criteria in FA score (see Table 3.3).

Table 3.3 Classification Standard of Learners

Learner Groups	Judgement Criteria
NN(near-native)	Average Foreign Accent Score $8<x\leq10$, or rated within the range $8<x\leq10$ by at least 2 judges
AD(advanced)	Average Foreign Accent Score $6<x\leq8$, or rated within the range $6<x\leq8$ by at least 2 judges
IN(intermediate)	Average Foreign Accent Score $4<x\leq6$, or rated within the range $4<x\leq6$ by at least 2 judges

3.3 Materials

Test sentences were designed to examine the effects of four key independent variables—Sentence Type, Focus Position, Sentence length (Syllables), and Final Word Type—on the dependent variables in both systemic and realizational dimensions. Moreover, we aim to investigate the interaction between these four sentence-level variables and variable Groups. The SentenceCode, defined by these four independent variables, is shown in the table below (see Table 3.4). The test sentences consist of **36 conditions (2 Sentence Types × 3 Focus Positions × 3 Syllables × 2 Final Word Types**) (see Table 3.5 for detailed materials). To facilitate extensive F0 extraction, we aim to words with sonorant onsets (preferably nasal) and, when possible, avoid coda consonants. Due to sentence length and semantic considerations, we can just select as many sonorants as possible. Narrow focus was realized in **bold** words while broad focus sentences served as controls. Each sentence was elicited using different prompts and repeated twice by each participant (and up to three or four times in cases of misreads or disruptions), resulting in approximately **2808 utterances (2 repetitions × 39 participants × 36 sentences)**. For the QFF and QIF prompts, they always contained a negative sentence, where the negated word was meant to contrast with the focus of the target sentence. For the remaining prompts, they were designed as questions that asked for specific information contained within the target sentences. This method has been successfully employed in previous studies (Cooper et al. 1985; Xu 1999; Xu 2005).

Table 3.4 Representation of SentenceCode

SentenceCode	Representation	Variables
Fw/CwXXXXX	Fw: Function word Cw: Content word	Final Word Type
XXSs/Ms/LsXXX	Ss: Short sentence (7 syllables) Ms: Medium sentence (14 syllables) Ls: Long sentence (21 syllables)	Syllables
XXXXQ/DXX	Q: Question D: Declarative	Sentence Type
XXXXXIF/FF XXXXXC	IF: Intermediate Focus Position FF: Final Focus Position C: Broad Focus (no narrow focus)	Focus Position

Table 3.5 Materials

Prompt	Sentence	SentenceCode
7 Syllables		
What did you say?	Manny murmured to herself.	FwSsDC
How did Manny talk to herself?	Manny **murmured** to herself.	FwSsDIF
Who did Manny murmur to?	Manny murmured to **herself**.	FwSsDFF

Table 3.5 (Continued)

Prompt	**Sentence**	**SentenceCode**
What is your question?	Manny murmured to herself?	FwSsQC
Manny did not **shout at** herself. What is your question?	Manny **murmured** to herself?	FwSsQIF
Manny did not murmur to **himself**. What is your question?	Manny murmured to **herself**?	FwSsQFF
What did you say?	Manny murmured melodies.	CwSsDC
How did Manny sing melodies?	Manny **murmured** melodies.	CwSsDIF
What did Manny murmur?	Manny murmured **melodies**.	CwSsDFF
What is your question?	Manny murmured melodies?	CwSsQC
Manny did not **hum** melodies. What is your question?	Manny **murmured** melodies?	CwSsQIF
Manny did not murmur **words**. What is your question?	Manny murmured **melodies**?	CwSsQFF
14 Syllables		
What did you say?	Ariana will worry about how to manage teams.	CwMsDC
What will Ariana feel about how to manage teams?	Ariana will **worry** about how to manage teams.	CwMsDIF
What kind of things will Ariana worry about how to manage?	Ariana will worry about how to manage **teams**.	CwMsDFF
What is your question?	Ariana will worry about how to manage teams?	CwMsQC

Table 3.5 (Continued)

Prompt	Sentence	SentenceCode
Ariana will not be **cool** about how to manage teams. What is your question?	Ariana will **worry** about how to manage teams?	CwMsQIF
Ariana will not worry about how to manage **individuals**. What is your question?	Ariana will worry about how to manage **teams**?	CwMsQFF
What did you say?	Ariana will worry about how to manage them.	FwMsDC
What will Ariana feel about how to manage them?	Ariana will **worry** about how to manage them.	FwMsDIF
What kind of things will Ariana worry about how to manage?	Ariana will worry about how to manage **them**.	FwMsDFF
What is your question?	Ariana will worry about how to manage them?	FwMsQC
Ariana will not be **cool** about how to manage them. What is your question?	Ariana will **worry** about how to manage them?	FwMsQIF
Ariana will not worry about how to manage **him**. What is your question?	Ariana will worry about how to manage **them**?	FwMsQFF
21 Syllables		
What did you say?	May launched herself into a journey of learning to reveal the wonders of sounds.	CwLsDC
What kind of journey did May launch herself into to reveal the wonders of sounds?	May launched herself into a journey of **learning** to reveal the wonders of sounds.	CwLsDIF

Table 3.5 (Continued)

Prompt	**Sentence**	**SentenceCode**
What kind of wonders did May launch herself into a journey of learning to reveal?	May launched herself into a journey of learning to reveal the wonders of **sounds**.	CwLsDFF
What is your question?	May launched herself into a journey of learning to reveal the wonders of sounds?	CwLsQC
May did not launch herself into a journey of **teaching** to reveal the wonders of sounds. What is your question?	May launched herself into a journey of **learning** to reveal the wonders of sounds?	CwLsQIF
May did not launch herself into a journey of learning to reveal the wonders of **syntax**. What is your question?	May launched herself into a journey of learning to reveal the wonders of **sounds**?	CwLsQFF
What did you say?	May launched herself into a journey of learning to reveal the wonders of ours.	FwLsDC
What kind of journey did May launch herself into to reveal the wonders of ours?	May launched herself into a journey of **learning** to reveal the wonders of ours.	FwLsDIF
What kind of wonders did May launch herself into a journey of learning to reveal?	May launched herself into a journey of learning to reveal the wonders of **ours**.	FwLsDFF
What is your question?	May launched herself into a journey of learning to reveal the wonders of ours?	FwLsQC

Table 3.5 (Continued)

Prompt	**Sentence**	**SentenceCode**
May did not launch herself into a journey of **teaching** to reveal the wonders of ours. What is your question?	May launched herself into a journey of **learning** to reveal the wonders of ours?	FwLsQIF
May did not launch herself into a journey of learning to reveal the wonders of **theirs**. What is your question?	May launched herself into a journey of learning to reveal the wonders of **ours**?	FwLsQFF

3.4 Procedure

Recording was conducted in a quiet office at xxx University. The participant was seated comfortably in front of a computer monitor, with the microphone positioned approximately one foot away from their lips. Prior to each trial, the experimenter provided general instructions about the recording procedure and briefly introduced the experiment's aim--facilitating second language learners' pronunciation. However, the specific research purpose was not disclosed to ensure that participants could produce the intonation naturally. The experimenter also guided the speaker through the computer interface, familiarizing them with the process on the screen. Once the speaker felt comfortable, they could start the experiment by clicking the "Start" button displayed on the screen.

The experimental interface was designed using JavaScript to fully meet the requirements of this study. Once started, participants were able to independently navigate the web page to complete tasks such as reviewing the experimental instructions, familiarizing themselves with the materials, and undergoing the pre-test and formal recording. Notably, in addition to using Adobe Audition for professional recordings, an automatic backup recording was also generated via the website for contingency purposes. The design of the experimental pages can be referenced in Figures 3.1 and 3.2. The experimental interface includes the following stages:

1. **Welcome Page**

2. **Instructions** (where the experiment's key points were emphasized. One of the most important points is that participants were instructed to read the target sentence fluently at the normal speaking rate, similar to that of prompts. They were also advised not to pause too long in the middle of a sentence.).

3. **Materials** (at this stage, speakers were given sufficient time to practice and get familiar with the material, to ensure as little reading mistakes as possible).

4. **Pre-test** (simple target sentences and corresponding prompts similar to those in the formal experiment, but with different content, to ensure a smooth transition into the main experiment).

5. **Formal Experiment**: During the formal experiment, the subject pressed the "Next" button on the screen to display the target sentence. At the same time, a pre-recorded AI-generated prompt was played. The subject then read aloud the displayed sentence in response to the prompt. A progress bar at the top displayed current reading progress, allowing participants to track their progress throughout the experiment. If a reading mistake was identified by the experimenter, the participant was asked to repeat the sentence. A total of 72 sentences (36 sentences × 2 repetitions) and their corresponding prompts were presented in random order, with a different order for each participant, so the speakers were unaware of the true purpose of the study.

6. End Page

For the formal experiment, the experimenter used a Shure MV51 microphone to record. The data were then digitized into WAV format on a Dell desktop using Adobe Audition 2024, with a 16kHz sampling rate.

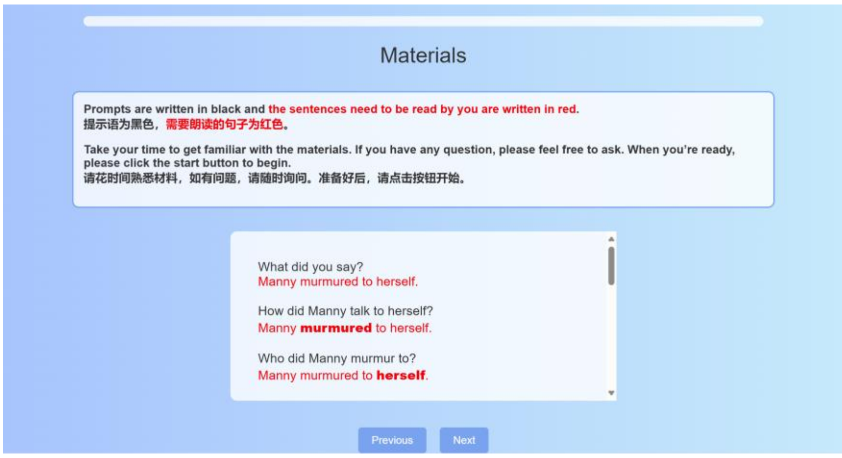


Figure 3.1 Materials Page

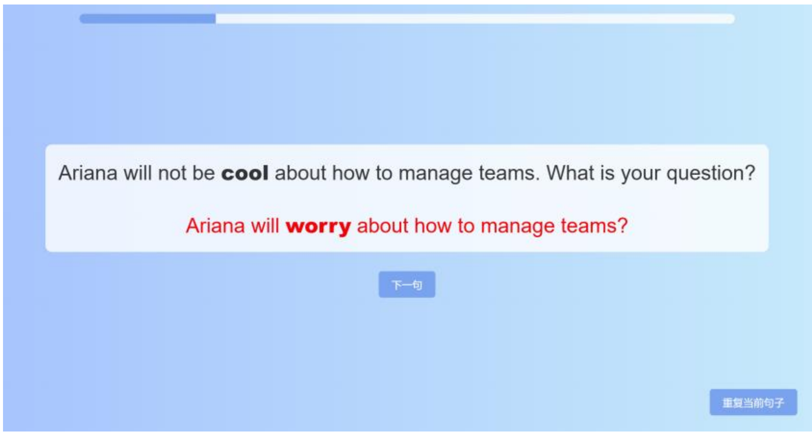


Figure 3.2 Formal Experiment Page

3.5 Data Analysis

The data analysis process includes the following steps: **Pre-step**: Audio recordings from Adobe Audition were segmented into target sentences, each labeled with the SpeakerCode and SentenceCode. One of the two repetitions was selected, and noise at the beginning and end of the sentence was removed. **Step 1**: F0 extraction was performed using ProsodyPro (Xu 2013), a Praat (Boersma 2001) script used for analysis of continuous prosodic events, which automatically generated F0 track for each sentence. The F0 curves were then manually corrected. Quantified measurements were taken from the F0 curves and then computed through some algorithms and python scripts. **Step 2**: Alignment was conducted using Montreal Forced Aligner (MFA) (McAuliffe et al. 2017), which automatically aligned the speech with its transcription at the phonetic level. **Step 3**: ToBI annotations were applied to identify prosodic features, such as focus and boundary tones, as qualitative data. Statistical analysis was performed using RStudio for realizational measurements.

It should be noted that our data analysis for **each participant**, excluding data entry and statistical analysis, took an average of **over 25 hours**. With **39 participants**, the total time amounted to over **1000 hours**. Most of the time and effort were spent on **Step 1** and **Step 3**. Therefore, the following section will provide a detailed explanation of the data analysis process. A tutorial for the analysis was created, including over **4 hours** of screen recordings and extensive notes, intended to assist with future data processing and support others working with the data.

3.5.1 Pre-step

In the **Pre-step** of the data analysis, the following four steps were performed:

Segmentation: Audio recordings from Adobe Audition were segmented into target sentences, with minimal noise retained at the beginning and end, which was later refined.

Coding: Each sentence was labeled with the SpeakerCode and SentenceCode.

Selection: One of the two repetitions was selected. The selection criteria were as follows: In most cases, the two repetitions were highly consistent. In such instances, the F0 curve, extracted using filtered autocorrelation method in Praat (Boersma 2001), was chosen based on which one exhibited a smoother pitch contour. If an inconsistency was noticed between the two repetitions, two conditions were considered:

A repetition was excluded if it was judged to have the wrong focus audibly, as determined by the author.

If the two repetitions appeared to differ in melodic pattern but both contained the correct focus, both were retained.

Noise Removal: Minimal noise at the beginning and end of the sentence was removed using Praat (Boersma 2001). This was typically done by examining the formants and soundwaves. The first or second formant points at the beginning of the sentence, as well as the last or second-last points at the end, were removed to eliminate extraneous noise.

3.5.2 Step 1: F0 Extraction and Quantified Measurements

In **Step 1**, the process was divided into two main stages. The first stage involved using ProsodyPro (Xu 2013) to automatically extract F0 curves and manually correct the pitch, generating sound, pitch tier, and pitch files for use in subsequent steps. The second stage utilized the pitch files from the previous step along with specific scripts and algorithms to extract the necessary quantified measurements, which corresponded to the three dependent variables in the realizational dimension of overall sentence intonation.

In terms of F0 extraction, due to the unsatisfactory results of automatic F0 extraction, manual correction of F0 was performed. The waveform and pitch markings and annotations were displayed together using the Praat script ProsodyPro (Xu 2013) for the correction process. The pitch markings were then manually adjusted to ensure the utmost accuracy of the F0 values. Although F0 curves automatically extracted by ProsodyPro (Xu 2013) was significantly more accurate than that of Praat (Boersma 2001)'s filtered autocorrelation algorithm and we tried our best to utilize as many sonorants as possible in target sentences, speaker-related factors such as creaky voice, breathy voice and glottalization and objective factors such as inevitable minor noise still caused some unavoidable pitch jumps (sudden, incorrect shifts in pitch), which added considerable workload to the manual correction process. Typical issues encountered in manual correction included:

a. **Incomplete Markings.** An exemplar of incomplete marking is shown in Figure 3.3, in which we can see some periodic soundwaves, but automatically they were not given any pitch markings.

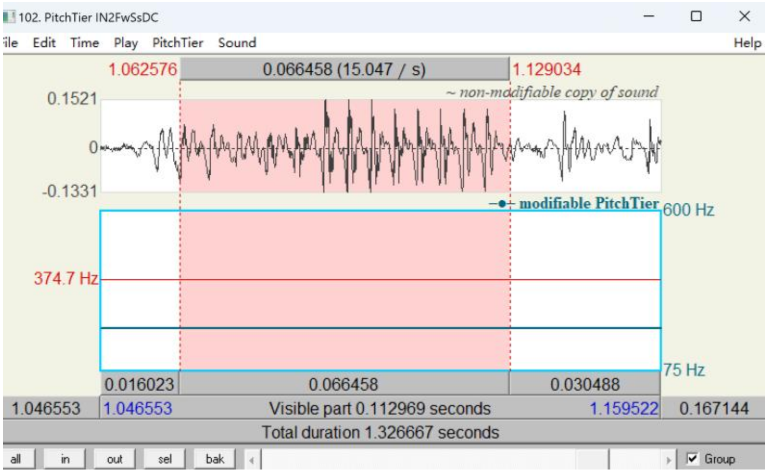


Figure 3.3 An Exemplar of Incomplete Markings

b. **Excessive Markings.** An exemplar of excessive marking is shown in Figure 3.4, where some irregular waveforms were assigned pitch markings.

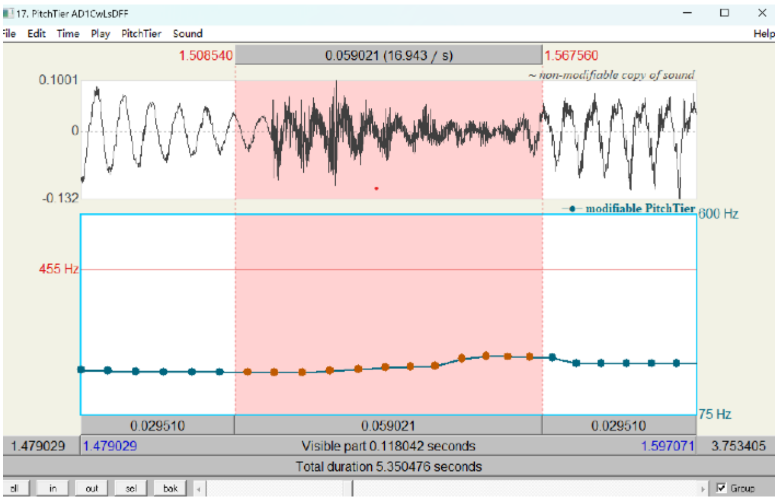


Figure 3.4 An Exemplar of Excessive Markings

c. **Pitch jumps (sudden periodic shifts).** Pitch jumps are one of the most significant and difficult issues to correct in pitch marking. A classic pitch jump occurs when a waveform, which exhibits consistent periodicity, suddenly shows a point where the cycle period is doubled, leading to a pitch shift of 0.5 times the expected value. An exemplar of pitch jumps is shown in Figure 3.5, where pitch suddenly dropped to half.

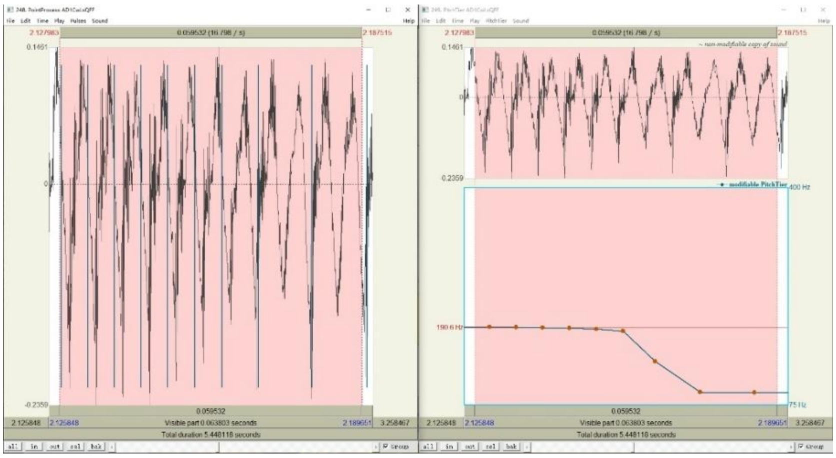


Figure 3.5 An Exemplar of Pitch Jumps

In the first pass, ProsodyPro (Xu 2013) was used to address these issues to the greatest extent possible. Following this, we performed a re-check using sound + pitch tier and sound + pointprocess files to detect any remaining pitch marking issues. Our workflow strictly adhered to the above procedure, with multiple rounds of verification. We tried our best to correct the voiced parts, ensuring that extreme F0 values were avoided and that smooth F0 contours were maintained for voice segments.

After having manually corrected the F0 extraction, we used Praat's (Boersma 2001) "Manipulation" function to resynthesize the original sound files with the corrected pitch tier and pointprocess files. This process was not entirely based on Pitch-Synchronous Overlap and Add (PSOLA) (Moulines and Laroche 1995), as using PSOLA in this case resulted in some pitch jumps. The resynthesis produced new sound files with the corrected F0. Subsequently, we applied Praat's (Boersma 2001) "Analyze Periodicity" function to generate pitch files with the corrected F0 values, using the pitch-filtered autocorrelation method. To generate continuous pitch contours, unvoiced pats in pitch files were interpolated through. Finally, the pitch contours were smoothed using Praat's (Boersma 2001) smoothing algorithm (frequency band = 10 Hz).

Quantified Measurements:

We used the pitch files with the adjusted F0 from the F0 extraction process, along with custom scripts and algorithms, to extract the necessary quantified measurements. These measurements correspond to the highlighted sections in the Table 3.6.

Table 3.6 Measurements with Quantified Ones in Highlights

Measurements				
	Systemic Dimension	Realizational Dimension		
Overall Sentence Intonation	Tone/None-tone language	**Pitch Change Amount on the Syllable Level(st)**	**Pitch Range (st)**	**Number of F0 Peaks and Valleys on the Utterance Level**
Focus Realization	Pitch Pattern	/	/	
Boundary Tone	Pitch Pattern	/	/	

To normalize male and female speakers, F0 measurements in Hertz were converted to semitones using the equation with a reference of 100 Hz. The conversion equation proposed by Fant et al. (2002) was applied in this study:

$$f(st) = 12\log_2\left(\frac{f(Hz)}{100(Hz)}\right)$$

This frequency normalization allows for representation of the intonation contour, making it easier to compare across speakers and calculate group averages. With the exception of the number of F0 peaks and valleys on the utterance level, all other F0 measurements discussed in this study are presented in semitones (st). Listed in the following are the quantified measurements. They were extracted from individual trimmed F0 curves instead of the mean F0 curves averaged across participants.

Pitch range on sentence level: absolute difference between maximum F0 and minimum F0 of voiced segments within a sentence **(in semitone).**

Pitch change amount on the syllable level: sum of the absolute pitch change amount of every voiced segment within a sentence **(in semitone)**/number of syllables within the sentence**.**

Number of F0 peaks and valleys on the utterance level: amount of F0 peaks and valleys in a sentence detected by **Prominence-Based F0 Peak-Valley Detection Method**.

As for the first two quantified measurements -- Pitch change amount on the syllable level(st) and Pitch range(st), they were taken by a custom python script, in which we inputted all values in a sentence and then the measurements were printed out.

But things were a little more complicated for Number of F0 peaks and valleys on the utterance level. Unfortunately, we failed to replicate the convex-hull method proposed by Yuan and Liberman (2014) for detecting significant F0 peaks and valleys. Therefore, we adopted our own method, named the Prominence-Based F0 Peak-Valley Detection Method. Our script began by defining a time-aligned F0 contour, ensuring that each F0 measurement corresponded to a precise moment in time. It then identified significant pitch peaks and valleys, where a peak was defined as a point that stood out relative to its neighboring values by a prominence threshold of 16 Hz. This threshold ensured that only acoustically relevant fluctuations were detected, filtering out minor variations that could result from micro-prosodic effects or measurement noise. Valleys were identified using the same method, but applied to an inverted version of the F0 contour.

To validate the appropriateness of the prominence threshold (16 Hz) for both peaks and valleys, we initially explored a range of prominence values between 10 and 20. Through iterative adjustments and parameter combinations, we fine-tuned the threshold to best match both perceptual judgments and the overall shape of the F0 contours. The parameter adjustment process involved sentences from various Participants, Syllables, Focus Positions, and Sentence Types. Ultimately, a prominence threshold of **16 Hz** for both peaks and valleys was determined.

Once peaks and valleys were identified, their positions and values were extracted, and their distribution was assessed. If the number of detected peaks or valleys was insufficient (≤ 1), the process terminated, and only the raw F0 contour was visualized. Otherwise, the script employed cubic spline interpolation to generate upper and lower envelopes, connecting the detected peaks and valleys, respectively. Finally, the original contour, peaks, valleys, and envelope curves were visualized to facilitate further interpretation (see Figure 3.6 as an exemplar). The number of F0 peaks and valleys was also printed out.

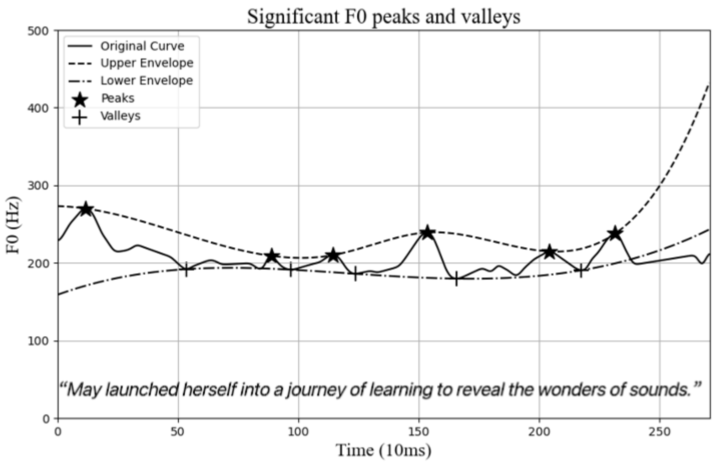


Figure 3.6 An Exemplar of F0 Peaks and Valleys Detected by Prominence-Based F0 Peak-Valley Detection Method

3.5.3 Step 2: Alignment

Approximately 2800 sentences were initially annotated automatically using the Montreal Forced Aligner (MFA) (McAuliffe et al. 2017). Only a very few cases of misalignment were identified, which were then manually corrected using the Praat program (Boersma 2001).

3.5.4 Step 3: ToBI Annotations

ToBI annotations corresponded to the systemic dimension of the measurements (see Table 3.7) and were considered qualitative data. Using the ToBI system, we annotated the pitch patterns of focus realization and boundary tones. The types of pitch accents and boundary tones used in this study are listed below (see Table 3.7). For a detailed introduction to the ToBI system, see Section 2.2.2.

Table 3.7 Measurements with Qualitative Ones in Highlights

Measurements				
	Systemic Dimension	Realizational Dimension		
Overall Sentence intonation	Tone/None-tone language	Pitch Change Amount on the Syllable Level(st)	Pitch Range (st)	Number of F0 Peaks and Valleys on the Utterance Level
Focus Realization	Pitch Pattern	/	/	
Boundary Tone	Pitch Pattern	/	/	

Table 3.8 Pitch Accents and Boundary Tones in ToBI System

Pitch Accents (for focus realization)	Boundary Tones
H*(!H*), L*, L+H*(L+!H*), L*+H, H+!H*	L-L%, H-H%, L-H%,H-L%

In this study, a total of 2800 sentences were annotated using the ToBI system. All sentences were annotated on Textgirds, generated in step 2, for boundary tones, while pitch accents were marked only for sentences with narrow focus. Break Indices in the ToBI system were not the focus of this study, and therefore, their annotations results are not presented. An exemplar can be seen in Figure 3.8. The sentence was represented by three tiers: the top tier contained the text of the sentence itself, the middle tier displayed the results of the MFA alignment, where words were segmented into phones, and the bottom tier showed the ToBI annotation results. In this case, "learning" served as the focus of the sentence and was marked with an L+H* pitch accent based on auditory perception. The boundary tone for the sentence was H-H%, which was fixed at the end of the sentence.

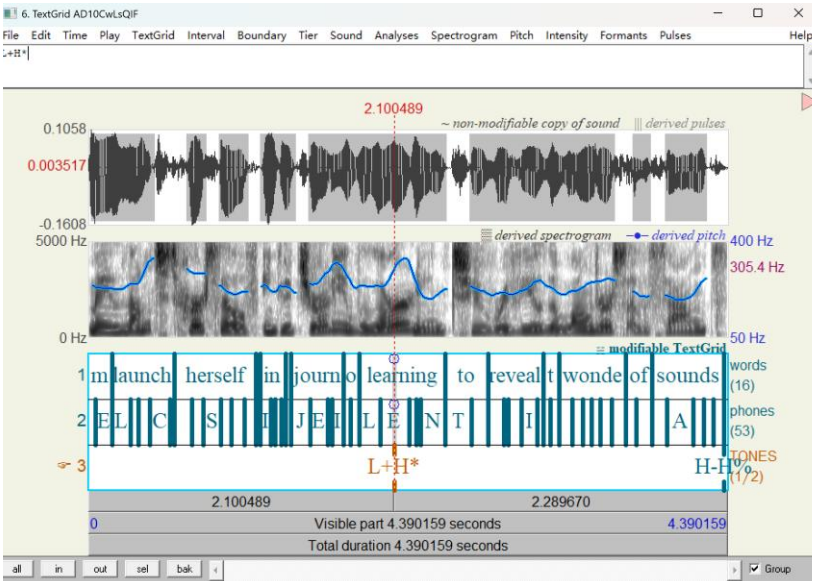


Figure 3.7 An Exemplar of ToBI Annotations

In all, the study investigates the intonational features in native speakers and learners' sentence productions by using both quantitative and qualitative data. A total of 2800 sentences were analyzed, with 5 native English speakers (GroupNA) and 34 learners (GroupNN, GroupAD, GroupIN) participating in the experiment. The materials were designed to assess the effects of Sentence Type, Focus Position, Syllables, and Final Word Type on pitch patterns, with Groups being another key variable in this study. Data collection was carried out in a controlled environment with a structured procedure, involving 2 repetitions of sentences by responding to prompts. As for data analysis, F0 extraction was performed using ProsodyPro (Xu 2013), followed by manual correction. Alignment was done using MFA (McAuliffe et al. 2017). Pitch peaks and valleys were detected using the Prominence-Based F0 Peak-Valley Detection Method. ToBI annotations identified focus realization and boundary tones.

Chapter 4 Results

In this chapter, we will first present the results of pre-experiment and then demonstrate our main experiment thoroughly.

4.1 Pre-Experiment

The pre-experiment mainly aims to categorize participants into distinct proficiency levels based on their Foreign Accent Scores. A total of 37 learners were recruited for the study, of which 34 participants were retained. One participant was excluded due to excessive creaky voice, while two others were excluded for failing to meet the fluency criteria, as they made frequent misreadings and omissions. **5 native speakers**, who were also participants in the native group, were recruited as judges for the pre-experiment. However, only three of them were retained due to the deviant Foreign Accent Scores of the other two. According to the classification standard described in the research design section, the 34 learners were divided into three groups. The **near-native** group consisted of 11 participants, who either had a Foreign Accent Score ranging from 8 to less than 10 or were rated as near-native by at least two native speaker judges (out of three qualified judges). The **advanced** group, comprising 12 learners, included those with a Foreign Accent Score ranging from 6 to less than 8 or who received an advanced rating from at least two native speaker judges within this range. The **intermediate** group also consisted of 11 participants, whose Foreign Accent Scores ranged from 4 to less than 6, or who were rated as intermediate by at least two native speaker judges within this range. Table 4.1 presents the learner groups, judgement criteria and corresponding codes of the learners.

Table 4.1 Learner Group Classification, Judgement Criteria, and Speaker Codes

Learner Groups	Judgement Criteria	Speaker Code
NN (near-native)	Average Foreign Accent Score $8<x\leq10$, or rated within the range $8<x\leq10$ by at least 2 judges	NN1-NN11
AD (advanced)	Average Foreign Accent Score $6<x\leq8$, or rated within the range $6<x\leq8$ by at least 2 judges	AD1-AD12
IN (intermediate)	Average Foreign Accent Score $4<x\leq6$, or rated within the range $4<x\leq6$ by at least 2 judges	IN1-IN11

The pre-experiment also investigated the Intonation Scores across different learner groups, which were evaluated by the same three qualified native judges. The criteria for classification are based on the method described in the Research Design section. Table 4.2 presents the average Foreign Accent Scores alongside the corresponding Intonation Scores for each of the three groups.

Table 4.2 Average Foreign Accent Scores and Corresponding Intonation Scores

Learner Groups	Average Foreign Accent Scores	Average Intonation Scores
NN (near-native)	8.42	8.35
AD (advanced)	6.64	6.52
IN (intermediate)	4.32	4.39

The Spearman's rank correlation between Foreign Accent Score and Intonation

Score is 0.96 (p< 2.2e-16), indicating a strong positive correlation (see Figure 4.1). This suggests a robust relationship, where an increase in Foreign Accent Score is associated with an increase in Intonation Score.

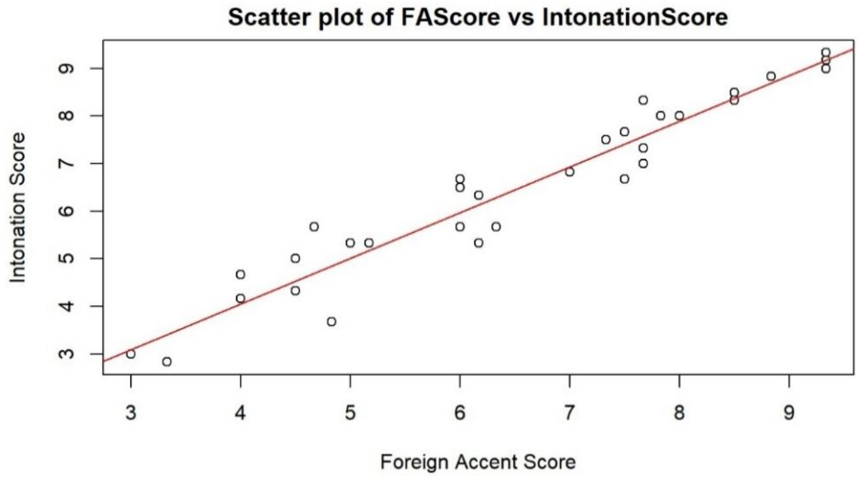


Figure 4.1 Scatter Plot of FA Score vs Intonation Score

Paired t-tests were also conducted to compare the Foreign Accent Scores and Intonation Scores within each learner group (NN, AD, and IN). No significant differences were found between the Foreign Accent Scores and intonation in any of the three groups. For the NN (Near-native) group, t=0.754, p=0.468; for the AD (Advanced) group, t=0.784, p=0.451; and for the IN (Intermediate) group, t=−0.426, p=0.679. The paired t-tests demonstrate the consistency between learners' accent and intonation levels.

4.2 Main Experiment

Table 4.3 Measurements in Main Experiment

Measurements				
	Systemic Dimension	Realizational Dimension		
Overall Sentence Intonation	Tone/None-tone language	Pitch Change Amount on the Syllable Level(st)	Pitch Range(st)	Number of F0 Peaks and Valleys on the Utterance Level

Table 4.3 （Continued）

Measurements		
Focus Realization	Pitch Pattern /	/
Boundary Tone	Pitch Pattern /	/

In the main experiment, we developed our framework (as shown in Table 4.3) based on Mennen's (2015) L2 Intonation Learning theory (LILt) framework and the Autosegmental-Metrical (AM) approach (Pierrehumbert 1980; Pierrehumbert and Beckman 1998). Mennen's framework, in exploring L1 and L2 intonation, suggests four dimensions. We selected the two most crucial dimensions for our study: the systemic and realizational dimension. These two dimensions distinguish between phonological (systemic) representation and phonetic (realizational) representation.

Systemic dimension: Data are qualitative and based on ToBI annotations.

Realizational dimension: Data are quantitative and will undergo statistical testing and mixed-effects regression modeling to identify significant predictors of the dependent variables.

The AM approach, which views English as an intonation language, identifies three key pitch events—pitch accent, phrase accent, and boundary tone—which contrast typologically with Mandarin's tonal language system.

To summarize the structure of the Results section, the findings will be presented in the following order:

Overall Sentence Intonation: This section focuses on the realizational dimension, represented by quantitative data.

Focus Realization: This section presents qualitative data based on ToBI annotations.

Boundary Tone: This section also uses qualitative data from ToBI annotations.

This structure follows the integration of the AM approach and Mennen's LILt framework, while adapting the design of our experiment. By clearly separating the systemic and realizational dimensions, the study aims to examine the relationship between phonological representation and phonetic realization across these key intonation features.

4.2.1 Overall Sentence Intonation

In the Overall Sentence Intonation section, we measured three acoustic features: **pitch change amount on the syllable level (st), pitch range (st), and the number of F0 peaks and valleys on the utterance level**. Each of these features was analyzed using mixed effects regression models. Prior to modeling, the necessary independent variables were visualized to ensure proper assessment.

4.2.1.1 Pitch Change Amount on the Syllable Level(st)

Before conducting the modeling, we examined the influence of some necessary factors on pitch change amount and visualized the data to better understand the relationships between these factors and the outcome variable.

4.2.1.1.1 Pre-Modeling Data Visualization and Analysis

The mean pitch change amount on the syllable level(st) across different **Groups** was as follows: the NA group had a mean of 2.43 (SD = 0.847), the NN group had a mean of 2.72 (SD = 0.883), the AD group had a mean of 3.04 (SD = 1.23), and the IN group had a mean of 3.09 (SD = 1.07). These values suggest that the pitch change amount increased as the proficiency level of the groups decreased, with the IN and AD groups exhibiting higher pitch change amounts compared to the NA and NN groups (see Figure 4.2).

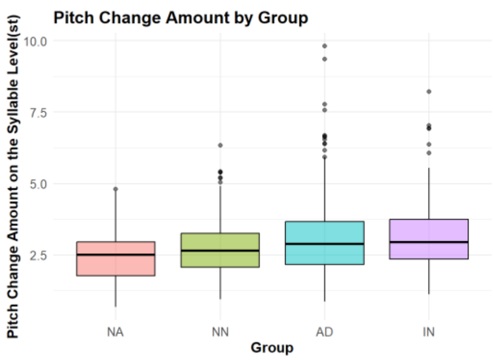


Figure 4.2 Pitch Change Amount by Group

The mean pitch change amount on the syllable level(st) for **Sentence Type** was as follows: declarative sentences had a mean of 2.74 (SD = 0.991), while questions had a mean of 3.03 (SD = 1.13). This indicates that pitch change amount was greater in questions compared to declarative sentences (see Figure 4.3).

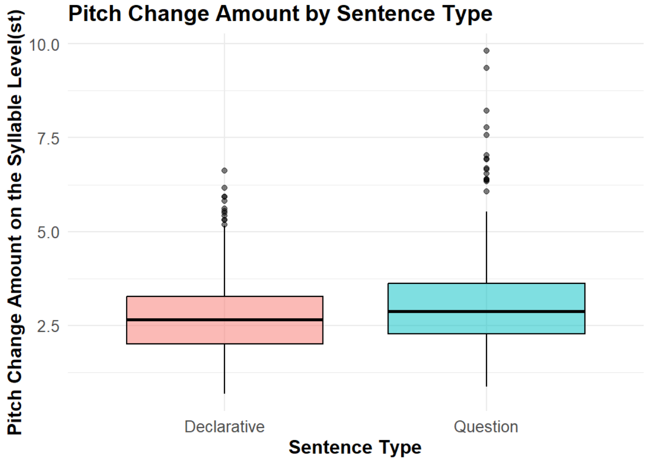


Figure 4.3 Pitch Change Amount by Sentence Type

The mean pitch change amount on the syllable level(st) for **Focus Position** was as follows: broad focus had a mean of 2.60 (SD = 0.948), intermediate focus had a mean of 3.10 (SD = 1.13), and final focus had a mean of 2.95 (SD = 1.07). This suggests that pitch change amount was highest for intermediate focus, second for final focus, with broad focus showing the lowest mean (see Figure 4.4).

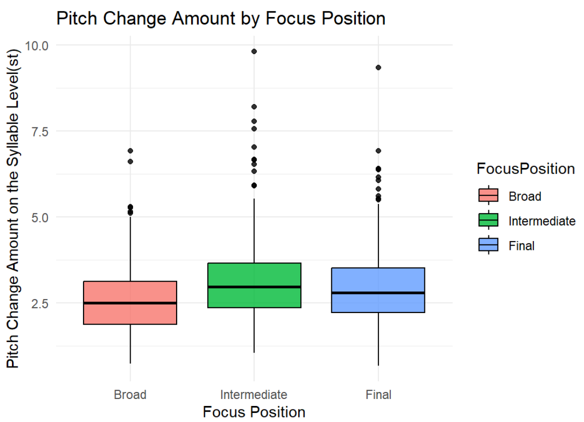


Figure 4.4 Pitch Change Amount by Sentence Type

The mean pitch change amount on the syllable level(st) for **Syllables** was as follows: for 7 syllables, the mean was 3.33 (SD = 1.27), for 14 syllables, the mean was 2.56 (SD = 0.874), and for 21 syllables, the mean was 2.76 (SD = 0.870). This indicates that the pitch change amount was greatest for 7 syllables, with a decrease observed for 14 syllables, followed by a slight increase at 21 syllables (see Figure 4.5).

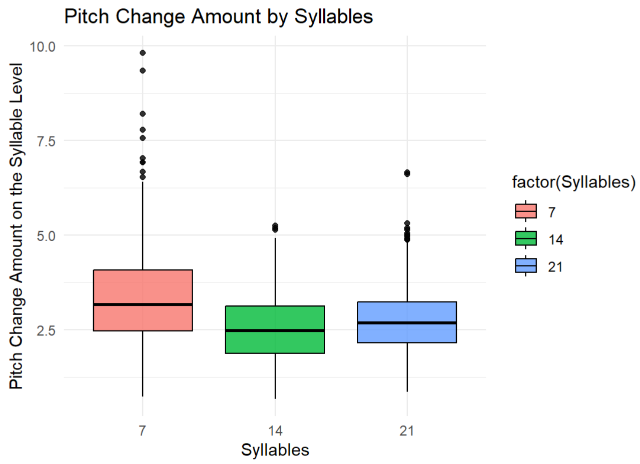


Figure 4.5 Pitch Change Amount by Syllables

To sum up, based on the visualization and the mean and SD values, we observed that **IN** and **AD** groups showed higher pitch change amounts compared to **NA** and **NN** groups. Pitch change was higher in questions than in **declarative** sentences, and **intermediate focus** exhibited the greatest pitch change, followed by **final focus**, with **broad focus** showing the least. Additionally, pitch change was highest for **7 syllables**, decreased for **14 syllables**, and slightly increased for **21 syllables**. These patterns will be further examined for statistical significance through modeling.

4.2.1.1.2 Linear Mixed-Effects Modeling

In RStudio, we constructed two models using the lmer function: one with interaction terms and one without. After running the models, we found that the model with interaction terms provided a better fit. The linear mixed-effects model was fitted to examine the relationship between various predictors (Group, Gender, Sentence Type, Focus Position, Syllables, and Final Word Type) and the outcome variable, log-transformed Pitch Change (Log transformation was applied to pitch change amount on the syllable level(st) to ensure normality and meet the assumptions of the linear mixed model). The reference group for this analysis was defined as GroupNA (Native), Gender Female, Sentence Type Declarative, Focus Position Broad, Syllables 7, and Final Word Type Content. All other groups and variables were compared against this baseline. The model included both fixed and random effects, with random intercepts for SpeakerCode and SentenceCode to account for variability between individual speakers and sentences.

Model Formula:

The model was specified as follows:

$$\begin{aligned}\text{logPitchChange} &\\ &\sim \text{Group} * \text{Gender} + \text{Group} * \text{Sentence Type} + \text{Group} \\ &* \text{Focus Position} + \text{Group} * \text{Syllables} + \text{Group} * \text{Final Word Type} \\ &+ (1|\text{SpeakerCode}) + (1|\text{SentenceCode})\end{aligned}$$

Model Fit Assessment:

REML Criterion: The REML criterion at convergence was -24.4, which suggests that the model fit the data well, accounting for both fixed and random effects.

Scaled Residuals: The scaled residuals ranged from -5.11 to 6.64, with a median of 0.01. These values indicate that there is no severe departure from the normality assumption, though some extreme residuals were present. This is typical of complex models with multiple interaction terms.

Random Effects:

The model accounted for random effects of SpeakerCode and SentenceCode to address variability between speakers and sentences. The random effect for SpeakerCode showed more variance (variance = 0.062667, SD = 0.2503) compared to SentenceCode (variance = 0.002034, SD = 0.0451), with residual variance of 0.046758 (SD = 0.2162). The inclusion of random effects helps control for variability between speakers and sentences, allowing the fixed effects to more accurately capture their influence on pitch change.

Fixed Effects:

The model results reveal several significant and non-significant effects on pitch change. Below is a summary of the fixed effects (see Table 4.4):

Table 4.4 Fixed Effects

Predictors	Estimate	Std. Error	t-value	p-value
(Intercept)	0.865821	0.183772	4.711	< 0.001***
GroupNN	0.149619	0.203071	0.737	0.461
GroupAD	0.311829	0.205099	1.520	0.129
GroupIN	0.167498	0.201288	0.832	0.405
GenderMale	-0.254689	0.230727	-1.104	0.270
SentenceTypeQuestion	0.165295	0.033934	4.871	< 0.001***
FocusPositionIntermediate	0.235832	0.041471	5.687	< 0.001***
FocusPositionFinal	0.130288	0.041979	3.104	0.002**
Syllables14	-0.241061	0.042233	-5.708	< 0.001***
Syllables21	-0.035731	0.041415	-0.863	0.388
FinalWordTypeFunction	-0.012970	0.033915	-0.382	0.702
GroupNN:GenderMale	-0.085452	0.303834	-0.281	0.779
GroupAD:GenderMale	-0.096145	0.277841	-0.346	0.729
GroupIN:GenderMale	0.126937	0.351426	0.361	0.718
GroupNN:SentenceTypeQuestion	-0.100900	0.038011	-2.655	0.008**
GroupAD:SentenceTypeQuestion	-0.057837	0.037344	-1.549	0.121
GroupIN:SentenceTypeQuestion	-0.050996	0.037811	-1.349	0.177
GroupNN:FocusPositionIntermediate	-0.051205	0.046473	-1.102	0.270
GroupAD:FocusPositionIntermediate	-0.081732	0.045755	-1.786	0.074
GroupIN:FocusPositionIntermediate	-0.086079	0.046226	-1.862	0.063
GroupNN:FocusPositionFinal	0.003774	0.046785	0.081	0.936

Table 4.4 (Continued)

Predictors	**Estimate**	**Std. Error**	**t-value**	**p-value**
GroupAD:FocusPositionFinal	0.009871	0.045920	0.215	0.830
GroupIN:FocusPositionFinal	-0.026244	0.046547	-0.564	0.573
GroupNN:Syllables14	0.006921	0.046963	0.147	0.883
GroupAD:Syllables14	-0.054830	0.046169	-1.188	0.235
GroupIN:Syllables14	0.017845	0.046755	0.382	0.702
GroupNN:Syllables21	-0.157118	0.046374	-3.388	< 0.001***
GroupAD:Syllables21	-0.185338	0.045517	-4.072	< 0.001***
GroupIN:Syllables21	-0.114487	0.046071	-2.485	0.013*
GroupNN:FinalWordTypeFunction	0.009691	0.037998	0.255	0.799
GroupAD:FinalWordTypeFunction	0.007195	0.037325	0.193	0.847
GroupIN:FinalWordTypeFunction	0.054439	0.037784	1.441	0.150

The analysis reveals significant effects of **Sentence Type**, with **Questions** showing greater pitch variation compared to **Declarative** sentences ($t = 4.871$, $p < 0.001$). Both **Focus Position Intermediate** ($t = 5.687$, $p < 0.001$) and **Final** ($t = 3.104$, $p < 0.05$) were significantly associated with pitch change, while **Syllables14** also showed a significant relationship with pitch variation ($t = -5.708$, $p < 0.001$). Significant interactions were observed between **Group and Sentence Type**, particularly for the NN group in questions ($t = -2.655$, $p < 0.01$), as well as between **Group and Syllables**, with notable effects for the NN ($t = -3.388$, $p < 0.001$), AD ($t = -4.072$, $p < 0.001$), and IN ($t = -2.485$, $p < 0.05$) groups at 21 syllables, indicating that syllable length significantly influences pitch variation across groups.

To sum up, in terms of Pitch Change Amount on the Syllable Level(st), the model results reveal several significant and non-significant effects on it. Notably, significant effects were found for **Sentence Type**, **Focus Position**, and **Syllables**. Although **Group** itself did not show significant main effects in the model (with p-values generally higher than 0.05), significant interactions were observed between **Group** and **Sentence Type**, as well as between **Group** and **Syllables**, emphasizing the nuanced relationship between learner proficiency and intonation patterns.

4.2.1.2 Pitch Range on the Utterance Level (st)

Similar to our analysis of pitch change, we began by exploring the impact of several relevant factors on pitch range, visualized the data, and examined the relationships between these factors and the outcome variable before proceeding with the modeling.

4.2.1.2.1 Pre-Modeling Data Visualization and Analysis

The mean pitch range across different **Groups** was as follows: the NA group had a mean of 9.44 (SD = 3.22), the NN group had a mean of 9.15 (SD = 2.88), the AD group had a mean of 9.30 (SD = 4.76), and the IN group had a mean of 9.04 (SD = 3.53). Based on these values, there does not appear to be a clear difference in pitch range across the groups, as the means are relatively close to each other and the standard deviations overlap (see Figure 4.6).

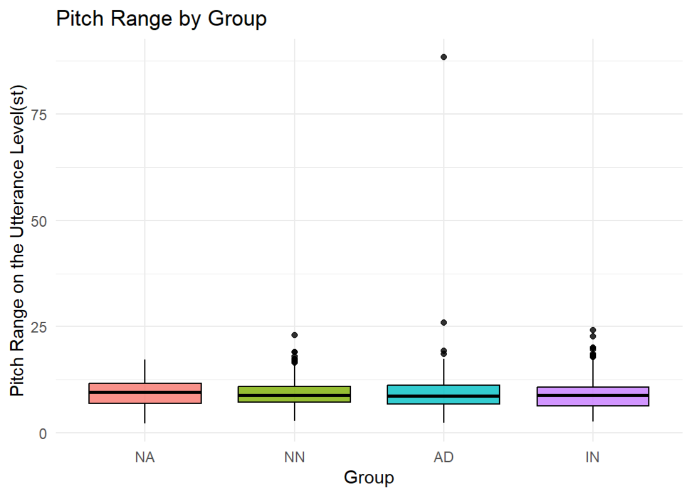


Figure 4.6 Pitch Range by Group

The mean pitch range for **Sentence Type** was as follows: declarative sentences had a mean of 8.73 (SD = 4.36), while questions had a mean of 9.65 (SD = 3.03). The observed difference in mean pitch range between question and declarative sentences will be further examined in the modeling to determine whether this difference is statistically significant (see Figure 4.7).

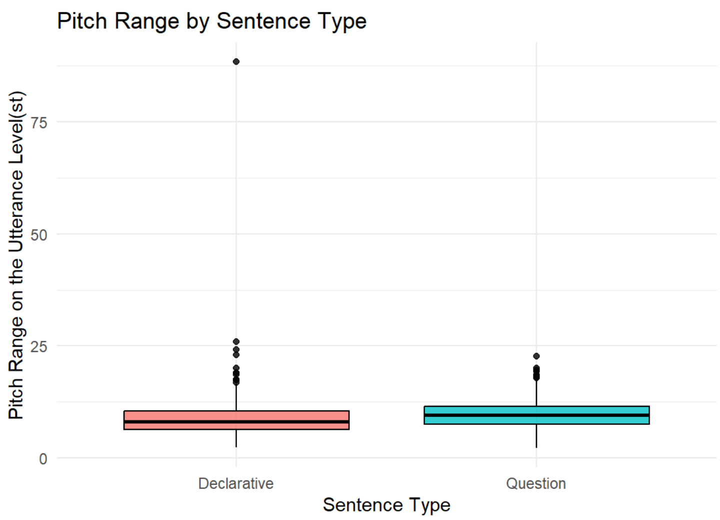


Figure 4.7 Pitch Range by Sentence Type

The mean pitch range for **Focus Position** was as follows: broad focus had a mean of 8.41 (SD = 3.01), intermediate focus had a mean of 9.91 (SD = 4.75), and final focus had a mean of 9.27 (SD = 3.09). This indicates that both intermediate focus and final focus had higher pitch ranges compared to broad focus, with intermediate focus showing the highest pitch range (see Figure 4.8).

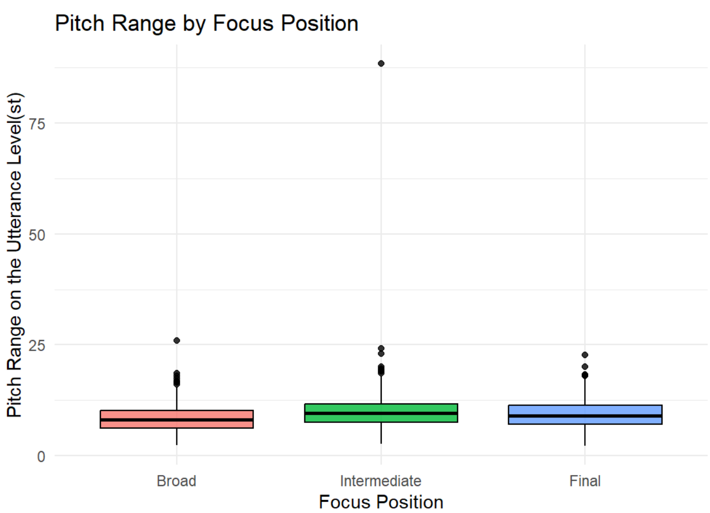


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Figure 4.8 Pitch Range by Focus Position

The mean pitch range for Syllables was as follows: for 7 syllables, the mean was 8.77 (SD = 3.20), for 14 syllables, the mean was 9.13 (SD = 3.09), and for 21 syllables, the mean was 9.70 (SD = 4.71). These values suggest that pitch range increased as the number of syllables grew, with 21 syllables showing the highest mean pitch range, followed by 14 syllables and 7 syllables (see Figure 4.9).

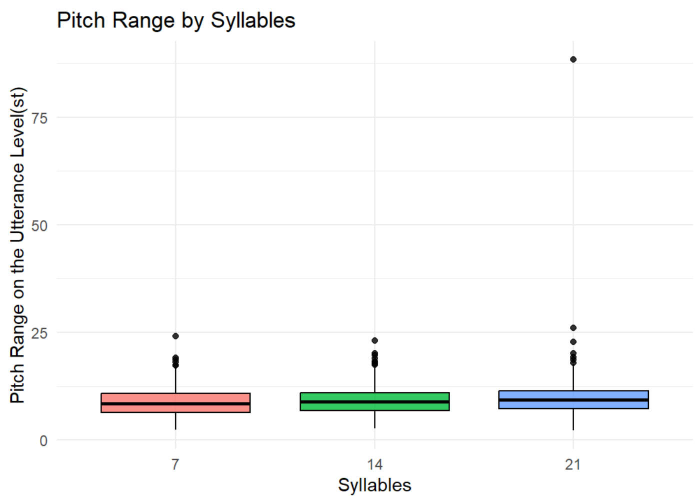


Figure 4.9 Pitch Range by Focus Position

In conclusion, from the visualization and the mean and standard deviation values, we found that there does not appear to be a clear difference in pitch range across the **groups**. Pitch range was greater in **questions** compared to **declarative sentences**, and **intermediate focus** showed the largest pitch range, followed by **final focus**, with **broad focus** showing the smallest range. Moreover, pitch range was greatest for **7 syllables**, decreased for **14 syllables**, and slightly increased for **21 syllables**. These observed trends will be further analyzed for statistical significance through modeling.

4.2.1.3 Linear Mixed-Effects Modeling

For the analysis of pitch range, similar to the approach used for pitch change amount, we constructed two models using the lmer function: one with interaction terms and one without. After running both models, the model with interaction terms provided a better fit. The reference group for the fixed effects and the random effects structure remained unchanged, with the baseline group defined as GroupNA (Native), Gender Female, Sentence Type Declarative, Focus Position Broad, Syllables 7, and Final Word Type Content. Random intercepts for SpeakerCode and SentenceCode were included to account for variability between individual speakers and sentences. As for the outcome variable, log_Pitch Range, log transformation was also applied to ensure normality for the linear mixed model)

Model Formula:

The model was specified as follows:

$$\begin{aligned}\text{log\_PitchRange} \\ &\sim \text{Group} * \text{Gender} + \text{Group} * \text{Sentence Type} + \text{Group} \\ &* \text{Focus Position} + \text{Group} * \text{Syllables} + \text{Group} * \text{Final Word Type} \\ &+ (1|\text{SpeakerCode}) + (1|\text{SentenceCode})\end{aligned}$$

Model Fit Assessment:

REML Criterion: The REML criterion at convergence was 276.8, indicating that the model provides a reasonable fit to the data, considering both fixed and random effects.

Scaled Residuals: The scaled residuals ranged from a minimum of -6.7202 to a maximum of 9.1982. The first quartile was -0.5409, the median was -0.0069, and the third quartile was 0.5330. These values suggest that the residuals are generally centered around zero, with some extreme values present, which is common in complex models.

Random Effects:

The model includes random effects for SpeakerCode and SentenceCode to account for variability between speakers and sentences. The random effect for SpeakerCode showed a variance of 0.069246 (SD = 0.2631), indicating moderate variability in pitch range across different speakers. The random effect for SentenceCode showed a variance of 0.001176 (SD = 0.0343), suggesting minimal variability in pitch range between sentences. Additionally, the residual variance was 0.058447 (SD = 0.2418). Given that the residual variance is relatively small, it suggests that the model fits the data well, with most of the variability in pitch range accounted for by the fixed and random effects.

Fixed Effects:

The results of the model show both significant and non-significant effects on pitch range. A summary of the fixed effects is provided below (see Table 4.5):

Table 4.5 Fixed Effects

Predicators	**Estimate**	**Std. Error**	**t-value**	**p value**
(Intercept)	1.971288	0.193507	10.187	< 0.0001 ***
GroupNN	0.012130	0.214451	0.057	0.955
GroupAD	0.084898	0.216559	0.392	0.695
GroupIN	-0.013881	0.212573	-0.065	0.948
GenderMale	-0.163788	0.242834	-0.674	0.500
SentenceTypeQuestion	0.328683	0.036379	9.035	< 0.0001 ***
FocusPositionIntermediate	0.203299	0.044482	4.570	< 0.0001 ***
FocusPositionFinal	0.126001	0.044905	2.806	0.005 **
Syllables14	0.017116	0.045209	0.379	0.705
Syllables21	0.106940	0.044367	2.410	0.016 *
FinalWordTypeFunction	-0.039611	0.036359	-1.089	0.276
GroupNN:GenderMale	-0.001318	0.319789	-0.004	0.997
GroupAD:GenderMale	-0.058290	0.292421	-0.199	0.842
GroupIN:GenderMale	0.164195	0.369864	0.444	0.657
GroupNN:SentenceTypeQuestion	-0.137948	0.042455	-3.249	0.001 **
GroupAD:SentenceTypeQuestion	-0.282056	0.041713	-6.762	< 0.0001 ***
GroupIN:SentenceTypeQuestion	-0.277950	0.042223	-6.583	< 0.0001 ***
GroupNN:FocusPositionIntermediate	-0.051672	0.051902	-0.996	0.319
GroupAD:FocusPositionIntermediate	-0.070217	0.051107	-1.374	0.170
GroupIN:FocusPositionIntermediate	-0.053990	0.051631	-1.046	0.296

Table 4.5 (Continued)

Predicators	**Estimate**	**Std. Error**	**t-value**	**p value**
GroupNN:FocusPositionFinal	-0.008806	0.052269	-0.168	0.867
GroupAD:FocusPositionFinal	0.007604	0.051311	0.148	0.882
GroupIN:FocusPositionFinal	-0.073316	0.051986	-1.410	0.159
GroupNN:Syllables14	0.017748	0.052470	0.338	0.735
GroupAD:Syllables14	0.053021	0.051581	1.028	0.304
GroupIN:Syllables14	0.063817	0.052211	1.222	0.222
GroupNN:Syllables21	-0.055742	0.051798	-1.076	0.282
GroupAD:Syllables21	0.023238	0.050847	0.457	0.648
GroupIN:Syllables21	0.015751	0.051442	0.306	0.760
GroupNN:FinalWord TypeFunction	0.030457	0.042438	0.718	0.473
GroupAD:FinalWord TypeFunction	0.042571	0.041694	1.021	0.307
GroupIN:FinalWord TypeFunction	0.033295	0.042190	0.789	0.430

There were no significant main effects for **Group** (NN, AD, IN compared to NA), indicating that, in isolation, the **Group** variable itself did not have a significant direct effect on pitch range, but its interaction with **Sentence Type** was important. As for **Sentence Type** (Question vs. Declarative), the model results indicate that it significantly influenced pitch range ($t = 9.035$, $p < 0.0001$), with questions showing greater pitch range. Additionally, **Focus Position** had a significant effect, with both Focus Position Intermediate ($t = 4.570$, $p < 0.0001$) and Focus Position Final ($t = 2.806$, $p < 0.05$) showing greater pitch variation compared to other positions. **Syllables21** was significantly associated with pitch range ($t = 2.410$, $p = 0.016$), indicating that longer syllables (21 syllables) resulted in a greater pitch range. The interaction between **Group** and **Sentence Type** was also significant, with GroupNN:SentenceTypeQuestion ($t = -3.249$, $p < 0.01$), GroupAD:SentenceTypeQuestion ($t = -6.762$, $p < 0.0001$), and

GroupIN:SentenceTypeQuestion (t = -6.583, $p < 0.0001$) showing significant effects, suggesting that the interaction between group and Sentence Type significantly influenced pitch variation, especially in questions compared to declarative sentences.

To sum up, significant effects were found for **Sentence Type**, **Focus Position** and **Syllables 21**. The **Group** variable itself did not have a significant direct effect on pitch range, but its interaction with **Sentence Type** was significant across all the groups.

4.2.1.3 Number of F0 Peaks and Valleys on the Utterance Level

As with our analysis of pitch change and pitch range, we first investigated the effect of several key factors on the number of F0 peaks and valleys, visualized the data, and assessed the relationships between these factors and the outcome variable prior to conducting the modeling.

4.2.1.3.1 Pre-Modeling Data Visualization and Analysis

The mean number of F0 peaks and valleys on the utterance level for each **Group** was as follows: the NA group had a mean of 3.53 (SD = 3.12), the NN group had a mean of 4.63 (SD = 3.17), the AD group had a mean of 5.42 (SD = 3.78), and the IN group had a mean of 5.69 (SD = 3.56). These values suggest that the number of F0 peaks and valleys increased as the proficiency level decreased, with the IN and AD groups showing higher values compared to the NA and NN groups (see Figure 4.10).

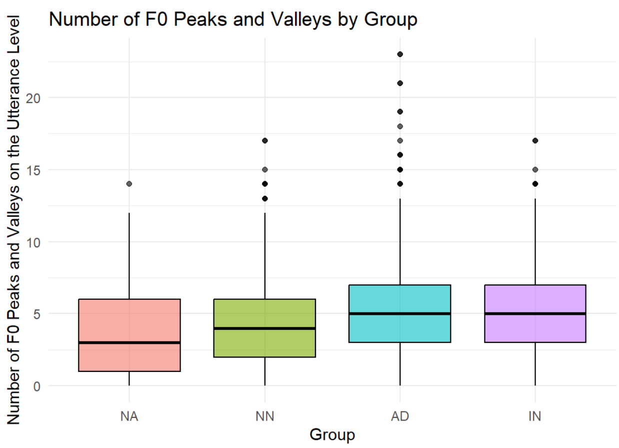


Figure 4.10 Number of F0 Peaks and Valleys by Group

The mean number of F0 peaks and valleys for **Sentence Type** was as follows:

Declarative sentences had a mean of 4.97 (SD = 3.47), while Questions had a mean of 5.10 (SD = 3.61). The difference between the two Sentence Types is relatively small, suggesting that Sentence Type has a minimal impact on the number of F0 peaks and valleys (see Figure 4.11).

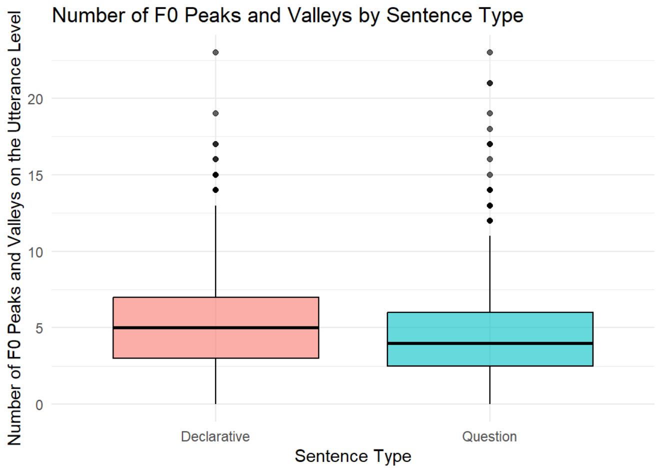


Figure 4.11 Number of F0 Peaks and Valleys by Sentence Type

The mean value of number of F0 peaks and valleys for **Focus Position** increases from the Broad position (mean = 4.77, SD = 3.66) to the Intermediate position (mean = 5.08, SD = 3.34) and further to the Final position (mean = 5.25, SD = 3.62), with the Final position having the highest mean. The standard deviations are relatively consistent across all three levels, ranging from 3.34 to 3.66, indicating similar variability in the data for each Focus Position. However, the significance of these differences has yet to be tested, and further analysis is needed to determine whether the observed patterns are statistically meaningful (see Figure 4.12).

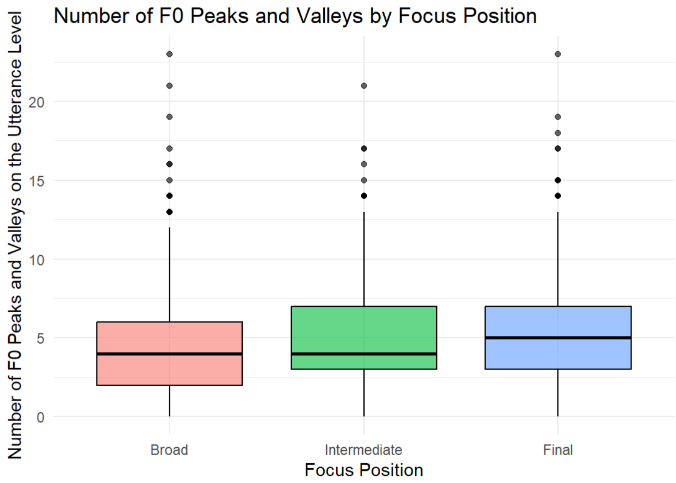


Figure 4.12 Number of F0 Peaks and Valleys by Sentence Type

The mean number of F0 peaks and valleys significantly increased with **Syllables**: for 7 Syllables, the mean was 2.73 (SD = 1.62), for 14 syllables, the mean was 4.35 (SD = 2.20), and for 21 syllables, the mean was 7.98 (SD = 3.93). This shows a clear upward trend, with the 21 Syllables showing notably higher values compared to the 7 and 14 Syllable conditions (see Figure 4.13).

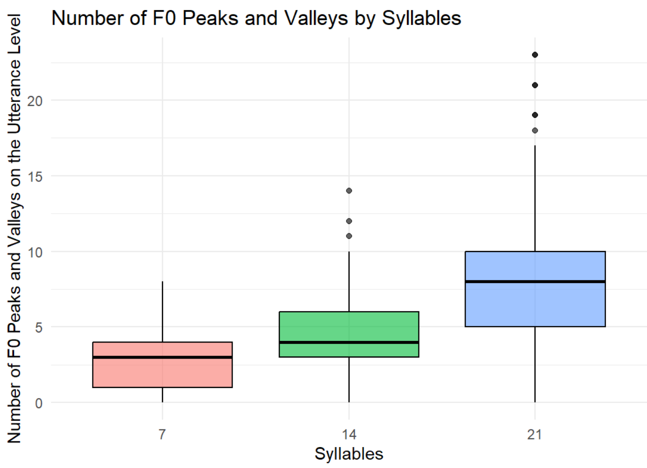


Figure 4.13 Number of F0 Peaks and Valleys by Syllables

In summary, the number of F0 peaks and valleys increased as proficiency level decreased, with the **IN** and **AD** groups showing higher values than the **NA** and **NN** groups. **Sentence Type** had a minimal impact, with only a slight difference between

Declarative sentences and **Questions**. **Focus Position** showed a gradual increase in the number of F0 peaks and valleys from **Broad** to **Intermediate**, and then to **Final**, although the statistical significance of these differences requires further testing. Finally, syllable length had a clear upward trend, with **21 Syllables** showing notably higher values compared to **7** and **14 Syllables**.

4.2.1.3.2 Linear Mixed-Effects Modeling

In the analysis of the number of F0 peaks and valleys, we followed a methodology analogous to that employed for pitch change and pitch range. Two models were constructed using the lmer function: one incorporating interaction terms and another without them. To ensure normality for the linear mixed model, the outcome variable was standardized. Upon evaluation, the model including interaction terms demonstrated a superior fit. The reference group for both fixed and random effects remained consistent, with the baseline defined as GroupNA (Native), Gender Female, Sentence Type Declarative, Focus Position Broad, Syllables 7, and Final Word Type Content. To address variability across individual speakers and sentences, random intercepts for SpeakerCode and SentenceCode were included.

Model Formula:

The model was specified as follows:

$$\begin{aligned}\text{Numstandard} \sim\ & \text{Group} * \text{Gender} + \text{Group} * \text{Sentence Type} + \text{Group} \\ & * \text{Focus Position} + \text{Group} * \text{Syllables} + \text{Group} * \text{Final Word Type} \\ & + (1|\text{SpeakerCode}) + (1|\text{SentenceCode})\end{aligned}$$

Model Fit Assessment

REML Criterion: The REML value at convergence is 2650.6, indicating the overall fit of the model.

Scaled Residuals:

The scaled residuals range from a minimum of -3.5651 to a maximum of 4.9177, with the first quartile at -0.6230, the median at -0.0343, and the third quartile at 0.5896, indicating that the residuals are roughly symmetric around zero but include a few extreme values, such as the maximum residual close to 5.

Random Effects Variance: The random effects reveal that the between-sentence variability (SentenceCode) is negligible, with a variance of 0.0002556 (SD = 0.016), while the between-speaker variability (SpeakerCode) is moderate, with a variance of 0.177 (SD = 0.421). The residual variance is 0.316 (SD = 0.562), accounting for the majority of the total variance.

Fixed Effects:

The results of the model demonstrate both significant and non-significant effects on number of F0 peaks and valleys. Here is the summary of the fixed effects (see Table 4.6):

Table 4.6 Fixed Effects

Predictors	Estimate	Std. Error	t-value	p-value
(Intercept)	-0.6624404	0.3205737	-2.066	0.039*
GroupNN	0.1084128	0.3578788	0.303	0.762
GroupAD	0.2718042	0.3609041	0.753	0.451
GroupIN	-0.0018414	0.3548489	-0.005	0.996
GenderMale	-0.7626344	0.3929923	-1.941	0.052
SentenceTypeQuestion	0.1361917	0.0812058	1.677	0.094
FocusPositionIntermediate	0.0171142	0.0993476	0.172	0.863
FocusPositionFinal	0.1167304	0.1000447	1.167	0.243
Syllables14	0.4287545	0.1007788	4.254	< 0.001***
Syllables21	1.2698654	0.0989909	12.828	< 0.001***
FinalWordTypeFunction	-0.0001882	0.0811687	-0.002	0.998
GroupNN:GenderMale	-0.0470961	0.5176783	-0.091	0.928
GroupAD:GenderMale	-0.0927231	0.4732507	-0.196	0.845
GroupIN:GenderMale	0.1978557	0.5985429	0.331	0.741
GroupNN:SentenceTypeQuestion	-0.2429717	0.0985732	-2.465	0.014*
GroupAD:SentenceTypeQuestion	-0.0159117	0.0968619	-0.164	0.870
GroupIN:SentenceTypeQuestion	-0.0538963	0.0980143	-0.550	0.582

Table 4.6 (Continued)

Predictors	Estimate	Std. Error	t-value	p-value
GroupNN:FocusPositionIntermediate	0.0662531	0.1204969	0.550	0.582
GroupAD:FocusPositionIntermediate	0.0650381	0.1186640	0.548	0.584
GroupIN:FocusPositionIntermediate	0.0909756	0.1198796	0.759	0.448
GroupNN:FocusPositionFinal	0.0368023	0.1214019	0.303	0.762
GroupAD:FocusPositionFinal	0.0663288	0.1191950	0.556	0.578
GroupIN:FocusPositionFinal	0.0506643	0.1207050	0.420	0.674
GroupNN:Syllables14	-0.0054903	0.1218678	-0.045	0.964
GroupAD:Syllables14	0.0992294	0.1197993	0.828	0.408
GroupIN:Syllables14	0.0096579	0.1212031	0.080	0.936
GroupNN:Syllables21	0.0575718	0.1202674	0.479	0.632
GroupAD:Syllables21	0.1871373	0.1180813	1.585	0.113
GroupIN:Syllables21	0.5059119	0.1194075	4.237	< 0.001***
GroupNN:Final WordTypeFunction	-0.0380384	0.0985277	-0.386	0.699
GroupAD:Final WordTypeFunction	-0.0303798	0.0968214	-0.314	0.754
GroupIN:Final Word TypeFunction	0.0377989	0.0979313	0.386	0.699

The analysis reveals that **Group** itself has no significant main effect, indicating no substantial differences across groups in isolation; among all single factors, only **Syllables** show significant effects, with both Syllables21 ($t = 12.828$, $p < 0.001$) and Syllables14 ($t = 4.254$, $p < 0.001$) having strong positive impacts on the dependent variable, particularly for Interaction between **Syllables21** and **GroupIN** ($t = 4.237$, $p < 0.001$), where the effect is notably stronger. Additionally, significant interactions are observed between **Group** and other variables, such as **Sentence Type** (e.g., GroupNN:SentenceTypeQuestion: $t = -2.465$, $p = 0.014$) and **Syllables**, suggesting that the influence of these factors varies across different groups, though the specific nature of these interactions requires further exploration.

In brief, while Group has no significant main effect, **Syllables21 and Syllables14** show strong positive effects, with GroupIN exhibiting a particularly strong interaction with Syllables21. Significant interactions between **Group and Sentence Type** further suggest group-specific variability, though their precise patterns require further exploration.

In all, in terms of overall sentence intonation, the study thoroughly analyzed intonation patterns through three realizational acoustic features—pitch change amount on the syllable level (st), pitch range on the utterance level (st) and number of F0 peaks and valleys on the utterance level. Each acoustic feature was analyzed using a separate interaction model. **Syllables** showed strong effects, particularly **Syllables21**, which significantly impacted both **pitch range** ($t = 2.41$, $p = 0.016$) and **number of F0 peaks and valleys** ($t = 12.83$, $p < 0.001$), while **Syllables14** significantly influenced **pitch change** ($t = -5.71$, $p < 0.001$) and **number of F0 peaks/valleys** ($t = 4.25$, $p < 0.001$). Specific main effects included **Sentence Type** and **Focus Position**, with **SentenceTypeQuestion** significantly affecting **pitch change** ($t = 4.87$, $p < 0.001$) and **pitch range** ($t = 9.04$, $p < 0.001$), and **FocusPositionIntermediate** showing significance for both **pitch change** ($t = 5.69$, $p < 0.001$) and **pitch range** ($t = 4.57$, $p < 0.001$). However, **number of F0 peaks/valleys** were mainly driven by **Syllables21**, with no significant effects from Sentence Type or Focus Position. While **Group** (**NN, AD, IN**) did not show significant main effects, its interactions with **Sentence Type** and **Syllables** revealed nuanced patterns, particularly for lower-proficiency groups (AD/IN), who showed stronger interactions with Sentence Type and Syllables. These results suggest proficiency-dependent adaptation to structural complexity. Visualization of group differences showed that **IN/AD groups** had higher means for **pitch change** and more **number of F0 peaks/valleys** compared to **NA/NN groups**, but the modeling results indicated that these differences were mediated by interactions rather than direct group effects. Moving forward, the systemic dimension (ToBI annotations) will be analyzed to clarify the phonological encoding of focus and boundary tone, bridging the gap in the realizational dimension and resolving ambiguities in the interaction effects.

4.2.2 Focus Realization

In the Focus Realization section, we analyzed the pitch patterns associated with different Focus Positions, Sentence Types, and Syllables. The focus realization data are mainly qualitative and based on ToBI annotations, which provide insight into the phonological encoding of focus. We excluded control sentences, leaving 24 sentences per participant for analysis (unless a sentence exhibited two distinct pitch patterns for the same focus). Within focus realization, no significant effect of Final Word Type on the dependent variable was found; consequently, the data were pooled, with each participant contributing 12 sets of sentences (**2 Focus Position * 2 Sentence Type * 3 Syllables**). Specifically, we measured the proportions of pitch patterns (L*, H*, !H*, L+H*, L+!H*, L*+H, and H+!H*) (Refer to Section 2.2.2 for tone visualizations and explanations) for each participant across the 12 sets of sentences (see Table 4.7 for details).

The analysis of this data will be presented in the following order: GroupNA, GroupNN, GroupAD and GroupIN, in terms of focus realization in the 12 sets of sentences. At the end of this section, typical group differences in the realization of different Sentence Types and Focus Positions will be illustrated through pitch contour figures, offering a visual representation of the phonological patterns across groups.

Table 4.7 Representation of SentenceCode in 12 Sets of Sentences

SentenceCode	**Representation**	**Predicators**
Fw/CwXXXXX	Fw: Function word Cw: Content word	Final Word Type
XXSs/Ms/LsXXX	Ss: Short sentence (7 syllables) Ms: Medium sentence (14 syllables) Ls: Long sentence (21 syllables)	Syllables
XXXXQ/DXX	Q: Question D: Declarative	Sentence Type
XXXXXIF/FF	IF: Intermediate Focus Position FF: Final Focus Position	Focus Position

To facilitate readability, we would like to clarify the coding system used for SentenceCode. The first two letters, Fw, represent function words, while Cw stands for content words. This pair corresponds to the Final Word Type independent variable. The middle two letters in the sentence code indicate the Syllables variable, with Ss representing short sentences (7 syllables), Ms representing medium-length sentences (14 syllables), and Ls representing long sentences (21 syllables). The third-to-last letter denotes the Sentence Type, where Q stands for questions and D stands for declarative sentences. Finally, the last two letters indicate the Focus Position, with IF representing intermediate focus and FF representing final focus. These 12 distinct combinations form the 12 sets used in the analysis.

4.2.2.1 Focus Realization in GroupNA

The following table shows the proportion of tones used by GroupNA across the 12 sets of sentences (see Table 4.8). The table will be divided into smaller ones, and the analysis will generally follow the order of Sentence Type, Focus Position, Final Word Type, Syllables, and Exceptional cases.

Table 4.8 Pitch Pattern in GroupNA

Tones / SentenceCode	L*(%)	H*(%)	L+H*(%)	L*+H(%)	!H*(%)	Sum (%)
FwSsDIF/CwSsDIF	0	50	30	10	10	100
FwSsDFF/CwSsDFF	0	90	0	0	10	100
FwSsQIF/CwSsQIF	100	0	0	0	0	100
FwSsQFF/CwSsQFF	100	0	0	0	0	100
FwMsDIF/CwMsDIF	0	70	30	0	0	100
FwMsDFF/CwMsDFF	0	70	0	0	30	100
FwMsQIF/CwMsQIF	36.36	9.09	18.19	36.36	0	100
FwMsQFF/CwMsQFF	100	0	0	0	0	100
FwLsDIF/CwLsDIF	0	45.45	36.36	18.19	0	100
FwLsDFF/CwLsDFF	0	72.73	9.09	9.09	9.09	100
FwLsQIF/CwLsQIF	14.29	14.29	7.14	64.28	0	100
FwLsQFF/CwLsQFF	100	0	0	0	0	100

4.2.2.1.1 Pitch pattern by Sentence Type in GroupNA

For GroupNA, the majority of focus in Declarative sentences is realized with tones related to H* (H*, L+H*, !H*), accounting for 93.79% of cases, with a small proportion (6.21%) using L* and L*+H. In contrast, for Questions, the focus is predominantly realized with tones related to L* (L*, L*+H), making up 91.88% of cases, with only a minimal occurrence (8.12%) of H* and L+H* tones (see Table 4.9).

Table 4.9 Pitch Pattern by Sentence Type in GroupNA

Sentence Type	L* (L* + L*+H)	H* (H* + L+H* + !H*)	Total
Declarative (D)	6.21%	93.79%	100%
Question (Q)	91.88%	8.12%	100%

4.2.2.1.2 Pitch Pattern by Focus Position in GroupNA

For GroupNA, focus realization in Questions varies based on Focus Position and Syllables: in Final Focus (QFF), L* is used exclusively (100%) across all sentence lengths (7, 14, and 21 syllables), while in Intermediate Focus (QIF), L* dominates in 7-syllable sentences (100%), but longer sentences (14 and 21 syllables) show increased use of L*+H (36.36% and 64.28%, respectively) alongside minimal occurrences of H* (9.09% and 14.29%) and L+H* (18.19% and 7.14%). These patterns indicate a strong preference for L* and its variants in questions, with final focus strictly adhering to L* and intermediate focus exhibiting greater variability, particularly in longer sentences (see Table 4.10).

Table 4.10 Pitch Pattern by Question Focus Position in GroupNA

Sentence Type	Focus Position	Syllables	L*	L*+H	H*	L+H*	!H*	Total
Question (Q)	Final Focus (FF)	7 (Ss)	100%	0%	0%	0%	0%	100%
	Final Focus (FF)	14 (Ms)	100%	0%	0%	0%	0%	100%
	Final Focus (FF)	21 (Ls)	100%	0%	0%	0%	0%	100%

Table 4.10 (Continued)

Sentence Type	**Focus Position**	**Syllables**	**L***	**L*+H**	**H***	**L+H***	**!H***	**Total**
Question (Q)	Intermediate Focus (IF)	7 (Ss)	100%	0%	0%	0%	0%	100%
	Intermediate Focus (IF)	14 (Ms)	36.36%	36.36%	9.09%	18.19%	0%	100%
	Intermediate Focus (IF)	21 (Ls)	14.29%	64.28%	14.29%	7.14%	0%	100%

For GroupNA, focus realization in Declarative sentences also varies by Focus Position and Syllables: in Final Focus (DFF), H* (including H*, !H*, and L+H*) is used exclusively (100%) in 7 and 14-syllable sentences, while in 21-syllable sentences, L*+H shows up at a small proportion (9.09%). In Intermediate Focus (DIF), H* dominates in 14-syllable sentences (100%), but shorter (7 syllables) and longer (21 syllables) sentences show minor occurrences of L*+H (10% and 18.19%, respectively). These patterns highlight a strong preference for H* -related in declarative sentences, with final focus more strictly adhering to H*-related and intermediate focus exhibiting a little more variability, particularly in shorter and longer sentences (see Table 4.11).

Table 4.11 Pitch Pattern by Declarative Focus Position in GroupNA

Sentence Type	**Focus Position**	**Syllables**	**L***	**L*+H**	**H***	**L+H***	**!H***	**Total**
Declarative (D)	Final Focus (FF)	7 (Ss)	0%	0%	90%	0%	10%	100%
	Final Focus (FF)	14 (Ms)	0%	0%	70%	0%	30%	100%
	Final Focus (FF)	21 (Ls)	0%	9.09%	72.73%	9.09%	9.09%	100%

Table 4.11 (Continued)

Sentence Type	Focus Position	Syllables	L*	L*+H	H*	L+H*	!H*	Total
Declarative (D)	Intermediate Focus (IF)	7 (Ss)	0%	10%	50%	30%	10%	100%
	Intermediate Focus (IF)	14 (Ms)	0%	0%	70%	30%	0%	100%
	Intermediate Focus (IF)	21 (Ls)	0%	18.19%	45.45%	36.36%	0%	100%

In brief, in both QFF and DFF, the Focus Position is the primary determinant of tone realization, with Final Focus showing more consistency (L*-related for questions, H*-related for declaratives), while Intermediate Focus displays more variability.

4.2.2.1.3 Pitch Pattern by Syllables and Final Word Type in GroupNA

No clear trends are observed in GroupNA regarding the influence of Final Word Type on the pitch patterns used in focus realization. As for Syllables, the case is consistent with that of Final Word Type.

4.2.2.1.4 Exceptional Cases in GroupNA

GroupNA only exhibits a few anomalies in FwSsQFF, where the final word, "herself," requires the pitch accent to align with the stressed syllable of the word. Typically, L* should align with "self." However, in three cases, L* is incorrectly aligned with "her."

4.2.2.2 Focus Realization in GroupNN

The table below presents the distribution of tones used by GroupNN across the 12 sentence sets (see Table 4.12). This table will be broken down into smaller tables, with the analysis organized by Sentence Type, Focus Position, Final Word Type, Syllables, and Exceptional Cases, as in GroupNA.

Table 4.12 Pitch Pattern in GroupNN

SentenceCode \ Tones	L* (%)	H* (%)	L+H* (%)	L*+H (%)	!H* (%)	L+!H* (%)	H+!H* (%)	Sum(%)
FwSsDIF/CwSsDIF	5	25	65	0	0	5	0	100
FwSsDFF/CwSsDFF	0	55	35	0	5	5	0	100
FwSsQIF/CwSsQIF	66.67	14.28	19.05	0	0	0	0	100
FwSsQFF/CwSsQFF	100	0	0	0	0	0	0	100
FwMsDIF/CwMsDIF	0	5	85	5	0	5	0	100
FwMsDFF/CwMsDFF	0	45	30	0	20	5	0	100
FwMsQIF/CwMsQIF	19.05	4.76	33.33	42.86	0	0	0	100
FwMsQFF/CwMsQFF	90	5	5	0	0	0	0	100
FwLsDIF/CwLsDIF	0	0	85	0	5	5	5	100
FwLsDFF/CwLsDFF	0	55	45	0	0	0	0	100
FwLsQIF/CwLsQIF	4.35	0	34.78	60.87	0	0	0	100
FwLsQFF/CwLsQFF	100	0	0	0	0	0	0	100

4.2.2.2.1 Pitch Pattern by Sentence Type in GroupNN

For GroupNN, focus realization in Declarative sentences is predominantly achieved using tones related to H* (including H*, L+H*, !H*, L+!H*, and H+!H*), accounting for 96.67% of cases, with only a minimal proportion (3.33%) using L* and L*+H. In contrast, for Questions, focus is primarily realized with tones related to L* (including L* and L*+H), making up 80.63% of cases, while H*-related tones still account for a notable proportion (19.37%) (see Table 4.13).

Compared with GroupNA, the performance of GroupNN is similar. Both groups share with a higher proportion of H*-related tones in Declarative sentences compared to the L*-related tones used in Questions. Interestingly, GroupNN exhibits an even higher proportion of H*-related tones in Declarative sentences than GroupNA, while in Questions, GroupNN uses more H*-related tones than GroupNA.

Table 4.13 Pitch Pattern by Sentence Type in GroupNN

Sentence Type	L* (L* /L*+H)	H* (H* + L+H* + !H* + L+!H* + H+!H*)	Total
Declarative (D)	3.33%	96.67%	100%
Question (Q)	80.63%	19.37%	100%

4.2.2.2.2 Pitch Pattern by Focus Position in GroupNN

For GroupNN, focus realization in Questions varies significantly based on Focus Position and Syllables: in Final Focus (QFF), L* is used exclusively in short (7 syllables, 100%) and long (21 syllables, 100%) sentences, while medium-length sentences (14 syllables) show a minor presence of H*-related tones (5% H* and 5% L+H*). In Intermediate Focus (QIF), H*-related tones, particularly L+H*, increase with sentence length: in short sentences (7 syllables), L+H* accounts for 19.05%, rising to 33.33% in medium sentences (14 syllables) and 34.78% in long sentences (21 syllables). Meanwhile, L*+H becomes dominant in longer sentences (42.86% in 14 syllables and 60.87% in 21 syllables), indicating a shift toward more complex tonal patterns as sentence length increases. These patterns suggest that GroupNN relies heavily on L* for final focus but incorporates more H*-related tones, especially L+H*, in intermediate focus as sentences grow longer (see Table 4.14).

Table 4.14 Pitch Pattern by Question Focus Position in GroupNN

Sentence Type	Focus Position	Syllables	L*	H*	L+H*	L*+H	Total
Question (Q)	Final Focus (FF)	7(Ss)	100%	0%	0%	0%	100%
	Final Focus (FF)	14 (Ms)	90%	5%	5%	0%	100%
	Final Focus (FF)	21 (Ls)	100%	0%	0%	0%	100%

Table 4.14 (Continued)

Sentence Type	**Focus Position**	**Syllables**	**L***	**H***	**L+H***	**L*+H**	**Total**
Question (Q)	Intermediate Focus (IF)	7 (Ss)	66.67%	14.28%	19.05%	0%	100%
	Intermediate Focus (IF)	14 (Ms)	19.05%	4.76%	33.33%	42.86%	100%
	Intermediate Focus (IF)	21 (Ls)	4.35%	0%	34.78%	60.87%	100%

For GroupNN, focus realization in Declarative sentences is characterized by a strong preference for H*-related tones: in Final Focus (DFF), H*-related tones (including H*, L+H*, !H*, L+!H*, and H+!H*) account for 100% of cases across all sentence lengths (7, 14, and 21 syllables). In Intermediate Focus (DIF), H*-related tones dominate in long sentences (21 syllables, 100%), while short (7 syllables) and medium (14 syllables) sentences show minor use of L*-related tones (5% L* in 7 syllables and 5% L*+H in 14 syllables). These patterns highlight a robust reliance on H*-related tones in declaratives, with only minimal variability in intermediate focus for short and medium-length sentences (see Table 4.15).

Table 4.15 Pitch Pattern by Sentence Type in GroupNN

Sentence Type	**Focus Position**	**Syllables**	**L***	**H***	**L+H***	**L*+H**	**!H***	**L+!H***	**H+!H***	**Total**
Declarative (D)	Final Focus (FF)	7 (Ss)	0%	55%	35%	0%	5%	5%	0%	100%
	Final Focus (FF)	14 (Ms)	0%	45%	30%	0%	20%	5%	0%	100%
	Final Focus (FF)	21 (Ls)	0%	55%	45%	0%	0%	0%	0%	100%

Table 4.15 (Continued)

Sentence Type	**Focus Position**	**Syllables**	**L***	**H***	**L+H***	**L*+H**	**!H***	**L+!H***	**H+!H***	**Total**
Declarative (D)	Intermediate Focus (IF)	7 (Ss)	5%	25%	65%	0%	0%	5%	0%	100%
	Intermediate Focus (IF)	14 (Ms)	0%	5%	85%	5%	0%	5%	0%	100%
	Intermediate Focus (IF)	21 (Ls)	0%	0%	85%	0%	5%	5%	5%	100%

In terms of Focus Position, GroupNN shows similar patterns to GroupNA, with Final Focus demonstrating more consistency (L*-related for questions, H*-related for declaratives), while Intermediate Focus exhibits more variability. However, unlike GroupNA, GroupNN prefers a wider range by using L+H* in DIF, while GroupNA shows a stronger preference for H*. In DFF, both groups are dominated by H*, but GroupNN shows a noticeable increase in the proportion of L+H* compared to GroupNA. Additionally, in QIF, although both groups prefer wider-range L*+H rather than L* in long sentences, GroupNN shows an increasing trend in the use of L+H* with greater sentence length, a pattern not observed in GroupNA.

4.2.2.2.3 Pitch Patterns by Syllables and Final Word Type in GroupNN

Similar to GroupNA, GroupNN does not show a significant influence of Final word Type on focus realization, and Syllables does not appear to have a notable impact either. The only exception is in QIF, where GroupNN exhibits an increasing trend in the use of wider-range L+H* and L*+H as sentence length increases.

4.2.2.2.4 Exceptional Cases in GroupNN

As in GroupNA, GroupNN exhibits most of its anomalies in FwSsQFF, where L* incorrectly aligns with "her" instead of "self" in four cases. Additionally, in FwSsQIF, where the intermediate word "murmur" requires the pitch accent to align with the stressed syllable (the first "mur"), there is one instance where the stress is incorrectly placed on the second "mur."

4.2.2.3 Focus Realization in GroupAD

The table below shows the tone distribution used by **GroupAD** across the 12 sentence sets (see Table 4.16). It will be divided into smaller tables for analysis, following the same structure as for GroupNA and GroupNN, with an order of Sentence Type, Focus Position, Final Word Type, Syllables, and Exceptional Cases.

Table 4.16 Pitch Pattern in GroupAD

Tones / SentenceCode	L* (%)	H* (%)	L+H* (%)	L*+H (%)	!H* (%)	L+!H* (%)	H+!H* (%)	Sum (%)
FwSsDIF/CwSsDIF	4.55	31.82	59.09	4.55	0	0	0	100
FwSsDFF/CwSsDFF	7.69	50	26.92	0	11.53	0	3.86	100
FwSsQIF/CwSsQIF	13.64	18.18	54.55	13.64	0	0	0	100
FwSsQFF/CwSsQFF	75	14.29	10.71	0	0	0	0	100
FwMsDIF/CwMsDIF	0	4.35	91.30	4.35	0	0	0	100
FwMsDFF/CwMsDFF	0	22.73	45.45	0	31.82	0	0	100
FwMsQIF/CwMsQIF	13.64	9.09	63.64	9.09	0	4.55	0	100
FwMsQFF/CwMsQFF	100	0	0	0	0	0	0	100
FwLsDIF/CwLsDIF	0	26.09	60.87	13.04	0	0	0	100
FwLsDFF/CwLsDFF	0	50	40.91	0	9.09	0	0	100
FwLsQIF/CwLsQIF	7.69	7.69	65.38	15.38	3.85	0	0	100
FwLsQFF/CwLsQFF	95.45	4.55	0	0	0	0	0	100

4.2.2.3.1 Pitch Pattern by Sentence Type in GroupAD

For GroupAD, focus realization in Declarative sentences is dominated by H*-related tones (94.31%), with only a small proportion (5.70%) using L*-related tones (L* and L*+H). In Questions, the patterns differ significantly between Focus Positions: Final Focus (QFF) is predominantly realized with L*-related tones (90.15%), while

Intermediate Focus (QIF) relies heavily on H*-related tones (75.64%), particularly L+H*. Notably, this contrasts with GroupNA/NN, where L*-related tones dominate in both QFF and QIF (see Table 4.17).

Table 4.17 Pitch Pattern by Sentence Type in GroupAD

Sentence Type	L* (L*, L*+H)	H* (H*, L+H*, !H*, L+!H*, H+!H*)	Total
Declarative (D)	5.70%	94.31%	100%
Question Final Focus (QFF)	90.15%	9.85%	100%
Question Intermediate Focus (QIF)	24.36%	75.64%	100%

4.2.2.3.2 Pitch Pattern by Focus Position in GroupAD

For GroupAD in QFF, only 14-syllable sentences exclusively use L* (100%). In 7-syllable (Ss) and 21-syllable (Ls) sentences, there is a mix, with H*-related tones appearing in smaller proportions.

For QIF, GroupAD shows a shift compared to GroupNN and GroupNA, moving away from the dominance of L*-related tones. Instead, H*-related tones dominate (75.64%), with L+H* being the most prevalent across all Syllables. The proportion of wider-range L+H* rises as Syllables increases (see Table 4.18).

Table 4.18 Pitch Pattern by Question Focus Position in GroupAD

Sentence Type	Focus Position	Syllables	L*	H*	L+H*	L*+H	!H*	L+!H*	Total
Question (Q)	Final Focus (FF)	7 (Ss)	75%	14.29%	10.71%	0%	0%	0%	100%
	Final Focus (FF)	14 (Ms)	100%	0%	0%	0%	0%	0%	100%

Table 4.18 (Continued)

Sentence Type	Focus Position	Syllables	L*	H*	L+H*	L*+H	!H*	L+!H*	Total
Question (Q)	Final Focus (FF)	21 (Ls)	95.45%	4.55%	0%	0%	0%	0%	100%
	Intermediate Focus (IF)	7 (Ss)	13.64%	18.18%	54.55%	13.64%	0%	0%	100%
Question (Q)	Intermediate Focus (IF)	14 (Ms)	13.64%	9.09%	63.64%	9.09%	0%	4.55%	100%
	Intermediate Focus (IF)	21 (Ls)	7.69%	7.69%	65.38%	15.38%	3.85%	0%	100%

Similar to GroupNA/NN, DFF in GroupAD is almost entirely realized with H*-related tones (92.31–100%), with only short sentences (7 syllables) showing a minor presence of L* (7.69%)

While GroupAD largely follows GroupNA/NN in using H*-related tones for DIF, it exhibits a higher proportion of L*-related tone, particularly in long sentences (13.04% L*+H), suggesting a unique adaptation in tonal complexity (see Table 4.19).

Table 4.19 Pitch Pattern by Declarative Focus Position in GroupAD

Sentence Type	Focus Position	Syllables	L*	H*	L+H*	L*+H	!H*	H+!H*	Total
Declarative (D)	Final Focus (FF)	7 (Ss)	7.69%	50.00%	26.92%	0%	11.53%	3.86%	100%
	Final Focus (FF)	14 (Ms)	0%	22.73%	45.45%	0%	31.82%	0%	100%
	Final Focus (FF)	21 (Ls)	0%	50.00%	40.91%	0%	9.09%	0%	100%

Table 4.19 (Continued)

Sentence Type	Focus Position	Syllables	L*	H*	L+H*	L*+H	!H*	H+!H*	Total
	Intermediate Focus (IF)	7 (Ss)	4.55%	31.82%	59.09%	4.55%	0%	0%	100%
Declarative (D)	Intermediate Focus (IF)	14 (Ms)	0%	4.35%	91.30%	4.35%	0%	0%	100%
	Intermediate Focus (IF)	21 (Ls)	0%	26.09%	60.87%	13.04%	0%	0%	100%

Similar to GroupNN/NA in Final Focus, the pitch patterns for GroupAD show good tonal consistency in both DFF and QFF, though slightly less so than GroupNN/NA. Most tones are L*-related in QFF and H*-related in DFF. Additionally, as seen in GroupNN/NA, tonal consistency is higher in Final Focus than in Intermediate Focus.

The most notable difference between GroupAD and GroupNN/NA appears in QIF, where GroupAD moves away from the dominance of L*-related tones. Instead, H*-related tones dominate (75.64%) in QIF, with L+H* being the most frequent across all Syllables. The proportion of the wider-range L+H* increases as Syllables grow.

4.2.2.3.3 Pitch Pattern by Syllables and Final Word Type in GroupAD

As in GroupNA/NA, there does not appear a significant influence of Final word Type on focus realization. Syllables may have some influence to some extent. It is worth noting that In QIF, GroupNN exhibits an increasing trend in the use of wider-range L+H* and L*+H as Syllables grow, but its dominant pitch pattern is still L*+H, but in GroupAD, the dominant pitch pattern has shifted to L+H*.

4.2.2.3.4 Exceptional Cases in GroupAD

Unlike GroupNA/NN, who only show minor exceptional cases, those exceptional ones account for 10% in GroupAD. All the exceptional cases lie in 7-syllable final focus (SsQFF/SsDFF) sentences. In GroupAD, exceptional cases predominantly occur with multi-syllable final words such as "herself" and "melodies." The variety and number of exceptional tones in these cases are notably higher compared to GroupNN/NA. One significant pattern observed in these cases is the frequent occurrence of double pitch accents on a single word. For example, in the word "herself," there is an instance where L+H* is placed on "her," followed by L* on "self." This type of double pitch accents is absent in GroupNA/NN, where each word typically receives a single pitch accent. Below is a summary of exceptional cases in GroupAD (see Table 4.20).

Table 4.20 Exceptional Cases in GroupAD

Exceptional Cases	Count
L*(her)	4
L*(dies)	4
H*(her)	3
L+H*(her)	2
L*(her+self)	2
L+H*(her) L*(self)	1
!H*(her) H*(self)	1
H*(mel) L*(dies)	1
H*(her+self)	1
L+H*(her) L*(self)	1
L*(her) H*(self)	1
H*(her) H+!H*(self)	1

4.2.2.4 Focus Realization in GroupIN

The table below presents the distribution of tones used by GroupIN across the 12 sentence sets (see Table 4.21). This table will be broken down into smaller sections for analysis, following the same structure applied to GroupNA, GroupNN, and GroupAD.

The analysis will be organized according to Sentence Type, Focus Position, Final Word Type, Syllables, and Exceptional Cases.

Table 4.21 Pitch Pattern in GroupIN

Tones / SentenceCode	L* (%)	H* (%)	L+H* (%)	L*+H (%)	!H* (%)	L+!H* (%)	H+!H* (%)	Sum (%)
FwSsDIF/CwSsDIF	0	27.27	63.64	4.55	4.55	0.	0	100
FwSsDFF/CwSsDFF	4.17	45.83	25	4.17	16.67	4.17	0	100
FwSsQIF/CwSsQIF	4.17	29.17	50	16.67	0	0	0	100
FwSsQFF/CwSsQFF	66.67	13.33	10	6.67	0	3.33	0	100
FwMsDIF/CwMsDIF	0	17.39	65.22	13.04	0	4.35	0	100
FwMsDFF/CwMsDFF	0	36.36	22.73	0	40.91	0	0	100
FwMsQIF/CwMsQIF	4	12	64	20	0	0	0	100
FwMsQFF/CwMsQFF	86.36	9.09	0	0	4.55	0	0	100
FwLsDIF/CwLsDIF	0	8	60	16	12	4	0	100
FwLsDFF/CwLsDFF	0	31.82	22.73	0	36.36	4.55	4.55	100
FwLsQIF/CwLsQIF	0	12.50	45.83	29.17	8.33	4.17	0	100
FwLsQFF/CwLsQFF	80	12	4	0	4	0	0	100

4.2.2.4.1 Pitch Pattern by Sentence Type in GroupIN

For GroupIN, focus realization in Declarative sentences is predominantly achieved using H*-related tones (93.02%), with a small proportion (6.98%) using L*-related tones (L* and L*+H), similar to GroupNA/NN/AD. However, in Question Intermediate Focus (QIF), GroupIN exhibits a unique pattern: QIF relies heavily on H*-related tones (75.67%), particularly L+H*. This contrasts sharply with GroupNA/NN, but is similar to GroupAD (see Table 4.22).

Table 4.22 Pitch pattern by Sentence Type in GroupIN

Sentence Type	L* (L*, L*+H)	H* (H*, L+H*, !H*, L+!H*, H+!H*)	Total
Declarative (D)	6.98%	93.02%	100%
Question Final Focus (QFF)	77.68%	22.32%	100%
Question Intermediate Focus (QIF)	24.33%	75.67%	100%

4.2.2.4.2 Pitch pattern by Focus Position in GroupIN

For GroupIN in QFF, although sentences still prefer L*-related tones (77.68%), there does not appear any exclusive L* in any Syllables.

For QIF, GroupIN mirrors GroupAD but contrasts with GroupNA/NN, with L+H* dominating instead of L*-related tones (see Table 4.23).

Table 4.23 Pitch Pattern by Question Focus Position in GroupIN

Sentence Type	Focus Position	Syllables	L*	H*	L+H*	L*+H	!H*	L+!H*	Total
Question (Q)	Final Focus (FF)	7 (Ss)	66.67%	13.33%	10%	6.67%	0%	3.33%	100%
	Final Focus (FF)	14 (Ms)	86.36%	9.09%	0%	0%	4.55%	0%	100%
	Final Focus (FF)	21 (Ls)	80%	12%	4%	0%	4%	0%	100%
Question (Q)	Intermediate Focus (IF)	7 (Ss)	4.17%	29.17%	50%	16.67%	0%	0%	100%
	Intermediate Focus (IF)	14 (Ms)	4%	12%	64%	20%	0%	0%	100%
	Intermediate Focus (IF)	21 (Ls)	0%	12.50%	45.83%	29.17%	8.33%	4.17%	100%

For GroupIN, DFF is dominated by H*-related tone, with minor L*-related tone, which follows a pattern similar to all other Groups.

As for DIF in Group IN, L+H* is the primary tone, with L*+H increasing in longer sentences (16.00% in 21 syllables) (see Table 4.24).

Table 4.24 Pitch Pattern by Declarative Focus Position in GroupIN

Sentence Type	Focus Position	Syllables	L*	H*	L+H*	L*+H	!H*	L+!H*	H+!H*	Total
	Final Focus (FF)	7 (Ss)	4.17%	45.83%	25.00%	4.17%	16.67%	4.17%	0.00%	100%
Declarative (D)	Final Focus (FF)	14 (Ms)	0.00%	36.36%	22.73%	0.00%	40.91%	0.00%	0.00%	100%
	Final Focus (FF)	21 (Ls)	0.00%	31.82%	22.73%	0.00%	36.36%	4.55%	4.55%	100%
	Intermediate Focus (IF)	7 (Ss)	0.00%	27.27%	63.64%	4.55%	4.55%	0.00%	0.00%	100%
Declarative (D)	Intermediate Focus (IF)	14 (Ms)	0.00%	17.39%	65.22%	13.04%	0.00%	4.35%	0.00%	100%
	Intermediate Focus (IF)	21 (Ls)	0.00%	8.00%	60.00%	16.00%	12.00%	4.00%	0.00%	100%

GroupIN shows tonal patterns similar to GroupNA/NN/AD, with good consistency in Final Focus, particularly in DFF and QFF, though a little less so than all other Groups. However, in QIF, similar to GroupAD, GroupIN differs from GroupNN/NA, as it shifts from L*-dominant to H*-dominant tones, with L+H* being most frequent.

4.2.2.4.3 Pitch Pattern by Syllables and Final Word Type in GroupIN

No clear trends are observed in GroupIN regarding the effect of Final Word Type on the pitch patterns used in focus realization. As for Syllables, we can only tell that the proportion of the L*+H increases as Syllables grow in DIF.

4.2.2.4.4 Exceptional Cases in GroupIN

In GroupIN, most exceptional cases lie in 7-syllable final focus (SsQFF/SsDFF) sentences with three occurring in LsQFF. Similar to GroupAD, GroupIN frequently use double pitch accents on a single word. Additionally, GroupIN and GroupAD both have various and quite a number of exceptional tones. Below is a summary of exceptional cases in GroupIN (see Table 4.25).

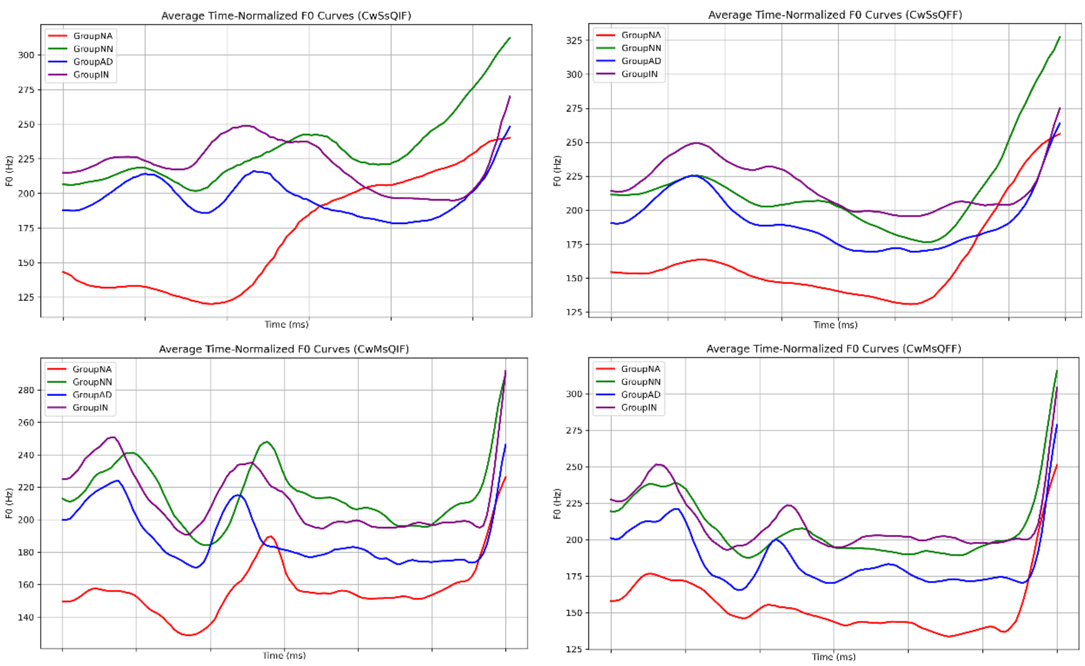
Table 4.25 Exceptional Cases in GroupIN

Exceptional Cases	Count
L*(dies)	7
L*(her)	2
H*(her+self)	2
L+H*(her) L*(self)	1
L*+H(mel) L*(dies)	1
L+!H*(mel) L*(dies)	1
L*(her) H*(self)	1
!H*(her)	1
H*(her) L*(self)	1
L+H*(her)	1
L*(her+self)	1
Aw:L* er:L*	1
H*(her)	1

4.2.2.5 F0 Curves Across Different SentenceCodes

Considering the fact that we did not extract specific data for realization dimension in terms of focus realization, in this section, we will present time-normalized F0 curves across groups for all narrow focus sentences as a compensatory measure.

In Figure 4.14, we present six question sentences of varying lengths (short, medium, and long) and different Focus Positions (intermediate focus on the left and final focus on the right), all of which have a Content Word as the Final Word Type. Since the proportion of male speakers in GroupNA is higher than in other groups, the overall F0 mean appears lower. However, we do not conduct intergroup comparisons of absolute F0 values; rather, we examine the F0 curves at different focus positions to identify potential group differences. On the one hand, when considering only the F0 curves at the Focus Position, Group NA generally exhibits deeper pitch valleys, and the alignment of focus appears to occur slightly earlier compared to other groups. On the other hand, when disregarding the focus position, GroupNA's overall F0 curve is smoother, with smaller pitch variations than those observed in the learner groups. Notably, among the learner groups, GroupNN's F0 trajectory most closely resembles that of GroupNA.



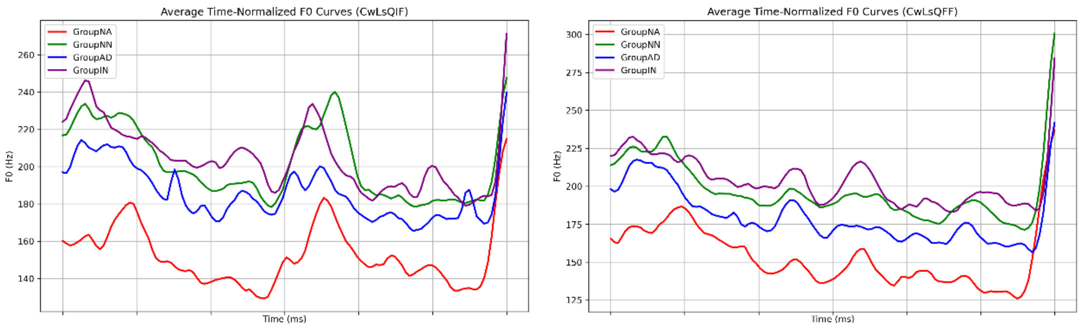
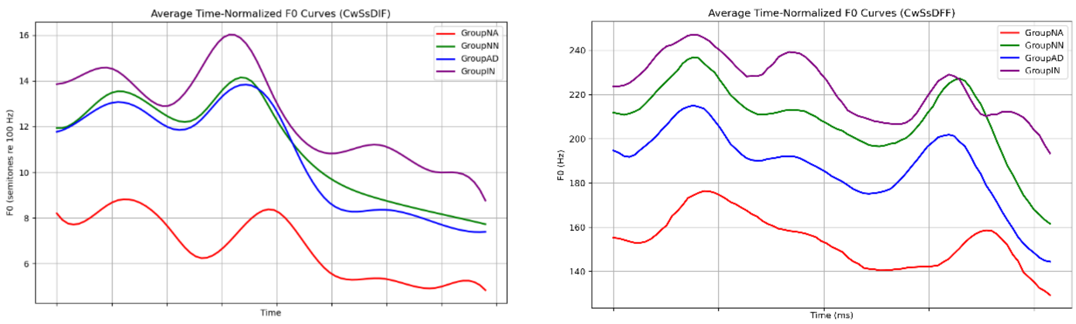


Figure 4.14 F0 Curves for Final Content Word Questions

In Figure 4.15, all variables remain unchanged from Figure 4.14 except for Sentence Type, as question sentences are replaced with declarative sentences. The declarative sentences provide a clearer distinction between GroupNA and the learner groups, as the pitch accent types tend to be more uniform across speakers, facilitating direct comparisons. On the one hand, when examining the F0 curves at the focus position in declarative sentences, GroupNA exhibits a noticeably smaller peak range than the learner groups, particularly in final focus positions. However, unlike in question sentences, we do not observe a consistent difference in peak alignment between GroupNA and the learner groups. On the other hand, when disregarding the focus position, the overall F0 curve of Group NA appears to contain fewer pitch accents than those observed in the learner groups.



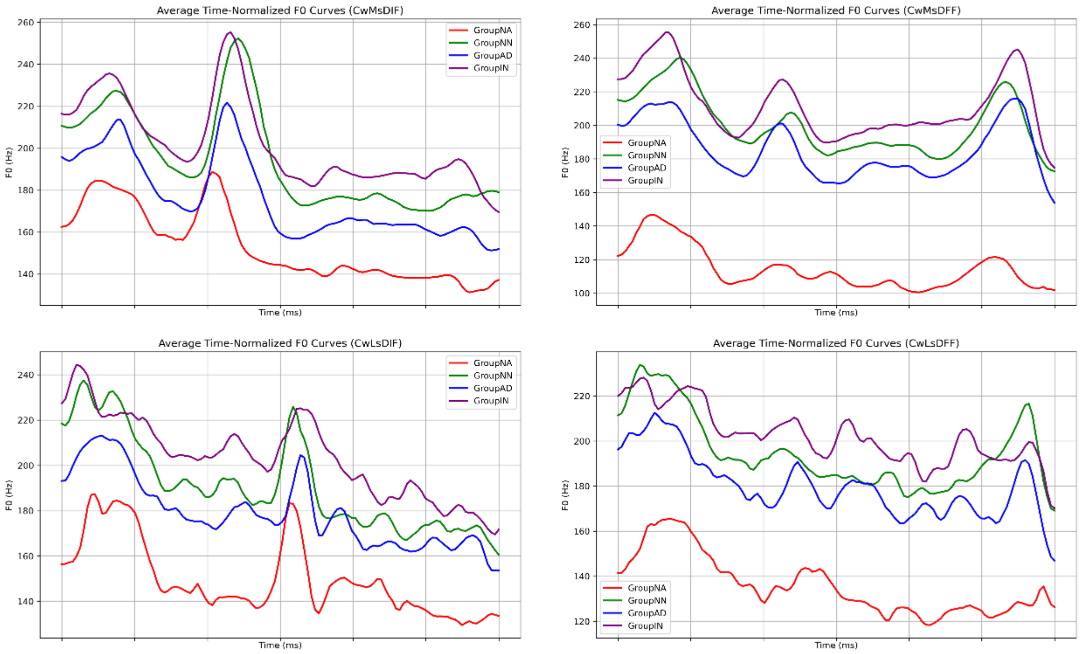
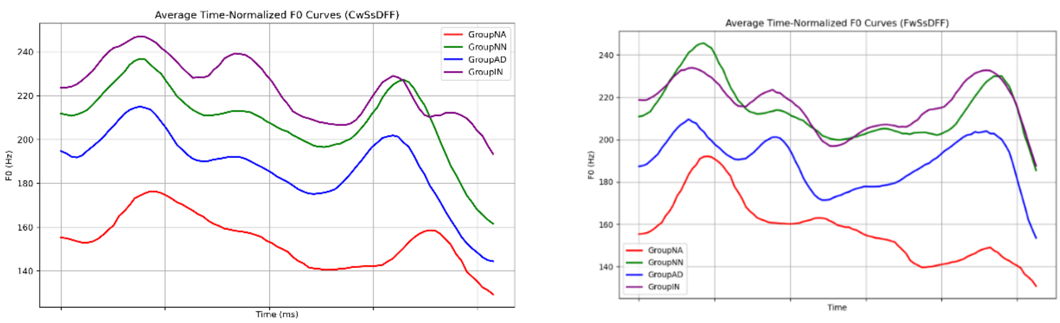


Figure 4.15 F0 Curves for Final Content Word Declaratives

We did not observe a significant trend difference in F0 curves based on Final Word Type. However, it is noteworthy that when the Final Word Type is a Function Word in focus, its pitch variation is considerably smaller than that of Content Words. This pattern is particularly evident in GroupNA, where, in cases where the Final Word Type is a Function Word in focus, the F0 curve in declarative final focus does not exhibit a distinct peak but rather remains relatively smooth. In contrast, learners tend to produce more pronounced peaks (see Figure 4.16 for exemplars).



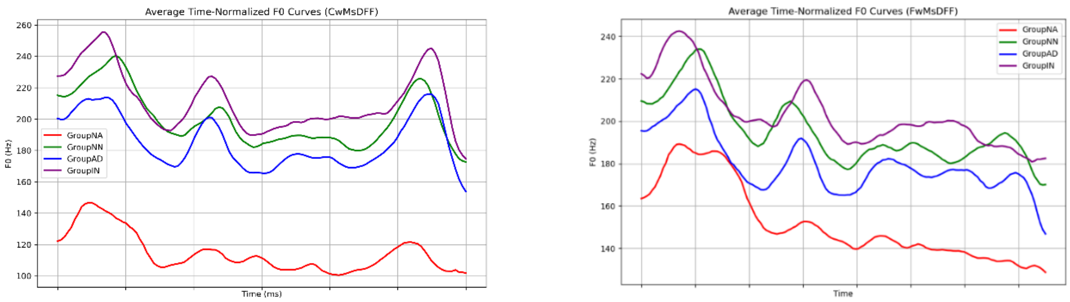


Figure 4.16 Exemplars of F0 Curves for Final Content and Function Words in Declarative Sentences

4.2.2.6 Summary of Focus Realization Across Groups

In the section on focus realization, we have already examined the characteristics of focus realization across different groups (GroupNA, GroupNN, GroupAD, and GroupIN) in relation to Sentence Type, Focus Position, Syllables, and Final Word Type. Starting from GroupNN, we also compared the groups with the previous ones, identifying both similarities and differences. For example, GroupIN is compared with not only GroupNA , but also GroupNN and GroupAD. Thus, in the summary of focus realization, we aim to outline the general patterns, rather than describing specific characteristics in details.

In general, in terms of Groups, GroupNN resembles GroupNA and GroupIN mirrors GroupAD across all other variables. As proficiency level decreases, tonal consistency (will be explained later) declines. As for the other four variables—Sentence Type, Focus Position, Syllables, and Final Word Type—the main factors at play are Sentence Type and Focus Position, with Syllables having a slight effect (which will be presented separately later) and Final Word Type having no significant impact.

When we combine the variables of Groups, Sentence Type, and Focus Position, a general pattern emerges, as shown in the heatmap (see Figure 4.17). Whether viewed from the Z-axis, Y-axis, or X-axis, the colors progressively change along the coordinate axes. Three conclusions on tonal consistency across the three variables can be first presented before explanations:

- **Group NA ≥ Group NN > Group AD ≥ Group IN**
- **Declaratives> Questions**
- **Final focus > Intermediate focus**

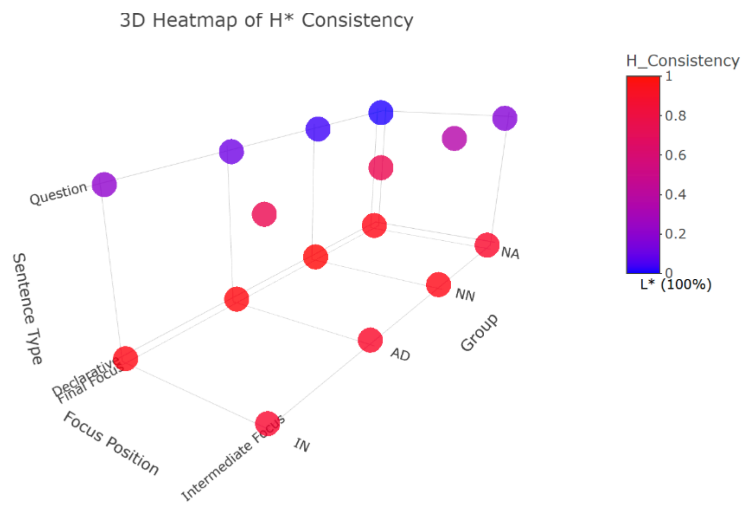


Figure 4.17 3D Heatmap of H* Consistency

We summarized the tonal consistency of different groups in terms of QFF, QIF, DFF, and DIF, specifically focusing on the proportion of H*-related tones. The higher the proportion of H*-related tones (the same as H_Consistency in the heatmap) approaching 100%, the higher the tonal consistency. Conversely, when the proportion of H*-related tones approaches 0%, it indicates that the proportion of L*-related tones is close to 100%, which also suggests high tonal consistency. This can be observed in the color bar on the right side of the heatmap. It's important to note that when the H_Consistency is at 50% on the color bar, it indicates the lowest tonal consistency, meaning that the proportions of H*-related and L*-related tones are equal.

From the heatmap, we can observe that the colors of QFF and QIF change noticeably from blue to red as proficiency level decreases, indicating that an increasing amount of H*-related tones are mixed into the L*-related tones. QIF shows a more pronounced shift compared to QFF. While QFF remains dominated by L*-related tones across all groups, in QIF, as proficiency levels decrease to AD and IN, H*-related tones become more prominent, with the color showing a shift toward H*-related dominance.

Although DFF and DIF also show a trend of decreasing tonal consistency with lower proficiency levels, both consistently outperform Question sentences. Across all groups, DFF and DIF maintain H*-related dominance, with deeper red colors indicating higher consistency. Additionally, DFF outperforms DIF in all groups, and the specific proportions can be seen in the bar charts for each group (see Figure 4.18).

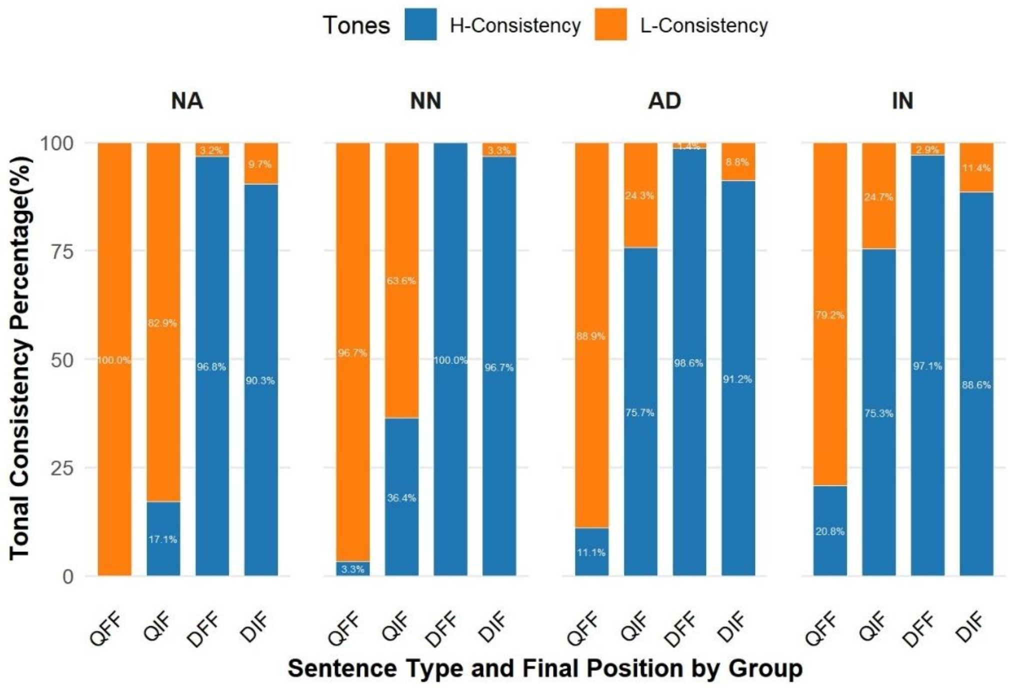


Figure 4.18 Tonal Consistency Across Three Variables

In addition to the general pattern reflected in the heatmap, here are three side observations:

- Overall, learners (**GroupNN/AD/IN**) tend to use a wider-range tones in focus realization compared to **Group NA**. For example, sometimes the dominant tone is L*+H rather than L*, and L+H* rather than H*, especially in **intermediate focus**.
- As **Syllables** increase, it sometimes promotes the use of wider-range tones. For instance, in **GroupIN**, the use of L*+H increases as **Syllables** grow in

DIF. As for **QIF** in **GroupAD**, the proportion of the wider-range L+H* also increases as **Syllables** grow.

- In **exceptional cases**, the variety and number of abnormal tones follow this order: **Group NA ≤ NN < AD ≤ IN.**

It is important to note here that in the Systemic dimension, we have clearly observed inter-group differences in Focus realization. This complements our results in the **overall sentence intonation** (realizational dimension), where we initially used interaction models and found significant group interactions, but no significant main effects of **Groups** alone. Here, the focus realization analysis fills the gap, providing more direct evidence for the group differences. As for whether these inter-group differences in the systemic dimension align with the three dependent variables—**pitch change amount at the syllable level (st), pitch range at the utterance level (st), and the number of F0 peaks and valleys at the utterance level**—we will address this in the discussion section.

To conclude this section in a metaphorical way, tonal consistency in focus realization can be likened to the mixing paint on a canvas. **GroupNA** is like a strong, pure color—bold and well-defined. As we move from **GroupNN** to **GroupIN**, the colors start blending more, with **GroupIN** being the most mixed, where the original tones become harder to distinguish. **Sentence Type** works like the brushstroke: **Declarative** is a firm, precise stroke, keeping the tonal consistency sharp, while **Question** is a lighter, more subtle stroke, causing the tone to blend and lose some clarity. **Focus Position** is like the placement of the strokes on the canvas. **Final Focus** is like a clear, bold stroke at the end of the sentence, preserving the tonal consistency, while **intermediate focus** is softer and more dispersed, leading to a less defined tonal effect.

4.2.3 Boundary Tone

In the Boundary Tone section, we analyzed the pitch patterns related to different Focus Position, Sentence Type, Syllables, and Final Word Type, similar to the approach taken in the Focus realization section. However, unlike focus realization, we found that only **Sentence Type** had a clear effect, while no significant effects were observed from the other independent variables. The boundary tone data are also qualitative and based on ToBI annotations, which provide insight into the phonological encoding of boundary tones.

We included control sentences in this section, ensuring that all 36 experimental sentences were examined. Boundary tones are categorized into four types: L-L%, H-H%, L-H%, and H-L% (Refer to Section 2.2.2 for tone visualizations and explanations). In Declarative sentences, native English speakers typically use L-L%, while in Yes-No Questions, H-H% is preferred. Specifically, we measured the proportions of each pitch pattern (L-L%, H-H%, L-H%, H-L%) for each participant across all 36 sentences.

If we consider L-H% and H-L% as abnormalities to Declarative and Question sentences, all the Groups have great performance, with **GroupNA** > **GroupNN≈GroupAD** > **GroupIN** (see Table 4.26). This order is similar to that in focus realization. Additionally, in terms of abnormalities (H-L% and L-H%), we did not find out clear relationship between Sentence Type and abnormalities, that is to say, H-L% can be assigned not only to Questions but also Declaratives, the same is true for L-H%. However, one observation worth noting is that DFF (declarative final focus) is the least likely to exhibit abnormalities across all sentences. Specifically, there is only one single abnormality of L-H% was detected in the CwMsDFF for the GroupIN. Across all **Groups** and **Syllables**, **DFF** consistently adhered to the L-L% pattern. The good performance of **DFF** in boundary tone is consistent with that in focus realization.

Table 4.26 Pitch Pattern in Boundary Tone

Boundary Tone / Groups	H-H%	L-L%	H-L%	L-H%
NA (193 Samples)	51.30%	47.15%	0.52%	1.04%
	98.45%		1.55%	

Table 4.26 (Continued)

Groups \ Boundary Tone	H-H%	L-L%	H-L%	L-H%
NN (366 Samples)	47.27%	46.72%	3.01%	3.01%
	93.99%		6.01%	
AD (408 Samples)	47.30%	47.79%	2.21%	2.70%
	95.10%		4.90%	
IN (418 Samples)	40.91%	46.65%	1.20%	11.24%
	87.56%		12.44%	

4.3 Summary of Results

In the Results section, we first presented our pre-experiment, which aim to categorize participants into **Groups** based on their Foreign Accent and Intonation Scores. A strong positive correlation was found between accent and Intonation Scores across groups, but no significant differences were observed within each group between accent and intonation. As for **overall sentence intonation**, it was first visualized and then analyzed through three main dependent variables in interaction model: pitch change amount at the syllable level (st), pitch range at the utterance level (st), and the number of F0 peaks and valleys at the utterance level. Significant effects were found for independent variables such as **Sentence Type** (Declaratives vs Questions), **Focus Position** (Broad vs Final vs Intermediate), and **Syllables**. **Groups**, while not showing direct effects, were found to interact significantly with **Sentence Type** and **Syllables**. However, the **Group** effect was testified in systemic dimensions, especially in **focus realization**, where three variables—**Groups, Sentence Type** and **Focus Position** all played a role on tonal consistency**,** which can be likened to the mixing paint on a canvas**.** In terms of **boundary tone**, Sentence Type was the only significant factor affecting boundary tones, with **Declarative** sentences predominantly showing L-L%

boundary tones and **Questions** favoring H-H%. Abnormalities in boundary tone patterns were observed more frequently in lower-proficiency groups, particularly in GroupIN.

Chapter 5 Discussion

This section discusses the research findings based on the analysis of intonational patterns in both native English speakers and Mandarin learners of English, as presented in Chapter 4. By comparing the results with previous studies, we will highlight both similarities and differences, focusing on the influence of second language transfer, particularly through the lens of L2 Intonation Learning theory (LILt). Section 5.2 serves as the main body of the Discussion, organized around the three research questions. Prior to delving into the detailed analysis of these questions, Section 5.1 will provide a brief discussion of the results from the pre-experiment.

5.1 The Relationship of FA Score and Intonation Score

In the pre-experiment, we found a strong linear correlation between the Foreign Accent (FA) score and the Intonation Score (Spearman's rank correlation = 0.96, $p < 2.2e-16$), with no significant differences observed between the FA and Intonation Scores across all learner groups (Group NN, Group AD, Group IN). This result recalls previous research on the role of prosody in the perception of FA. Studies such as Munro (1995) and Trofimovich and Baker (2007), which used low-pass filtered methods, demonstrated that prosody alone is sufficient for identifying Foreign Accents. However, as mentioned earlier, intonation is a subset of prosody. In Munro's study, intonation contours were one of the prosodic cues for FA identification, while in Trofimovich and Baker's (2007) study, prosody was more related to duration. Both studies highlight the importance of prosody under the broader concept in FA perception.

If our native judges understood the request of the experimenter correctly, at least the experimenter ensured they did, such that the Intonation Scores were based solely on the melodic component of prosody (distinct from other prosodic cues like duration), our findings are partially consistent with those of Munro (1995) and Trofimovich and Baker (2007). Specifically, we agree that prosody can independently play a role in FA perception. However, there remains disagreement in the literature about the relative contribution of different prosodic cues to FA. Many studies (e.g., Trofimovich and Baker 2007; Gut 2009) argue that temporal features such as speech rate and pause duration are more crucial, while some, like Jilka (2000), suggest that intonation is the most important prosodic cue. Our findings align more closely with Jilka's (2000) conclusions, adding that intonation alone can serve as a significant cue for FA perception.

One limitation of our study, however, is that it did not explore other prosodic cues nor the intricate relationship between segments and prosody in FA perception, as discussed in studies like Ulbrich and Mennen (2016). Nevertheless, it is important to note that, in terms of intonation alone, our findings demonstrate that it can serve as an independent cue to FA perception. There is a strong alignment between a speaker's overall accent and their intonation patterns, suggesting that deviations in intonation are unlikely to occur independently from a speaker's overall accent. We support the idea that segmental features and other prosodic cues are balanced in correlation with the FA score, similar to that in Boula de Mareüil, Marotta and Adda-Decker (2004).

If we model the FA score as follows:

FA score ~ Intonation Score + Other prosodic cues + Segmental score,

then if the FA score changes in sync with the Intonation Score, it follows that other scores (such as those for prosodic cues and segments) should also change proportionally. This suggests that improvements or deviations in intonation are accompanied by similar changes in these other factors, reflecting a cohesive relationship in the perception of FA. However, since we did not independently manipulate and assess intonation, further data is needed to testify model.

5.2 Main Experiment

As the main body of Discussion, in this section we will first interpret results in overall sentence intonation, which consists of three types of measurements in realizational dimension, namely, pitch change amount on the syllable level (st), pitch range (st), and the number of F0 peaks and valleys on the utterance level, and then answer the second and third research question about focus realization and boundary tone.

5.2.1 Tonal Characteristics on L2 Sentence Intonation

5.2.1.1 Pitch Change Amount on the Syllable Level (st) and the Number of F0 Peaks and Valleys on the Utterance Level

We grouped pitch change amount on the syllable level (st) and the number of F0 peaks and valleys on the utterance level in the same section because they both contribute to determining whether L2 English is influenced by tonal language. Specifically, the measurements can reveal whether learners exhibit more peaks and valleys and more F0 movements, compared to native English speakers, as seen in line 3, instead of line 1 or line 2 (see Figure 5.1). To provide a thorough analysis, we first probe deeper into the results for each measurement separately before discussing them together.

Figure 5.1 Assumed F0 Curves Under Different Conditions

5.2.1.1.1 Pitch Change Amount on the Syllable Level (st)

To address the question of which variables influence pitch change amount at the syllable level, we modeled the data, which revealed that **Sentence Type**, **Focus Position**, and **Syllables** significantly affected pitch change amount. Prior to modeling, we examined the influence of five variables on pitch change: Group, Sentence Type, Focus Position, Syllables and Final Word Type. Our preliminary analysis showed that **Group**, **Sentence Type**, **Focus Position**, and **Syllables** played a significant role (see table 5.1 for summary of preliminary analysis).

Table 5.1 Summary of Preliminary Analysis on Pitch Change Amount

Variables	Pitch Change Amount
Group	IN & AD > NA & NN
Sentence Type	Questions > Declarative Sentences
Focus Position	Intermediate focus > Final focus > Broad focus
Syllables	7 syllables > 21 syllables > 14 syllables

Then through modeling, we identified that **Sentence Type**, **Focus Position**, and **Syllables** were the significant variables.

In terms of **Sentence Type**, the increased pitch change in questions compared to statements is an intuitive finding. Yuan (2006) highlighted that, in Mandarin, statement intonation is considered the default or unmarked type, whereas question intonation is a marked intonation. Yuan identified three mechanisms for question intonation in Mandarin: a generally higher phrase curve, stronger final tones, and a tone-dependent mechanism that flattens falling tones, compared to statements. These mechanisms can naturally result in a higher pitch change for questions than for statements in Mandarin, as illustrated in the Figure 5.2 by Yuan (2006)

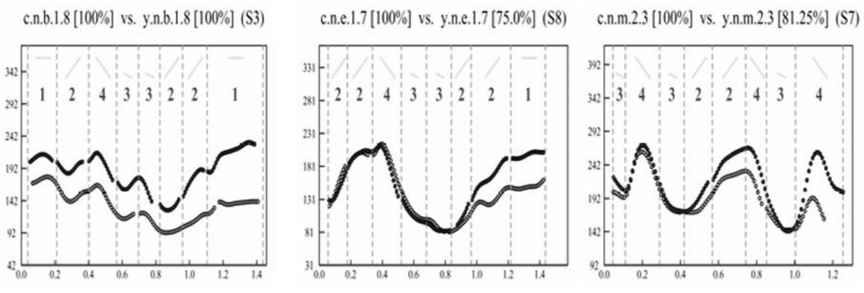


Figure 5.2 Question Curves (Dark Circles) and the Statement Curves (Open Circles) in Mandarin

Moreover, Liu and Xu (2007) found in their study of English question intonation that the pitch difference between declarative sentences and questions starts from the stressed syllable of the first content word, whether focused or not. Questions rise to quite high while statements have high or falling pitch target, as illustrated in the Figure 5.3 by Liu and Xu (2007).

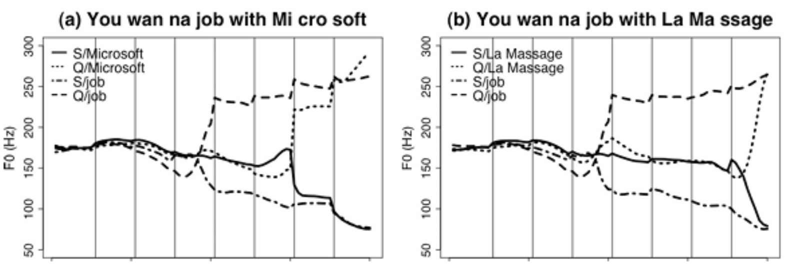


Figure 5.3 Question Curves and the Statement Curves in English

Then it is clear that, whether in Mandarin or English, question intonation involves more F0 variations. Therefore, it is inevitable that English, including both L2 English and native English, will show a higher pitch change amount in questions than in statements, regardless of whether L2 English is more influenced by Mandarin or English.

In terms of **Focus Position**, it is expected that the pitch change amount for both types of narrow focus is higher than for broad focus. This aligns with Xu (2005), who stated that narrow focus is realized by expanding the pitch range of the focused syllables while suppressing the pitch range of post-focus syllables, with pre-focus syllables remaining mostly unchanged. The widening of the pitch range around the focus word naturally leads to an increase in pitch change amount. Notably, the pitch change amount for intermediate focus is greater than for final focus, both in the pre-modeling data and in the model. However, this pertains to focus realization, so I will address this in more detail in the upcoming section on focus realization.

In terms of **Syllables**, the pitch change amount on the syllable level for 14 and 21 syllables is smaller than for 7 syllables, which is consistent across both the pre-modelling and the modeling results. However, this leads us to take a step further in our analysis. The pitch change amount for 21 syllables is closer to that of 7 syllables (not significant in the modeling), resulting in a pitch change pattern of 7 > 21 > 14. This is somewhat unexpected. If sentence length were causing negative transfer from L1 to L2 English, we would expect the pattern to be 21 > 14 > 7. However, the data reveals a more complex relationship. We propose the following reasons:

For the unexpected result (7 > 21 and 14): The smaller pitch change amount for 14 and 21 syllables compared to 7 syllables can be explained by the grammatical structure of the shorter sentence (e.g., "Manny murmured melodies/to herself"). In this case, nearly every word in the 7-syllable sentence carries significant grammatical weight. For instance, "Manny" is the subject (DP), "murmured" is the main verb (V), and "herself" functions as a reflexive pronoun (DP). In shorter sentences, each stressed syllable tends to receive a pitch accent, as these words play a key syntactic role in the sentence structure. In contrast, longer sentences, especially those with more syllables, an intonational phrase can be divided into multiple intermediate phrases, with unstressed words and secondary stress occurring more frequently. In these longer sentences, each intermediate phrase is less likely to carry pitch accents across all syllables, leading to a reduction in the overall pitch change. Thus, the higher number of pitch-accented syllables in the 7-syllable sentence helps explain its greater pitch change amount at the syllable level.

As for the expected result (21 > 14): This pattern aligns with the expectation that longer sentences may be influenced by L1 transfer, resulting in more F0 variation. The longer sentence (21 syllables) likely causes greater influence from Mandarin, a tonal language, leading to more pitch variations. This phenomenon is consistent with the notion that L2 English learners, particularly those with Mandarin as their L1, might transfer tonal intonational patterns into their English speech when sentence length increase

To address is whether **Groups** affect pitch change amount at the syllable level, and if so, how, in our modeling, we did not find significant main effects for Groups, suggesting there were no significant differences between any of the learner groups and the baseline group (GroupNA). However, significant interactions between Groups and Sentence Type, as well as Groups and Syllables, were observed. Notably, while the main effects of Group and Syllables 21 were not significant, the interaction between Groups and Syllables 21 was significant for all three learner groups (GroupIN, GroupAD, and GroupNN). This suggests that the differences between groups in pitch change amount become more apparent when sentence length (specifically, 21 syllables) is taken into account, confirming that groups affect pitch change amount at the syllable level.

From the pre-modeling data, we observed that pitch change amount increased as proficiency decreased, which aligns with previous studies showing that less proficient learners tend to exhibit more pitch variations. However, this finding needs further exploration to statistically confirm specific inter-group differences. No significant effects were found for other variables, such as Gender and Final Word Type.

The finding that L2 English shows more pitch change than native English speakers is consistent with previous research, such as Ding et al. (2016), and supports studies on Mandarin intonation, which report more F0 movements in Mandarin than in English (e.g., Yuan and Liberman 2014; Eady 1982), suggesting negative transfer from Mandarin. However, some studies argue that L2 English speakers exhibit less pitch change than native speakers, as proposed by Yuan (2018). In English of Mandarin speakers, intonation is always flat (Juffs 1990). One explanation for this is that L2 learners may be "too cautious" in their pitch variation (Zimmerer et al. 2014). As Yuan (2018) suggests, L2 speakers often focus more on segmental accuracy, which can result in a more restrained pitch range and less variation, as they may lack the confidence to realize a pitch range that is more native-like.

We observed this characteristic in a very small number of intermediate learners (GroupIN). For example, IN10 (female), even in longer sentences, exhibited very little F0 variation, such as in the FwLsDC (function final word, long sentence and declarative control) sentence. The F0 curve for IN10 was relatively flat, as shown below (see Figure 5.4). However, when we compare this to the same sentence produced by a native female speaker, we observed more F0 variation, as illustrated in the figure (see Figure 5.5).

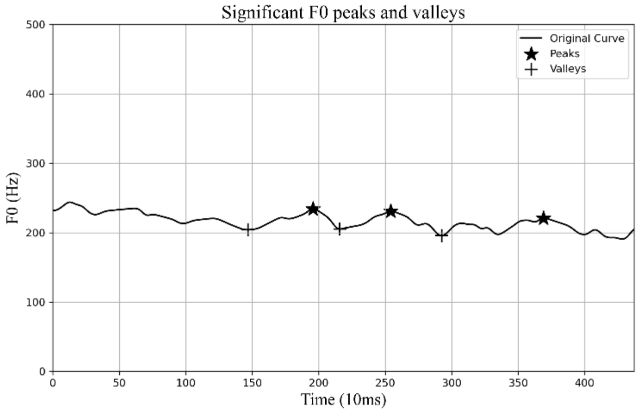


Figure 5.4 F0 Curve in FwLsDC by IN10

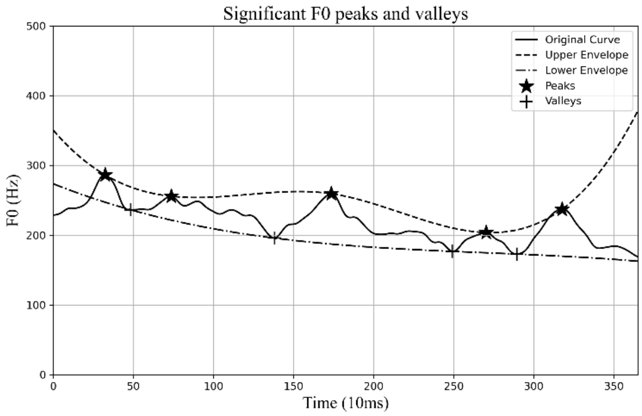


Figure 5.5 F0 Curve in FwLsDC by NA5

Therefore, while a minority of learners, particularly those with lower proficiency, seem to be influenced by non-linguistic factors, such as psychological factors (e.g., confidence), leading to less pitch variation, the overall trend shows that L2 learners tend to exhibit more F0 movements due to the influence of Mandarin.

5.2.1.1.2 The Number of F0 Peaks and Valleys on the Utterance Level

To address the question of which variables influence the number of F0 peaks and valleys on the utterance level, we modeled the data, which revealed that only **Syllables** significantly affected pitch change amount. Before modeling, we examined the influence of five variables on the number of F0 peaks and valleys: Group, Sentence Type, Focus Position, Syllables and Final Word Type. Our preliminary analysis showed that **Group** and **Syllables** played a significant role (see Table 5.2 for summary of preliminary analysis).

Table 5.2 Summary of Preliminary Analysis on the Number of F0 Peaks and Valleys

Variables	Pitch Change Amount
Group	IN & AD > NN > NA
Syllables	21 syllables > 14 syllables > 7 syllables

However, through modeling, we identified that only **Syllables** was a significant variable influencing the number of F0 peaks and valleys.

In terms of **Syllables**, longer sentences naturally caused more F0 fluctuations than shorter sentences, as longer sentences tend to have more intermediate phrases and pitch accents, both of which contribute to the creation of peaks and valleys in the F0 curves. Other factors, such as Sentence Type and Focus Position, did not result in significant differences, which aligns with our expectation. While Sentence Type and Focus Position influence pitch change amount, they do not significantly increase or decrease the number of pitch accents or intermediate phrases, which are the primary contributors to the F0 peaks and valleys.

To address is whether **Groups** affect the number of F0 peaks and valleys on the utterance level, and if so, how, in our modeling, we did not reveal significant main effects for Groups, indicating no significant differences between any of the learner groups and the baseline group (GroupNA). However, we did observe significant interactions between Groups and Syllables. This may suggest that Groups affect the number of F0 peaks and valleys on the utterance level. That said, further investigation is needed to conduct a more in-depth statistical analysis of inter-group effects. Regarding the "how" aspect, our preliminary analysis indicates that IN and AD groups show more F0 peaks and valleys compared to NN and NA groups. However, a more comprehensive statistical test is required to fully address this question.

5.2.1.2.3 Summary and Discussion of Pitch Change Amount and Number of

F0 Peaks and Valleys

In summary, for **pitch change amount on the syllable level (st)**, **Sentence Type**, **Focus Position**, and **Syllables** all played significant roles in the model. While **Groups** did not show significant main effects, the significant interactions between **Groups** and multiple factors suggest a noteworthy influence, which is further supported by the significant differences observed in the preliminary box plots. The results generally show that as proficiency decreases, the pitch change amount increases, consistent with most previous studies. However, some learners with lower proficiency, particularly in longer sentences, displayed reduced pitch variation, suggesting the influence of non-linguistic factors in lower-level learners.

For the **number of F0 peaks and valleys**, only **Syllables** was a significant factor in the model. Both the interaction effects and the preliminary analysis suggest that **Group** likely has a significant influence, showing the same trend of increasing pitch change amount as proficiency decreases. However, further in-depth statistical analysis is needed to confirm this.

Overall, we can infer that the majority of English learners are primarily influenced by Mandarin's tonal characteristics. These learners tend to show more pitch change amount and more F0 peaks and valleys than native English speakers, reflecting a negative transfer from Mandarin, a tone language, as seen in line 3. However, a small subset of learners, particularly some intermediate speakers, may exhibit a flatter intonation pattern (Line 1). Native speakers, when considering overall sentence intonation, are likely to fall somewhere in between these two patterns, as seen in Line 2. A hypothetical illustration of these patterns, based on actual data, is shown in Figure 5.6.

Figure 5.6 Hypothesized F0 Pitch Patterns

5.2.1.2 Pitch Range on the Utterance Level (st)

To address the question of which variables influence pitch range on the utterance level (st), we modeled the data, which revealed that **Sentence Type**, **Focus Position**, and **Syllables21** significantly affected pitch range. Prior to modeling, we examined the influence of five variables on pitch change: Group, Sentence Type, Focus Position, Syllables and Final Word Type. Our preliminary analysis showed that **Sentence Type**, **Focus Position**, and **Syllables** played a significant role (see Table 5.3 for summary of preliminary analysis), the same as in modeling.

Table 5.3 Summary of Preliminary Analysis on Pitch Range

Variables	Pitch Range
Sentence Type	Questions > Declarative Sentences
Focus Position	Intermediate focus > Final focus > Broad focus
Syllables	21 syllables > 7 & 14 syllables

Through modeling, we identified that **Sentence Type**, **Focus Position**, and **Syllables21** were the significant variables.

Interestingly but also Intuitively, the modeling results are roughly the same between pitch change amount and pitch range, with both showing significant effects from Sentence Type, Focus Position, and Syllables, especially Sentence Type and Focus Position. It seems that pitch range is closely aligned with pitch change amount, particularly in relation to Sentence Type and Focus Position. In fact, this alignment can be explained by the inherent characteristics of English intonation.

This observation brings us back to AM theory (Pierrehumbert 1980), where Sentence Type can be simplified to represent boundary tones, and Focus Position can be simplified to pitch accents. The overall F0 curve in an English sentence is influenced by both boundary tones and pitch accents, which collectively shape the intonational pattern.

To address whether **Groups** affect the number of F0 peaks and valleys on the utterance level, and if so, how, our modeling results did not reveal significant main effects for Groups, indicating no significant differences between any of the learner groups and the baseline group (GroupNA). Similarly, our preliminary analysis through visual box plots did not show any noticeable inter-group significance or patterns, which contrasts with the findings for pitch change amount and the number of F0 peaks and valleys, where clearer trends were observed in the pre-modeling data.

At first glance, it might seem logical to conclude that Groups do not have a significant effect on pitch range, given the lack of main effects in the modeling and the absence of group differences in the preliminary visual analysis. However, there might be some confusion here: while pitch change amount and number of peaks and valleys showed significant interactions between Groups and other factors in the modeling, we did suggest a significant effect of Groups.

The interpretation can be as follows: although Sentence Type, Focus Position, and Syllables were significant individually, the only significant interaction between Groups and Sentence Type in the pitch range data suggests that Groups may not have a direct effect. Rather, the impact of Sentence Type on pitch range is so strong that it drives the significant interaction. However, in interactions between Focus Position, Syllables, and Groups, we did not find significant results, which further supports the conclusion that Groups do not significantly influence pitch range.

These findings are somewhat inconsistent with the views of Chen (1972), Yuan (2014), and Yuan (2018), who suggest that L2 English should show a wider pitch range due to Mandarin transfer, or a narrower pitch range due to the generally "flat" nature of L2 English. However, our results indicate that L2 English does not show significant differences in pitch range compared to native English. This aligns with Keating and Kuo's view that there are no inherent differences in pitch range between Mandarin and English. Additionally, our findings are consistent with Ding et al. (2016)'s research on

L2 English and native English, as well as with Ding, Jokisch, and Hoffman (2012)'s study on L2 German. These studies suggest that the speech corpus or task, as well as acoustic measures, are more significant factors influencing pitch range than language itself.

In a nutshell, in overall sentence intonation, L2 English shows the influence of Mandarin, a tonal language, with more F0 movements and F0 fluctuations, while pitch range does not differ significantly from native speakers. These findings align with our predictions. Based on the modeling, Focus Position, Sentence Type, and Syllables all significantly affect pitch range and pitch change amount, while Syllables significantly influence the number of F0 peaks and valleys. These results largely reflect English intonation patterns, with some specific details to be revisited in the previous sections for further clarification.

5.2.2 Focus Realization is a Function of Groups, Sentence Types and Focus Position

To answer our second research question, in all, GroupNN is more similar to GroupNA, while GroupIN closely resembles GroupAD across all other variables. As the proficiency level decreases, there is a noticeable reduction in tonal consistency. Among the four variables—Sentence Type, Focus Position, Syllables, and Final Word Type—the influencing factors are **Sentence Type** and **Focus Position**, with **Syllables** showing a minor effect, and Final Word Type having no significant impact.

We created an intuitive 3D heatmap to visualize the main research findings, and observed that the tonal consistency of the three key variables—**Group, Sentence Type**, and **Focus Position**—tends to decrease along the x, y, and z axes (see Figure 5.7). More detailed results are presented in Figure 5.8. Our prediction is proved and will be discussed later.

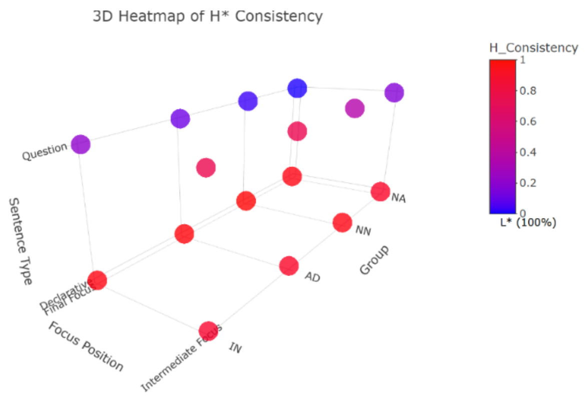


Figure 5.7 3D Heatmap of H* Consistency

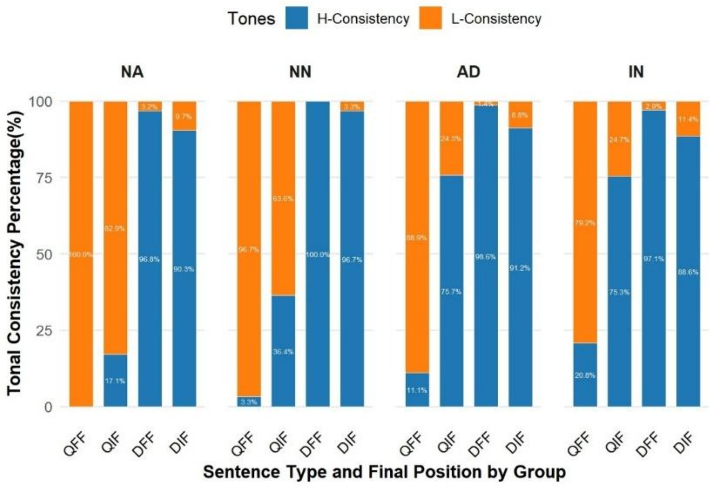


Figure 5.8 Tonal Consistency Across Three Variables

In terms of **Groups**, we observed significant differences between them. For GroupNA and GroupNN, the focus realization in question sentences, whether it was final focus or intermediate focus, was predominantly L*-related. However, for GroupAD and GroupIN, there was a noticeable increase in the use of H*-related, with H* becoming the dominant pitch accent, especially in intermediate focus. This supports our hypothesis that, as proficiency decreases, L2 English becomes more subject to negative transfer from Mandarin intonation, as Mandarin does not have L* and only uses H tones. These results are consistent with previous studies (e.g., McGory 1997; Shack 2000; Liu and Chen 2016). However, it is worth noting that while there were differences between GroupNA and GroupNN, the variation in Sentence Types and Focus Positions was minimal. This suggests that, at the systemic/phonological level, Near-native speakers have reached a native-like level and have overcome the negative transfer effect. In contrast, GroupAD and GroupIN still exhibit significant influence from negative transfer at the systemic level. This has implications for language teaching: only at super-high proficiency levels, learners have a better grasp of the phonological aspects, while advanced speakers do not show much difference from intermediate speakers. Therefore, for most Chinese English learners, teaching should focus on various pitch accents rather than just H*.

Additionally, we found that learners in GroupNN, GroupAD, and GroupIN tend to use a wider range of tones in focus realization compared to GroupNA. For example, dominant tones used were L+H* instead of L*, and L+H* instead of H*. This is somewhat inconsistent with findings from Hu et al. (2012) and Liu et al. (2021), who suggested that the focused word should have less pitch variation (Hu et al. 2012) or lower peaks and slopes (Liu et al. 2021). Although we did not directly extract data for the focused word in the realizational dimension, the wider range of pitch accents used by learners compared to GroupNA leads us to conclude that learners exhibit greater pitch variation in focus realization rather than a flatter pitch, which is more often seen in a small subset of intermediate speakers. However, our conclusion requires further data. Future studies can specifically address the relational dimension of focus realization. Previous research rarely differentiated between proficiency levels in detail and had small sample sizes, such as Hu et al.(2012), which only had 16 EFL learners and 2 native speakers. Thus, including more learner groups in future research is recommended.

In terms of **Sentence Type**, tonal consistency is better in declarative sentences than in yes-no questions, across all groups and Focus Positions. Two typical tonal elements are H*-related in declaratives and L*-related in questions. It is expected that all groups rarely use L*-related tones in declarative sentences, but lower proficiency learners, in particular, tend to use H*-related in questions. This could be due to two reasons: first, yes-no questions are marked compared to declarative sentences, and second, L*-related tones are less frequent than H*-related tones in general. We also observed that the use of H*-related tones in questions tends to occur more often in intermediate focus rather than in final focus, which will be discussed in terms of Focus realization.

In terms of **Focus Position**, final focus consistently performs better than intermediate focus, whether in question sentences or declarative sentences. This is consistent with Liu and Chen (2016), who explained this by the influence of tonal crowding, noting that learner performance resembles that of native speakers when an H% boundary tone is present. We found that not only learners but also native speakers exhibit better tonal consistency in final focus compared to intermediate focus. Additionally, syllables also have an effect. Short sentences with intermediate focus show higher tonal consistency than longer sentences, and longer sentences are more likely to use wider-range tones in intermediate focus. This difference may be due to the fact that shorter sentences have fewer intermediate phrases, while longer sentences contain more, with intermediate phrases often being segmented at the end of the focused word. Consequently, longer sentences with intermediate focus are more affected by phrase accents and are farther from the boundary tone, thus reducing the impact of the boundary tone, while shorter sentences experience the opposite effect.

Therefore, we believe that boundary tone plays a positive role in focus realization. When the boundary tone and pitch accents are close, tonal consistency improves. We also argue that the two typical tonal elements—H*-related in declaratives and L*-related in questions—are generally well mastered by learners. However, there are many interfering factors, such as sentence length and learning experience, which could result in negative transfer. Nevertheless, the boundary tone acts like an "interference eliminator," ensuring that within its influence, declarative sentences use H*-related tones and questions use L*-related tones.

Additionally, in **exceptional cases**, the variety and frequency of abnormal tones follow this pattern: Group NA ≤ NN < AD ≤ IN, which is consistent with the overall trend across groups. In Group AD and Group IN, we observed instances of two pitch accents on a word and incorrect placement of pitch accent. We may interpret these issues as a result of insufficient proficiency, where pitch accents were used to implement boundary tones. In these cases, the boundary tone becomes an interfering factor. The presence of two pitch accents on a word and incorrect stress placement indicates that the learner has not fully mastered the correct pitch accent placement for the sentence or word.

Finally, in terms of **F0 curves** which served as a compensation measure for realizational dimension, our results not consistent with Hu et al. (2012) and Liu et al. (2021). For the focused words, we found that their F0 curves have more pitch variations in learner groups, rather than native groups. We still consider this to be an influence of Mandarin tonal characteristics.

In all, focus realization in L2 English is a function of Groups, Sentence Type and Focus Position. Final focus may have an intricate relationship with boundary tone. We present a formula as follows:

$$\begin{aligned} \text{L2 Focus Realization}_{\text{final}} & \\ & \sim \left(\text{Boundary Tone}_{\text{pos}} \cdot \text{Mastered}\right) \\ & + \left(\text{Boundary Tone}_{\text{neg}} \cdot (1 - \text{Mastered})\right) \end{aligned}$$

This formula represents how L2 focus realization is influenced by the mastery of pitch accents and the effects of boundary tones. If the pitch accent is mastered (Mastered = 1), the boundary tone has a positive effect; if not mastered (Mastered = 0), the boundary tone has a negative effect, affecting the focus realization in L2 English accordingly.

5.2.3 Boundary Tone is a Function of Groups and Final Focus

To answer our third research question, all groups had great performance, with **GroupNA > GroupNN ≈ GroupAD > GroupIN**. In English intonation, H-H% and L-L% are typical in yes-no questions and declarative sentences, respectively. **Sentence Type** was the only variable that had a clear effect on boundary tones. In GroupNA, over 98% of sentences correctly selected the boundary tone, while the least accurate group, GroupIN, still had over 87% accuracy. Our prediction has also been proved.

The strong performance in boundary tone in L2 English can be related to the similarities between Mandarin and English boundary tones. Although there is ongoing debate about whether the boundary tone exists in Mandarin intonation (e.g., Rumjancev 1972; Ho 1976; Shen X.N. 1990; Lin 2004), it is undeniable that the distinction between Mandarin statements and questions primarily relies on the sentence-final syllables (e.g., Lin 2004; Yuan 2006). Mandarin and English both use high or low boundary tones to distinguish between questions (H%) and statements (L%). This explains why L2 English learners show strong consistency and performance in boundary tone, similar to native English speakers.

However, Mandarin and English also differ in terms of boundary tones. In Mandarin, the boundary tone in yes-no questions does not alter the tone contour (Chao 1979). Whether the tone is yinping, yangping, shangsheng, or qusheng, the distinction between question and statement boundary tones comes from the starting and/or ending height of the boundary tone F0 curve. In contrast, the distinction in English between declarative and question boundary tones is determined by the rise or fall of the F0 curve. This difference explains why, as learners' proficiency decreases, the probability of boundary tone deviations beyond H-H% and L-L% increases. Our results show that lower proficiency learners are more prone to negative transfer from Mandarin boundary tones. This transfer occurs because Mandarin boundary tones require the tone of the final word in a sentence to match the expected tonal pattern. For instance, when the conflict of a tone 2 (rising) in a declarative sentence, or a tone 4 (falling) in a question emerges, this discrepancy may lead to a flattening effect on the boundary tone. The complexity of Mandarin boundary tone realization may thus be transferred into L2 English, causing learners to produce more flattened tones. Additionally, in exceptional cases in L2 English, we observed that learners not only used a flatter boundary tone (such as L-H% instead of H-H%) but also occasionally produced a boundary tone that reversed the expected direction, such as H-L% in yes-no questions. This tendency became more pronounced as proficiency decreased, showcasing the complexity of boundary tone realization in L2 English and reflecting negative transfer from Mandarin, a tonal language.

One notable observation is that deviations in boundary tone were less frequent in final focus contexts, especially in declarative sentences with final focus, similar to Tang and Zhu (2022). We interpret this phenomenon from a new perspective. As mentioned earlier, there are both similarities and differences between Mandarin and English intonation boundary tones, which have led to both positive and negative effects on L2 English. The negative effect arises from the tonal characteristics of Mandarin, which brings complexity in the realization of boundary tones.

However, we propose that pitch accents (focus) on the final word of a sentence can effectively neutralize the negative transfer from Mandarin tones. By placing a pitch accent on the final word, the role of Mandarin tones in shaping the boundary tone is replaced by the pitch accent, allowing L2 English speakers to focus on the prosodic feature of the pitch accent rather than the tonal properties of the final word in Mandarin. This means that L2 English can maintain only the positive transfer of Mandarin and correctly realize the boundary tone without the interference from Mandarin tonal patterns.

To conclude, we present a formula as follows:

$$\begin{aligned}
&\text{L2 English Boundary Tone} \\
&\qquad = \left(\text{Mandarin Tones}_{\text{neg}}\right) + \left(\text{Mandarin Boundary Tones}_{\text{pos}}\right) \\
&\qquad - \left(\text{Mandarin Tones}_{\text{neg}} \cdot \text{Pitch Accent}_{\text{final}}\right)
\end{aligned}$$

The negative transfer from Mandarin tones is subtracted if a pitch accent is applied to the final word. Pitch Accent final takes the value 1 if there is a pitch accent on the final word, and 0 if there is no pitch accent on the final word.

To conclude this section, we have discussed the findings based on the three research questions. Overall, our predictions have been largely confirmed, but further data is needed to more robustly support the differences between the groups in terms of realizational measurements. In terms of focus realization and boundary tones, not only did we validate our predictions, but we also uncovered several deeper insights. Additionally, we have synthesized the intricate relationship between boundary tones and final focus, shedding light on the underlying mechanisms and interactions at play.

Chapter 6 Conclusion

In this chapter, we will first summary the major findings of this study briefly and then give implications for intonation learning, intonation teaching and L2 Intonation Learning theory (LILt). The study is not without limitations, which will be pointed out before giving suggestions for future research.

6.1 Research Findings

The study provides valuable insights into how Mandarin learners of English acquire English intonation, particularly with respect to the influence of their tonal first language. The research aims to explore the factors that affect overall sentence intonation, focus realization, and boundary tone in L2 English and native English, and how these patterns vary across different proficiency groups. By applying the L2 Intonation Learning theory (LILt), this study investigates the systemic and realizational dimensions of intonation in Mandarin and English, and how Foreign Accent (FA) levels affect L2 learners in English intonation patterns.

In terms of overall sentence intonation, the study examined three key measurements: pitch change amount on the syllable level (st), pitch range on the utterance level (st), and the number of F0 peaks and valleys on the utterance level. Learners at lower proficiency levels showed greater pitch variation, which manifested in larger pitch change amounts and a higher number of F0 peaks and valleys compared to native English speakers. This pattern can be attributed to the tonal nature of Mandarin, which involves frequent pitch variations at the syllable level. The pitch range, however, did not show significant differences across groups.

In terms of focus realization, GroupNN is more similar to GroupNA, while GroupIN closely resembles GroupAD across all other variables. As the proficiency level decreases, there is a noticeable reduction in tonal consistency. Among the four variables—Sentence Type, Focus Position, Syllables, and Final Word Type—the influencing factors are Sentence Type and Focus Position, with Syllables showing a minor effect, and Final Word Type having no significant impact.

In terms of boundary tones, all groups have great performance, with GroupNA > GroupNN ≈ GroupAD > GroupIN. In English intonation, Sentence Type is the only variable that had a clear effect on boundary tones.

Overall, the study confirms that Mandarin learners of English are influenced by their tonal first language, in the acquisition of English intonation. The systemic and realizational dimensions of LILt provide a useful framework for understanding how these L1-L2 differences manifest in the English intonation, particularly in terms of Overall sentence intonation, Focus realization, and Boundary tones. GroupIN and GroupAD exhibit relatively large deviations from GroupNA across most variables, suggesting persistent differences in both phonological and phonetic aspects of English intonation. In contrast, GroupNN demonstrate only minor deviations, with their performance in the systemic dimension closely aligning with that of GroupNA. However, their performance on boundary tones within the systemic dimension is weaker compared to their performance on focus realization. Near-natives have largely acquired the phonological structure of English intonation. However, certain differences at the realizational dimension suggest that there is still room for improvement in phonetic implementation.

6.2 Implications

The findings contribute to our understanding of the challenges Mandarin learners face in learning English intonation and also have implications for language teaching.

Firstly, for the majority of Mandarin learners of English who have not yet reached a near-native level, Mandarin tonal characteristics exert a significant influence on their English intonation. To mitigate this, learners must first become aware of this influence and then engage in targeted practice to gradually reduce its effects. Despite the fact that all participants in this study had over 10 years of English learning experience and were highly proficient, many still exhibited a strong foreign-accented (FA) intonation without being consciously aware of it. This lack of awareness likely stems from the predominant emphasis on segmental features in traditional language instruction, which often overlooks the need for explicit training in intonation. In my view, the primary reason for the considerable intonational differences observed in GroupAD and GroupIN compared to native speakers is precisely this lack of awareness. Importantly, the influence of negative transfer is not insurmountable—it can be effectively addressed through practice. One of the most effective approaches to intonation training is to conceptualize it as the melody of a song. The English sound system can be deconstructed, allowing learners to focus exclusively on intonation without being distracted by consonants, vowels, or even other prosodic cues.

Secondly, for the near-native group, subtle yet perceptible deviations from native speakers were still observed. This suggests that while the influence of Mandarin tonal characteristics can be significantly reduced, it is difficult to eliminate entirely. In particular, boundary tones, which inherently conflict with Mandarin final word tones, continued to exhibit differences between near-native learners and native speakers at the systemic level. While further deliberate practice may not lead to complete native-like proficiency, raising awareness of the fundamental differences between Mandarin and English intonation can facilitate more naturalistic, everyday practice. Near-native learners can continue refining their intonation, and this study sheds light on how they may progressively approximate native speaker patterns.

Thirdly, in terms of intonation pedagogy, greater attention needs to be given to lower-level learners, including those at advanced stages, as they still require focused instruction on refining their use of pitch accents and boundary tones in English. Existing textbooks often oversimplify intonational instruction by primarily presenting H* as the default pitch accent, whereas a broader range of pitch accent types should be introduced. Additionally, instructional materials should incorporate explicit explanations and contextualized examples of different boundary tones and pitch accents. The observed deficiencies in learners' intonation should not be interpreted as a lack of competence but rather as an aspect of language acquisition that can be substantially improved through targeted teaching. Thus, intonation training should no longer be overlooked in English language education.

LILt serves as the theoretical foundation of this study, and the findings contribute to the comparative analysis of Mandarin and English intonation while applying this framework to the study of L2 intonation. These results also have certain implications for the theory itself.

Firstly, LILt provides a valuable framework for examining intonational similarities and differences across languages. However, the intricate relationships between different dimensions of intonation require further investigation and clarification. At present, LILt does not offer a detailed account of the interactions between these dimensions, which poses challenges for its predictive capacity in L2 intonation research.

Secondly, as LILt is grounded in PAM-L2 and SLM, it posits that cross-linguistic similarities can lead to greater deviations in L2 intonation. However, this perspective remains somewhat ambiguous and presents difficulties when applied in prediction. Mandarin can be analyzed as possessing pitch accents and boundary tones, making it similar to the English intonation system. Conversely, it can also be characterized as lacking pitch accents and boundary tones, thereby differing fundamentally from English. Therefore, instead of relying solely on general assumptions of similarity, I argue that the influence of the L1 should be analyzed based on the specific manifestations of L2 intonation and the concrete differences between the two languages.

6.3 Limitations and Suggestions

Despite the valuable insights generated by this study, several limitations should be acknowledged. First, the sample size could have been larger. We divided the learners into three groups, with no more than 15 participants in each group, and the number of native speakers was also limited. Given the enormous amount of data processing involved—over 25 hours per participant—the sample size was constrained.

Second, for each sentence code, we only used one experimental sentence. Introducing more sentences under the same variables would have allowed for more reliable data and a broader understanding of the intonational differences between L2

English and native English.

Third, the significant group differences in the realization dimension of overall sentence intonation were not directly derived from the model's main factors. The significant differences were only observed after more detailed statistical analysis, suggesting the need for a more in-depth exploration of group variations.

Fourth, we did not directly examine the realization level of focus realization and boundary tone. The original experiment design had to be streamlined due to the extensive data processing requirements, and as a result, we did not explore these aspects as thoroughly as we could have.

The limitations above also offer suggestions for future studies. Additionally, we believe that the deviations observed in L*-related pitch accents in questions among lower-proficiency learners may also be influenced by textbooks and teaching methods, as they rarely emphasize the use of various pitch accents. Furthermore, the intricate relationship between boundary tone and final focus is a fascinating topic, and the underlying mechanisms deserves further investigation.

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Appendix 1

The Rainbow Passage

When the sunlight strikes raindrops in the air, they act as a prism and form a rainbow. The rainbow is a division of white light into many beautiful colors. These take the shape of a long round arch, with its path high above, and its two ends apparently beyond the horizon. There is, according to legend, a boiling pot of gold at one end. People look, but no one ever finds it. When a man looks for something beyond his reach, his friends say he is looking for the pot of gold at the end of the rainbow.